DISTRIBUTION PLANNING STANDARD

Saudi Electricity Company

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INTRODUCTION

The objective of Power Distribution System is to deliver the Electrical power to customers in safe, reliable and most economical way such that the customer receives a supply of Electrical power required by him at the time and place at which he can use it.

Several parameters of an Electricity supply such as frequency, continuity of supply, voltage level, etc. should be within allowable limits to ensure that the Customer obtains satisfactory performance for his electrical equipment while ensuring that the demands of the Customers continue to be met, the capital and operating costs of doing so should be reduced minimum as possible. Saudi Electricity Company (SEC) has developed this guideline through an internal consensus development process, after bringing together varied viewpoints and interests. However, it is a live working document and viewpoint expressed will be from time to time, subject to change and/or revision, for stabilization, to reflect stages of development and changes to comply with legislation and good industry practice. Comments are welcome.

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GENERAL STANDARD PRINCIPLES OF NETWORK PLANNING

The general principles of network planning are derived from The Saudi Arabian Distribution Code published by Electricity and Water Regulatory Authority (WERA)[1].

The general principles of design are outlined as follows:

- All equipment will operate within normal ratings and within the operating conditions set by the Saudi Arabian Distribution Code[2] when the system is operating anywhere from the minimum load to the forecasted maximum peak load
- Planning is based on normal and emergency equipment ratings. Emergency ratings are those, which can safely exist for a specified number of hours
- All standard materials and equipment shall be designed and constructed for satisfactory operation under the appropriate set of Service Conditions. Where local conditions differ from these standard conditions, standard material ratings shall be modified. Where it is not possible to use standard materials, other materials of higher rating may be used. These standard Service Conditions, while representative of the major load regions, will be exceeded at some locations within the Kingdom. It is therefore necessary for the user to confirm whether local conditions exceed standard conditions and to take appropriate action. Special surveys to define environmental and soil conditions should be carried out prior to major engineering works.

[1] WERA published The Saudi Arabian Distribution Code in 2021. In case of any updates to the aforementioned code, this document will need to be reviewed so that relevant changes are reflected

DEFINITIONS

Name	Definition
Voltage	The root mean square (rms) value of power frequency voltage, on a three-phase alternating current electrical system. This is measured between phases, unless otherwise indicated.
System Voltage	A value of voltage used within the Utilities Power System. It is generally expressed as a nominal voltage with an upper limit only. This upper limit defines the rated voltage for equipment
Service Voltage	The voltage value at the Customer's interface, declared by the Power Utility. This is generally expressed as a voltage range, in terms of a nominal voltage with plus and minus percentage variations.
Nominal Voltage	The voltage value, by which a system is designated and to which certain operating characteristics of the system are related.
Utilization Voltage	The voltage value at the terminals of utilization equipment, for example, domestic appliances. It is generally expressed as a voltage range, in terms of a nominal voltage with plus and minus percentage variations.
Highest Voltage	The highest effective value of voltage, which occurs under normal operating conditions at any time and at any point on the System. The term does not include transient voltages due to fault or switching
Low Voltage (LV)	A voltage used for the supply of electricity, the upper limit of nominal RMS value of which does not exceed 1kV.
Medium Voltage (MV)	A voltage used for the supply of electricity, the nominal value of which is between 1kV and 69 kV.
High Voltage (HV)	A voltage, used for the supply of electricity, the lower limit of nominal RMS value of which is greater than 100kV
Extra-high Voltage	An Extra-high Voltage: A voltage level exceeding 230kV.
Voltage Drop	The difference in voltage between one point in a power system and another, typically between the supply substation bus and the extremities of a network. This is generally expressed as a percentage of the nominal voltage.

Name	Definition
Network	The aggregate of Cable and or overhead line and associated equipment, used to transport electrical power between Substations or between Substations and Customer loads.
Substation	The aggregate of electrical equipment and facilities, by which electrical power is transformed in bulk from one voltage to another.
Distribution Substation	Transformed electrical power from medium voltage to low voltage
Main Distribution Substation	Transformed electrical power from medium voltage to another medium voltage.
Grid station	Transformed electrical power from high voltage to medium voltage
Phase Un-Balance	A measure of asymmetry between phase parameters in terms of magnitude, phase angle or both. This is generally expressed as a ratio of negative and or zero sequence values to the positive sequence value.
Ambient Temperature	The surrounding temperature (in the absence of the equipment) of the immediate environment in which equipment is installed. This temperature normally varies. A derived constant value is taken for the purposes of designing or rating equipment.
Ampacity	The maximum amount of electric current a conductor or cable can carry before sustaining immediate or progressive deterioration. The RMS electric current which a conductor or cable can continuously carry while remaining within its temperature rating.
Effectively Earthed System	A Power System in which the Neutral is connected to Earth either directly or through a Neutral Resistor.
Power System	The aggregate of all electrical equipment used to supply electrical power to a Customer, up to the Customer interface.
Distribution System	The aggregate of electrical equipment and facilities used to transfer electrical power to the Customer. Distribution Systems typically operate at voltages in the medium and low voltage ranges
Power Utility	Any entity that generates and supplies electrical power for sale to Customers.

Name	Definition
Urban	For Power supply purposes an urban area shall be interpreted as any town or city.
Rural	A rural area shall be interpreted as any location outside an urban area.
Supply Request	It is the request applied by the customer to get electric power supply from SEC's power system. It can contain a single building or multiple buildings and subsequently it can contain a single unit or multiple units and subsequently it can contain a single KWH Meter or multiple KWH Meters.
Unit	It is intended for the building's unit. Each unit should be used by one customer. Each building can contain a single unit or multiple units. One KWH Meter according to SEC regulations should supply each unit.
Customer	Any entity that purchases electrical power from a power utility, where each kWh meter represents customer. It is the owner of the supply request submitted to SEC to get electrical power.
Customer Interface	The point at which a customer's load is connected to the SEC's power system (location of energy meter). This shall normally be taken as the load side of the customer metering installation. The SEC shall normally be responsible for operating and maintaining all equipment on the supply of this interface. The customer shall be responsible for all equipment on the load side of this interface.
Power	The rate (in kilowatts) of generating, transferring or using energy.
Active Power	The product of R.M.S value of the voltage and R.M.S value of the in-phase component of the current. It is usually given in (K.W).
Apparent Power	The product of R.M.S value of the voltage and R.M.S value of the current. It is usually given in (K.V.A).
Reactive Power	The product of Voltage and current and the sine of the phase angle between them. Normally measured in (KVAR)
Power Factor	The ratio of active power to apparent power.
Demand Load	It is the maximum load drawn from the power system by a customer at the customer’s interface (either estimated or measured). Demand load = Connected load x Demand factor
Connected Load	The sum of the nameplate ratings of all present and future electrical equipment installed by a customer. Connected load is measured in Volt-amperes (VA).

Name	Definition
Contracted Load	It is the capacity of power supply (in Volt-Amperes)) which provided to the customer's KWH Meter.
Demand Factor (DF)	It is the ratio of the Demand Load of a customer's building's unit to the Connected Load of that customer's building's unit.
Individual Equipment Demand Factor (IEDF)	It is the demand factor used to calculate the demand of a specific piece of equipment. Value of IEDF generally varies between 0.1 and 1.0
Coincidence Factor	It is the ratio of the Coincident Demand Load of a customer's building with group of units (KWH Meters) to the Total Demand Load of that customer's building both taken at the same point of supply. $Coincidence\ Factor = \frac{Coincident\ Demand\ Load}{Total\ Demand\ Load}$
Diversity Factor	It is the inverse of the Coincidence Factor.
Coincident Demand Load (CDL)	It is the maximum (coincident) demand load of a customer's building with multiple units over a specified interval of time. It must be calculated from the Total Demand Load of that customer's building multiplying by the approved coincidence factor of that customer's building. It is expressed in Volt-Amperes (VA).
Outage	Any loss of supply to a Customer
Fault Outage	A loss of supply to a Customer due to some un-planned event in the Power System.
Frequency	The rate of oscillation of the AC supply. This is generally expressed as a frequency range, in terms of a nominal frequency in Hz (cycles per second), with plus and minus percentage limits.
Fundamental Frequency	The operating or system frequency of the Power System. Parameters whose frequency is the same as the fundamental frequency are referred to as fundamental parameters.
Interharmonic Frequency	Any frequency which is not an integer multiple of the fundamental frequency. By extension from harmonic order, the interharmonic order is the ratio of an interharmonic frequency to the fundamental frequency. This ratio is not an integer. (Recommended notation: m)
Harmonic Frequency	Frequency which is an integer multiple of the fundamental frequency. The ratio of the harmonic frequency to the fundamental frequency is the harmonic order (recommended notation: h)

Name	Definition
Harmonics	Parameters which vary at integer multiples of the nominal frequency of the Power System. The magnitudes of these quantities are generally expressed as percentage values of the fundamental parameter
Harmonic Distortion	The measure of a harmonic impressed on some fundamental quantity, which usually refers to voltage. This is generally expressed as the ratio of the magnitude of the relevant harmonic, to the fundamental value.
Even Harmonics	Harmonic quantities, at even multiples of the fundamental Frequency
Odd Harmonics	Harmonic quantities, at odd multiples of the fundamental frequency.
Total Harmonic Distortion (THD)	Is the aggregate of the Harmonic distortions at all Harmonic Frequencies. This is expressed as the root mean square value of Harmonic distortions, at all Harmonic Frequencies.
Disturbance	Any electromagnetic phenomenon which, by being present in the electromagnetic environment, can cause electrical equipment to depart from its intended performance .
Zero Sequence Voltage	A set of phase voltages of equal magnitude and zero phase angle, relative to each other. The 3-phase values are thus in phase with each other. The term zero sequence may also be applied, in the same sense, to AC currents, impedances, etc.
Negative Sequence Voltage	A set of symmetrical phase voltages (of equal magnitude and 120º phase angle) having the opposite phase sequence to that of the source. The term negative sequence may also be applied, in the same sense, to AC currents, impedances, etc.
Positive Sequence Voltage	A set of symmetrical phase voltages (of equal magnitude and 120º phase angle) having the same phase sequence as the source. The term positive sequence may also be applied, in the same sense, to AC currents, impedances, etc.

ABBREVIATIONS

A (Amp):	Ampere
AC:	Air Conditioning
ACSR:	Aluminum Conductor Steel Reinforced
AL:	Aluminum
CAPEX:	Capital Expenses
CF:	Coincidence Factor
CDL:	Coincident Demand Load
CB:	Circuit Breaker
CT:	Current Transformer
CYME:	Load flow software from Eaton
DED:	Distribution Engineering Department
DL:	Demand Load
DF:	Demand Factor
DP:	Distribution Pillar
DPS:	SEC Distribution Planning Standards
DOM:	SEC Distribution Operations Manual
WERA:	Electricity and Water Regulatory Authority
ED:	Electricity Department
GIS:	Geographical Information System
h:	Harmonic Order
HQ:	Saudi Electricity Company Headquarters
HV:	High Voltage
Hz:	Hertz
KA:	Kilo Ampere
KV:	Kilo Volt
KVA:	Kilo Volt Ampere
KW:	Kilo Watt
KWH:	Kilo Watt Hour
LBS:	Load Break Switch

LV:	Low Voltage
MCCB:	Moulded Case Circuit Breaker
MDN:	Main Distribution Substation
MRMU:	Metered Ring Main Unit
MV:	Medium Voltage
MVA:	Mega Volt Ampere
NOC:	No Objection Certificate
NOP:	Normal Open Point
O&M:	Operations & Maintenance
OH:	Overhead
PF:	Power Factor
PMT:	Pole Mounted Transformer
RMU:	Ring Main Unit
ROW:	Right of Way
SEC:	Saudi Electricity Company
SLD:	Single Line Diagram
SS:	Substation
THD:	Total Harmonic Distortion
UDS:	Unified Distribution System
UG:	Underground
V:	Volt
VA:	Volt Ampere
VD:	Voltage Drop
VIP:	Very Important Person
XLPE:	Cross linked Polyethylene

1 PLANNING STANDARDS

1.1 Design standards

Frequency

Standard Frequency: The standard system frequency shall have a nominal value of 60 Hz.

Operating Range: The maximum permissible frequency operating range shall be between 59.8 Hz and 60.2Hz .The preferred operating range should be between 59.9 Hz and 60.1 Hz.

Standard Distribution Voltages

The voltages listed in Table 1 shall be used as standard service voltages at the interface with power customers. The service voltage shall be maintained within the range defined by the indicated lowest and highest values, under steady state and normal system conditions and over the full loading range of the system. Where two voltages are listed e.g., 400/230 V the lower value refers to the phase to neutral voltage. All other values are phase-to-phase voltages.

Table 1: Standard Service Voltages

Nominal Voltage	Lowest Voltage	Highest Voltage
400/230 V	380/218.5 V	420/241.5 V
220/127 V	209/120 V	231/134 V
380/220 V	360/209 V	400/231 V
13.8kV	13.1 kV	14.5 kV
33kV	31.4 kV	34.7 kV
34.5kV*	32.78 kV	36.23 kV
69kV*	65.55 kV	72.45 kV
Percentage Limits	-5%	+5%

Note: * Existing but non-standard voltages

Harmonics

The maximum planning level of Harmonics in the power system are shown in following tables (2, 3, 4, and 5):-

Nominal Voltage	Total Harmonic Distortion (%)
230 – 400V	5.0
127 – 220V	5.0
11kV & 13.8kV	4.0
33kV-69kV	3.0

Table 2: Maximum continuous Total Harmonic Distortion levels expressed in % of Voltage at fundamental Frequency

Individual Harmonic planning level limits for LV Distribution Systems up to 1kV are shown in table 3

Odd Harmonics Non-multiple of 3)		Odd Harmonics (Multiple of 3)		Even Harmonics	
Order 'h'	Harmonic Voltage %	Order 'h'	Harmonic Voltage %	Order 'h'	Harmonic Voltage %
5	7.4	3	5.0	2	1.5
7	5.5	9	1.5	4	0.9
11	3.5	15	0.4	6	0.7
13	3.0	21	0.3	8	0.6
17	2.3	>21	0.2	10	0.5
19	2.0			>12	0.4
23	1.7				
25	1.5				
>25	(38.5/h-0.27				

Table 3: Distribution System at Voltages up to 1kV - maximum continuous individual Harmonic distortion planning levels expressed in % of Voltage at fundamental Frequency

Individual Harmonic planning level limits for the Distribution System >1kV and ≤35kV are shown in table 4

Odd Harmonics multiple of 3)		Odd Harmonics (Multiple of 3)		Even Harmonics	
Order 'h'	Harmonic Voltage %	Order 'h'	Harmonic Voltage %	Order 'h'	Harmonic Voltage %
5	6.3	3	4.0	2	1.5
7	4.4	9	1.2	4	0.8
11	2.7	15	0.3	6	0.6
13	2.3	21	0.2	8	0.5
17	1.7	>21	0.2	10	0.5
19	1.5			12	0.4
23	1.2			14	0.4
25	1.1			16	0.3
>25	(32.3/h) -.0.2			>16	(2.5/h)+0.22

Table 4: Distribution System at Voltages >1kV and ≤35kV - maximum continuous individual Harmonic distortion planning levels expressed in % of Voltage at fundamental Frequency

Individual Harmonic planning level limits for the Distribution System >35kV and ≤69kV are shown in table 5:

Odd Harmonics (Non-multiple of 3)		Odd Harmonics (Multiple of 3)		Even Harmonics	
Order 'h'	Harmonic Voltage %	Order 'h'	Harmonic Voltage %	Order 'h'	Harmonic Voltage %
5	4.1	3	2.0	2	1.1
7	2.9	9	1.0	4	0.6
11	1.9	15	0.3	6	0.5
13	1.6	21	0.2	8	0.4

17	1.2	>21	0.2	10	0.4
19	1.1			12	0.3
23	0.9			14	0.3
25	0.8			16	0.3
>25	20.4/h			>16	1.9/h+0.16

Table 5: Distribution **System** at **Voltages >35kV** and ≤69**kV** - maximum continuous individual **Harmonic** distortion planning levels expressed in % of **Voltage** at fundamental **Frequency**

Phase Unbalance

Under normal system conditions the three phase voltages shall be balanced at MV, and higher voltages in the system, such that the negative phase sequence voltage does not exceed 2% of the positive phase sequence voltage.

- Phase Unbalance for Users with a Dedicated Transformer

Users with a dedicated transformer shall ensure that their loads are so balanced that load unbalance which they create on the MV system meets the User negative-phase-sequence current criteria of 1%.

- Phase Unbalance for Users supplied at 13.8kV

  Users supplied at 13.8kV or a higher Voltage shall balance their loads, such that the load phase unbalance at the Customer User interface meets the User negative-phase-sequence current criteria of 1%.

- Phase Unbalance for all Other Users

  All other customers shall balance their loads over the three phases to the greatest degree possible. The SEC shall then balance these loads, within the power system, to meet the above criterion.

System Earthing

All the customers should apply only allowed method of Earthing of the distribution system and must ensure appropriate Earthing value mentioned in distribution code and in terms and conditions set at time of connection agreement.

Resistance Earth

The recommended resistance limits for different installations should be as shown in table 6 below.

Installation	Resistance
System Earthing	≤ 5 ohms
All distribution sub-station	≤ 5 ohms
Surge arrestors	≤ 5 ohms
LV Distribution panel	≤ 10 ohms

For details please refer to distribution Network grounding construction specifications SDCS-03 with its latest updates

Power Factor

Each customer shall maintain a power factor of 0.85 lagging[2] or higher at the interface.

$$\text{Power factor} = \cos\theta = \frac{P}{S} = \frac{P}{\sqrt{P^2 + Q^2}}$$

Where S: apparent power P: active power Q: reactive power

For industrial / government / commercial customers, having contracted load greater than 1.0 MVA, the minimum allowable power factor is 0.95 lagging. In case of deviation, a penalty will be imposed, as per Customer Services manual with latest updates.

Low power factor at customers end ultimately contributes to overall poor factor in SEC' distribution network, resulting in negative impacts of:

- Excessive voltage drops
- Technical losses
- Decrease in capacity of system

Short Circuit Levels

Short circuit levels on secondary bus bars for all grid stations should be evaluated by DED. Relevant studies need to be conducted using specialized software (such as CYME). Furthermore, The short-circuit rating of equipment at the connection point shall not be less than the design fault level of the Distribution System as shown in Table 7 below

Nominal Voltage	Load (KVA)	Short circuit level RMS symmetrical (kA)
220/127 V	<= 152	21
	>152	45
400/230 V	<=500	20
	>500	30
13.8 kV	All	21
33 kV	All	25
69 kV	All	31.5

[2] Power factor may be leading for small scale solar rooftop installations

1.2 Standard Conditions

Standard underground cables Rating Conditions

Table 8 Standard conditions overview

Condition	Value
Ground Temperature Direct Buried/ Underground Ducted, at depths of one (1) meter and more	35 °C
Soil Thermal Resistivity, at depths of one (1) meter and more	2.0 °C.m/w
Maximum Continuous Conductor Operating Temperature (XLPE)	90 °C
Maximum Short Circuit Conductor Temperature – 5 second Maximum Duration (XLPE)	250 °C
Loss Load Factor – Daily (Equivalent Load Factor = 0.88)	0.8
Burial Depth (to the center of the cable)	0.65 meter for LV & 0.8 meter for MV
Circuit Spacing (center to center)	0.30 m

However, depending on the conditions of usage, ratings of cables will need to be adjusted by applying de-rating factors:

$$Corrected\ \ Cable\ Rating = Cable\ Rating \times Burial\ Depth\ Correction\ Factor \times Soil\ Thermal\ Resistivity\ Correction\ Factor \times Ground\ Temperature\ Correction\ Factor$$

Table 9: Burial Depth Correction Factors

Burial Depth (m)	Correction Factor
0.5	1.03
0.6	1.01
0.65	1
0.8	0.98
1	0.96
1.25	0.93
1.50	0.91
1.75	0.90
2.0	0.89

Note: Burial depth refers to the distance from the center of the cable installation to the final grade (surface) level.

Table 10: Soil Resistivity Correction Factors

Soil Resistivity (degree K.m/w)	Correction Factor
0.7	1.44
1.0	1.33
1.2	1.25
1.5	1.13
2	1
2.5	0.90
3	0.81

Note: The value of soil thermal Resistivity chosen shall make full allowance for dry-out of the soil adjacent to the cables, due to heat emission from the cables. All soil within the 50°C. Isotherm surrounding the cables should be assumed to be dry. The soil at the ground surface should also be assumed dry. Thus, the value of soil thermal Resistivity chosen shall be higher than the background value derived from site measurements.

Table 11: Ground Temperature Correction Factors

Ground Temperature (degree C)	Correction Factor
25	1.08
30	1.04
35	1.00
40	0.95

Standard Overhead Lines Conductors Rating Conditions

Overhead Lines Conductors ratings based on the following standard conditions in the table below:

Table 12 Standard conditions overview

Condition	Value
Ambient Temperature	50 °C
Minimum wind velocity	0.6 m/sec
Altitude (above sea level)	1000m
Maximum continuous conductor operating temperature	80 °C
Emissivity (for Cu and Al)	0.5
Absorptive (of solar heat)	0.5

Furthermore, certain corrections need to be taken into account for use of conductors in various environment conditions using the following formula:

$$Corrected\ \ OH\ Conductor\ Rating = OH\ Conductor\ Rating \times Ambient\ Temperature\ Correction\ Factor \times Altitude\ Correction\ Factor \times Conductor\ Temperature\ Correction\ Factor$$

The correction factors are mentioned in the tables (13, 14, and 15.16) below:

Table 13

Ambient Air Temperature (degree C)	Correction Factor
45	1.10
50	1.00
55	0.88

Table 14

Altitude (in meters above sea level)	Correction Factor
0	1.05
1000	1.00
2000	0.95
3000	0.90

Table 15

Wind Velocity (in m/s)	Correction Factor
Natural convection	0.60
0.6	1.00
1.0	1.15
2.0	1.38
5.0	1.80

Table 16

Conductor Temperature (in degree C)	Correction Factor
75	0.88
80	1.00
85	1.10
90	1.19
120	1.60

EXAMPLE ON APPLYING CORRECTION FACTOR IN APPENDIX 2

1.3 Distribution Security Standards

Table 17 below sets out the recommended levels of supply for progressive of group demand for distribution network up to 33 KV:-

Table 17 :Levels of Distribution Security by WERA

Class	Group Demand	First Circuit Outage	Interpretation
A	Up to 2 MVA	As per repair time	N
B	> 2 to 12 MVA	• Within 3 hours (for sections of feeder with > 2 MVA demand) • As per repair time (for section of feeder with outage)	N-1 (Manual switched alternative) Group demand would normally be supplied on an open ring system
C	> 12 MVA	• Within 15 minutes	N-1 (Auto or remote switched alternative) Group demand will normally be supplied by at least two normally closed circuits or by one circuit with supervisory or automatic switching to an alternative circuit

1.4 Reliability Standards

SEC has set a number of supply standards for the customers. SEC makes all possible efforts to achieve these standards, evaluated in terms of following indices:

- System average interruption frequency index (SAIFI)
- System average interruption duration index (SAIDI)
- Customer average interruption duration index (CAIDI)
- Average system availability index (ASAI)
- Momentary average interruption frequency index (MAIFI)
- Customers minutes lost (CML)

For details, refer to Distribution Operation Manuals (DOM) with latest updates.

Standardization of recording of fault incidents, resulting outages and their duration must be established in order to achieve proper calculations of reliability indices.

2 Customer Load Estimation Methodology

2.1 Classification of customer facilities

In this chapter, Customers are classified according to the nature of use of their Facilities in reality and according to their connected and demand load estimation methodology. This shall not cause any conflict to any other customer classification used for financial and tariff purposes.

Customers' Facilities type should be determined according to the nature of the use of their Facilities in reality and according to an approved license or official document from the authority related to the nature of their use. In case there is a difference between the reality and the license, Customers' Facilities type should be determined according to the nature of the use of their Facilities in reality.

Table 18: Overview of customer facility categories

Category		Definition
C1 : Normal Residential Dwelling	Description	Any facility used as dwelling meant for private use.
	Includes	Houses, duplexes, apartments, villas, palaces, istrahat, etc.
C2 : Normal Commercial Shops	Description	Any facility designed for use as normal commercial shops.
	Includes	Commercial shops and stores, gold shops, pharmacies, boutiques, etc.
C3 : Furnished Flats	Description	Any facility designed for use as furnished flats (including labor housing)
	Includes	Furnished flats.
C4 : Hotels	Description	Any facility designed for use as hotels.
	Includes	Hotels, motels.
C5 : Malls	Description	Any facility designed for use as malls or shopping centers.
	Includes	Shopping centers, malls, supermarkets, hypermarkets.
C6 : Restaurants	Description	Any facility designed for use as restaurants.
	Includes	Restaurants, coffee shops, cafeteria.
C7 : Offices	Description	Any facility designed for use as work offices.
	Includes	Commercial offices, government offices, office complexes, offices, banks
C8 : Schools	Description	Any facility designed for use as schools.
	Includes	Schools, nursery, private training institute
C9 : Mosques	Description	Any facility designed for use as mosques.
	Includes	Mosques
C10 : Mezzanine in Hotel	Description	Any facility designed for use as mezzanine floor.
	Includes	Mezzanine in hotel
C11 : Common Area/Services in Buildings	Description	Any facility designed for use as common area/services in buildings.
	Includes	Roof, corridors, stairs, piazza
C12 : Public Services Facilities	Description	Any facility designed for use as public services facilities.
	Includes	Outdoor bath rooms, washing rooms

Category		Definition
C13 : Indoor Parking	Description	Any facility designed for use as indoor parking.
	Includes	Indoor parking
C14 : Outdoor Parking	Description	Any facility designed for use as outdoor parking
	Includes	Outdoor parking
C15 : Streets Lighting	Description	Any facility designed for use as streets lighting.
	Includes	Streets lights, roads lights
C16 : Parks & Gardens	Description	Any facility designed for use as parks & gardens.
	Includes	Parks & gardens
C17 : Open Spaces	Description	Any facility designed for use as open spaces.
	Includes	Open spaces
C18 : Hospitals\Medical Facilities	Description	Any facility designed for use as hospitals\medical facilities.
	Includes	Hospitals, medical centers
C19 : Medical Clinics	Description	Any facility designed for use as medical clinics (which is of smaller area and has limited medical facilities compared to a hospital)
	Includes	Medical clinics
C20 : Universities\High Educational Facilities	Description	Any facility designed for use as universities\high educational facilities.
	Includes	Universities, colleges, high educational institutes
C21 : Light Industries	Description	This includes all industries with load up to (4 MVA) inside designated industrial area or having industrial license.
	Includes	Small factories, livestock, poultry, dairy farms
C22 : Workshops	Description	Any facility designed for use as workshops.
	Includes	Workshops
C23 : Cooling Stores	Description	Any facility designed for use as cooling stores.
	Includes	Cooling stores
C24 : Warehouses	Description	Any facility designed for use as warehouses.
	Includes	Warehouses
C25 : Community Halls	Description	Any facility designed for use as community halls.
	Includes	Community halls, wedding party halls, auditorium
C26 : Recreational Facilities	Description	Any facility designed for use as recreational facilities.
	Includes	Clubs, theaters, cinemas, gymnasium
C27 : Farms\Agricultural Facilities	Description	This includes farms used for producing agricultural products (big one or small)
	Includes	Farms, green houses, production farms
C28 : Fuel Stations	Description	Any facility designed for use as fuel stations.
	Includes	Petrol pumps, fuel stations
C29 : Bulk Factories	Description	This includes all industries with load more than (4 MVA) inside designated industrial area or having industrial license.
	Includes	Big factories, manufacturing plants

2.2 Methodology

- The Residential/ Commercial customers (C1&C2) whose areas are defined the covered area tables in Appendix 1. Such customers are normally expected to have uniform behavior in terms of electrical requirements.
- Large residential/Commercial customers (C1&C2), where the covered area is beyond the limits given in the relevant tables in Appendix 1 OR customers who have a variety of load requirement irrespective of Floor Area or Lot Size, the power supply requirement of all such customers shall be estimated by using load density factor in table 19 Declared Load Method

$$C1 = 116 \text{ VA/ } m^2 \qquad C2 = 172 \text{ VA/}m^2$$

- Facilities Types (Customer categories C3 to C29) which their Connected Load can be estimated according to Area Load Density Method the power supply requirement of all such customers can be estimated by using load density factor in table 19.
- Facilities Types (Customer categories C3 to C29) which their Connected Load cannot be estimated according to Area Load Density Method the power supply requirement of all such customers shall be estimated y using Declared Load Method in KVA as 3phase system
  a. Notes For facility categories C1&C2 loads provided by customer can be more than loads estimation using by (Table VA/m2 Min. Loads) in this case customer should provide technical justifications for that loads
  b. If any type of customer does not provide his study. SEC will calculate the load by using (VA/m2 All Categories Min. Loads: Area-based types & Non Area-based type - one value) and the customer shall confirm that satisfy his requirements and no more loads required and such requirements should be documented.

2.3 Determining covered/built-up area

There are two types of Area used in Area Load Density Method as follows:

- Unit’s covered/built-up area

It is an individual built-up area for a customer's unit. It is calculated based on the drawings provided by the customer of its building or the project with approved municipal documents.

- Total covered/built-up area

It is the total built-up area for a plot land. It is the sum of covered or roofed areas excluding services areas which are open. Building's covered area shall be cross checked with approved municipal documents/permits to ascertain its correctness. Covered area is calculated based on the drawings provided by the customer of its building or the project.

Where total covered area is not mentioned in the building permits, it can be calculated by using the following equation :

$$Total\ Covered\ (Built-Up)Area\ = Plot\ Land\ Area\ \times Allowed\ Percentage\ of\ Construction\ on\ Plot\ Land \times Number\ of\ Floors.$$

2.4 Connected Load Estimation

Connected loads estimation for normal residential dwelling Commercial shops (Facility category C1,C2)

- Calculate the total connected load (KVA) according to the Unit covered/built-up area (square meter) Using the tables (1,2,3,4) in Appendix 1

Connected loads estimation for combined type customer (C1 & C2)

In case the customer building consists of both residential and commercial load e.g. the connected load shall be assessed separately corresponding to the areas associated with each using the respective tables. The total connected load shall be the sum total of the two values.

a. Determine floor area of the customer buildings separately for each category.

b. Read out from the appropriate tables the circuit breakers rating in each category.

c. Determine the total load by simple addition of circuit breakers ratings in each category.

Connected loads estimation for C1 & C2 with central AC

a. If the customer declared load for central AC happens to be less than unit AC load, central AC load shall be ignored.

b. Since AC load is already included in the values provided in the tables as customer minimum load, the same shall be subtracted AC load 70% from the connected load before adding central AC load at the following:

   Residential Customers = 81 VA/m²

   Commercial Customers = 120 VA/m²

- Determine covered area of the customer building.
- Determine the total connected load from the appropriate tables.
- Determine the unit AC connected load as per procedure given above.
- Subtract the estimated unit AC load from the total connected load and add customer declared central AC load to obtain total connected load of the customer provided
- Calculate the customer Connected load as follows :

Connected Load = Non-AC Connected Load + Central AC Load

Read out all other parameters of power supply from the tables against the computed total connected load, as obtained by above calculations.

If the connected load of the customer exceeds the circuit breaker rating provided for the slab, select the next higher size breaker which is adequate to provide for the connected load of the customer.

Connected loads estimation for customer categories C1 & C2 with built-up area exceeding table limits

For such customers (type C1 and type C2) an average load requirement VA/m² is considered as appropriate method for the load calculation as follows :

$$Total\ Connected\ Load\ (KVA) = Built\ up\ Area\ (m^2) \times Load\ Density\ (VA/m^2)\ /\ 1000$$

By using the following load density :

Residential Customers = 116 VA/m^2

Commercial Customers = 172 VA/m^2

Connected loads estimation for customer categories C1 and C2 with ceiling height above 3.5m

Assessment of AC Load for Mezzanine cases or for buildings with ceilings higher than the standard height of 3.5 meters shall be as follows:

Additional volume (m^3) = [Total Height (m) - Standard Height (3.5 m)] X Covered Area (m^2)

Additional AC Load (VA) = (24 VA/m^3) × Additional volume (m^3)

The calculated extra AC load by above formula shall be added as an additional load as follows:

Total Connected Load = Standard Connected Load from tables (1,2,3,4) Appendix 1 + Additional AC Load

Connected loads estimation for area-based types with additional special loads

Connected loads according to Area Load Density Method are only covering normal loads , any additional loads should be considered & added as additional special loads. Examples include swimming pool loads, additional elevators, Central AC, etc.

Connected loads estimation for other area-based customer facility types (C1 – C29)

For all such customers (from type C1 up to type C29) an average load requirement VA/m² is considered as appropriate method for the load calculation. This is illustrated in Table .

$$Total\ Connected\ Load\ (KVA) = Built\ up\ Area\ (m^2) \times Load\ Density\ (VA/m^2)\ /\ 1000$$

Table 19: Load Estimation for (other area-based facilities, facilities without AC and facilities in Winter Peak Area)

Code	Customer Category	Load Estimation for other area-based Facilities		Load Estimation for Facilities without AC (District Cooling)		Load Estimation for Facilities in Winter Peak Area (Without AC and with Heating)	
		Loads included*	VA/m^2	Loads included*	VA/m^2	Loads included*	VA/m^2
C1	Normal Residential Dwelling	(Lights + Air Conditioning + Power Sockets)	116	(Lights + Power Sockets)	35	(Lights + Air Heating + Power Sockets)	96
C2	Normal Commercial Shops	(Lights + Air Conditioning + Power Sockets)	172	(Lights + Power Sockets)	52	(Lights + Air Heating + Power Sockets)	144
C3	Furnished Flats	(Lights + Air Conditioning + Power Sockets)	140	(Lights + Power Sockets)	64	(Lights + Air Heating + Power Sockets)	116
C4	Hotels	(Lights + Air Conditioning + Power Sockets)	192	(Lights + Power Sockets)	76	(Lights + Air Heating + Power Sockets)	156
C5	Malls	(Lights + Air Conditioning + Power Sockets)	204	(Lights + Power Sockets)	60	(Lights + Air Heating + Power Sockets)	160
C6	Restaurants	(Lights + Air Conditioning + Power Sockets)	188	(Lights + Power Sockets)	76	(Lights + Air Heating + Power Sockets)	156
C7	Offices	(Lights + Air Conditioning + Power Sockets)	176	(Lights + Power Sockets)	72	(Lights + Air Heating + Power Sockets)	144
C8	Schools	(Lights + Air Conditioning + Power Sockets)	144	(Lights + Power Sockets)	64	(Lights + Air Heating + Power Sockets)	120
C9	Mosques	(Lights + Air Conditioning + Power Sockets)	148	(Lights + Power Sockets)	52	(Lights + Air Heating + Power Sockets)	120
C10	Mezzanine in Hotel	(Lights + Air Conditioning + Power Sockets)	80	(Lights + Power Sockets)	32	(Lights + Air Heating + Power Sockets)	64
C11	Common Area/Services in Buildings	(Lights + Power Sockets)	48	(Lights + Power Sockets)	48	(Lights + Power Sockets)	48
C12	Public Services Facilities	(Lights + Power Sockets)	40	(Lights + Power Sockets)	40	(Lights + Power Sockets)	40
C13	Indoor Parking	(Lights + Vans + Gates + Safety Systems)	24	(Lights + Vans + Gates + Safety Systems)	24	(Lights + Vans + Gates + Safety Systems)	24
C14	Outdoor Parking	(Lights)	4	(Lights)	4	(Lights)	4
C15	Streets Lighting	(Lights)	4	(Lights)	4	(Lights)	4
C16	Parks & Garden	(Lights + Water Distributor)	3.2	(Lights + Water Distributor)	3.2	(Lights + Water Distributor)	3.2

Code	Customer Category	Load Estimation for other area-based Facilities		Load Estimation for Facilities without AC (District Cooling)		Load Estimation for Facilities in Winter Peak Area (Without AC and with Heating)	
		Loads included*	VA/m^2	Loads included*	VA/m^2	Loads included*	VA/m^2
C17	Open Spaces	(Lights)	2.4	(Lights)	2.4	(Lights)	2.4
C18	Hospitals\Medical Facilities	(Lights + Air Conditioning + Power Sockets)	200	(Lights + Power Sockets)	92	(Lights + Air Heating + Power Sockets)	168
C19	Medical Clinics	(Lights + Air Conditioning + Power Sockets)	180	(Lights + Power Sockets)	80	(Lights + Air Heating + Power Sockets)	152
C20	Universities/ High Educational Facilities	(Lights + Air Conditioning + Power Sockets)	196	(Lights + Power Sockets)	100	(Lights + Air Heating + Power Sockets)	168
C21	Light Industries	(Lights + Motors + Power Sockets + AC)	224	(Lights + Motors + Power Sockets)	192	(Lights + Motors + Power Sockets + Heating)	212
C22	Workshops	(Lights + Power Sockets)	64	(Lights + Power Sockets)	64	(Lights + Power Sockets)	64
C23	Cooling Stores	(Lights +Chillers + Power Sockets)	208	(Lights + Power Sockets)	20	(Lights +Chillers + Power Sockets)	208
C24	Warehouses	(Lights + Vans + Power Sockets)	56	(Lights + Vans + Power Sockets)	56	(Lights + Vans + Power Sockets)	56
C25	Community Halls	(Lights + Air Conditioning + Power Sockets)	184	(Lights + Power Sockets)	92	(Lights + Air Heating + Power Sockets)	156
C26	Recreational Facilities	(Lights + Air Conditioning + Power Sockets)	160	(Lights + Power Sockets)	72	(Lights + Air Heating + Power Sockets)	132
C27	*Farms\Agricultural Facilities	(Lights + Power Sockets)	104	(Lights + Power Sockets)	92	(Lights + Power Sockets)	100
C28	Fuel Stations	(Lights + Power Sockets)	72	(Lights + Power Sockets)	56	(Lights + Power Sockets)	68
C29	Bulk Factories	Lights + Motors + Power Sockets)	236	(Lights + Motors + Power Sockets)	200	(Lights + Motors + Power Sockets + Heating)	224

* Load requirement for Agricultural land is calculate by using load declaration.

Table covers only normal loads. Any additional loads will be considered & added as special loads.

SEC – Load Declaration Form (SEC-LD)

- All types of customers having covered area beyond the table limits or having special power supply requirements for variety of load types such as industrial load, lighting loads, agricultural load and any other load must be assessed as per load declaration form 1 in Forms.

- Customers will be requested to fill this form at the time of filing request of supply application. This will be the basic information for the study of power supply requirement of the customer.

Service meter for normal residential dwelling and normal commercial shops types

- The service meter covers the common loads in the customer building includes service load such as (lighting of staircase, garden, water pump, swimming pool, elevators etc.).
- Minimum rating of a circuit breaker for customer facility types (C1& C2) is normally considered to be adequate for general services. If it does not meet customer requirement Declared Load method shall be used to determine service meter.

Examples for loads estimation in Appendix 2

2.5 Demand factors for all facility types

Table 20: Demand factors for all facility types

Code	Customer Category	DF
C1	Normal Residential Dwelling	0.5
C2	Normal Commercial Shops	0.6
C3	Furnished Flats	0.6
C4	Hotels	0.65
C5	Malls	0.6
C6	Restaurants	0.6
C7	Offices	0.6
C8	Schools	0.7
C9	Mosques	0.8
C10	Mezzanine in Hotel	0.65
C11	Common Area/Services in Buildings	0.7
C12	Public Services Facilities	0.65
C13	Indoor Parking	0.7
C14	Outdoor Parking	0.8
C15	Streets Lighting	0.8
C16	Parks & Garden	0.7
C17	Open Spaces	0.8
C18	Hospitals\Medical Facilities	0.7
C19	Medical Clinics	0.6
C20	Universities/High Educational Facilities	0.7
C21	Light Industries	0.8
C22	Workshops	0.8
C23	Cooling Stores	0.8
C24	Warehouses	0.6
C25	Community Halls	0.7
C26	Recreational Facilities	0.7
C27	Farms/ Agricultural Facilities	0.8
C28	Fuel Stations	0.6
C29	Bulk Factories	0.8

Individual equipment demand factors in table 5 in Appendix 1

2.6 Coincident factors

Table 21: Coincident factors

Number of Meters N	Coincident Factor CF(N)	Number of Meters N	Coincident Factor CF(N)	Number of Meters N	Coincident Factor CF(N)
1	1.000	34	0.581	67	0.568
2	0.723	35	0.581	68	0.568
3	0.688	36	0.580	69	0.568
4	0.668	37	0.579	70	0.568
5	0.654	38	0.579	71	0.567
6	0.644	39	0.578	72	0.567
7	0.636	40	0.578	73	0.567
8	0.629	41	0.577	74	0.567
9	0.624	42	0.577	75	0.566
10	0.619	43	0.576	76	0.566
11	0.616	44	0.576	77	0.566
12	0.612	45	0.575	78	0.566
13	0.609	46	0.575	79	0.566
14	0.607	47	0.575	80	0.566
15	0.604	48	0.574	81	0.565
16	0.602	49	0.574	82	0.565
17	0.600	50	0.573	83	0.565
18	0.598	51	0.573	84	0.565
19	0.597	52	0.573	85	0.565
20	0.595	53	0.572	86	0.564
21	0.594	54	0.572	87	0.564
22	0.592	55	0.572	88	0.564
23	0.591	56	0.571	89	0.564
24	0.590	57	0.571	90	0.564
25	0.589	58	0.571	91	0.564
26	0.588	59	0.570	92	0.564
27	0.587	60	0.570	93	0.563
28	0.586	61	0.570	94	0.563
29	0.585	62	0.570	95	0.563
30	0.584	63	0.569	96	0.563
31	0.583	64	0.569	97	0.563
32	0.583	65	0.569	98	0.563
33	0.582	66	0.568	99	0.563

2.7 Load Estimation for Special Cases

Table 22: Considerations for Load Estimation of Special Cases

	Special Case	Definition	Demand Factor	Coincidence Factor
1	Hajj area	Area which has permits from municipality to build buildings using for pilgrims lodging	0.9	1.0
2	Random area	Un-planned area according to the municipality and which has many buildings without construction permits from the municipality	1.0	0.8
3	Commercial center	Group of commercial shops which apply common working time so that all its shops are to be opened and closed at the same time	0.6	1.0
4	Commercial offices	Buildings with commercial offices that are open and closed at the same time	0.6	1.0
5	Workshop complex	Heavy load due to industrial / semi-industrial workshops that are open and closed at the same time	0.8	1.0

3 Procedure for coincident demand load (CDL) calculation

3.1 Low voltage Coincident demand load calculation (for 20A to 800A)

The procedure for calculating coincident demand load is as follows:

- Number of Individual units in customer's building should be determined according to SEC Customer Services Manual with its latest update.
- Connected Load (CL) in (KVA) for each Individual unit in customer's building should be estimated individually unit-by-unit
- For C1 and C2 Customer Type, Individual Circuit Breaker Rating (CBR) in (Amp) for the Individual unit in customer's building should be determined according to the estimated connected load (CL) tables in Appendix 1.
- For Customer Types (from C3 up to C29), Individual Circuit Breaker Rating (CBR) in (Amp) for the Individual unit by using (load density factor / declared load) and (CBR) should be determined to be the nearest up SEC standard (CBR).
- Number of Individual KWH Meters (N) required for the customer's building should be determined according to number of Individual units in customer's building and referring to SEC Customer Services Manual with its latest updates.
- Calculate the Coincident Demand Load (CDL) in (Amp) for the group of all KWH Meters of the customer's building as follows :

$$CDL = \left(\sum_{i=1}^{N}(CBR_i \times DF_i)\right) \times CF(N)$$

Where :

N = Number of Individual KWH Meters required for the customer's building.

$CBRi$ = Circuit Breaker Rating in (Amp) for the Individual KWH Meter no. (i).

DFi = Demand Factor for the Individual KWH Meter no. (i) which should be determined according to the utilization nature of the concerned Individual unit no. (i) in customer's building and referring to this Guideline.

$CF(N)$ = Coincident Factor for the group of all KWH Meters of the customer's building which should be determined according to Number of these KWH Meters (N) and referring to this Guideline. Use the following equation to calculate the Coincident Factor (N) :

$$CF(N) = \frac{(0.67 + \frac{0.33}{\sqrt{N}})}{1.25}$$

$For\ \mathrm{N} = 1 \Rightarrow CF(N) = 1$ $Diversity\ Factor\ (N) = 1/CF(N)$

- For a group of (N) KWH Meters in the customer's building where all of them have same Circuit Breaker Rating (CBR) in (Amp) and same Demand Factor (DF), the equation to calculate the Coincident Demand Load (CDL) in (Amp) for this group of KWH Meters could be simplified as follows : $CDL = N \times CBR \times DF \times CF(N)$
- For a group of (N) KWH Meters in the customer's building where any one of them has different Circuit Breaker Rating (CBR) in (Amp), the equation to calculate the Coincident Demand Load (CDL) in (Amp) for this group of KWH Meters will be as follows:

a. If all Circuit Breaker rating ≤160 (Amp) the equation to calculate the Coincident Demand Load (CDL) will be as follows:

$CDL = \sum_{i=1}^{N-1}(CBR_i \times DF_i) \times CF(N)$

b. If Circuit Breakers rating including one or more than 160 (Amp), then the equation to calculate the Coincident Demand Load (CDL) will be as follows:

$$CDL = \left[\sum_{i=1}^{g\ number\ of\ largest\ CB} CBR_i \times DF_i\right] CF(g) + \left[\sum_{g+1}^{N} CBR_i \times DF_i \times CF(N-g)\right]$$

g = number of circuit breaker(s) having largest rating

CDL Calculation form 2 in Forms

3.2 Low voltage Coincident demand load calculation for Private Substation (more than 800A)

- According to the calculated Coincident Demand Load (CDL) of the unit, the Connected Load (CL) in (KVA) for the KWH Meter for the unit should be determined to be the nearest up SEC standard Private Substation's (500– 1000 -1500) with main Circuit Breaker to that (CDL).

  CDL for private Substation = Sum of private Substation rating.

3.3 Medium voltage Coincident demand load calculation.

- Calculate the Coincident Demand Load (CDL) in (KVA) for the group of all Units of the customer's building as follows :

$$CDL = \left(\sum_{i=1}^{N} CL_i \times DF_i\right) \times CF(N)$$

Where :

N = Number of Individual Units required for the customer's building.

CL_i = Connected Load in (KVA) for the Individual Unit no. (i).

DF_i = Demand Factor $CF(N)$ = Coincident Factor

Note :

For a group of (N) unit in the customer's building where all of them have same connected load and same Demand Factor (DF), the equation to calculate the Coincident Demand Load (CDL) for this group could be simplified as follows :

$$CDL = N \times CL \times DF \times CF(N)$$

Note :

For a group of (N) unit in the customer's building where any one of them has different connected load, the equation to calculate the Coincident Demand Load (CDL) for this group of KWH Meters will be as follows :

$$CDL = \left[CL_{\substack{Largest\\Unit}} \times DF_{\substack{Largest\\Unit}}\right] + \left[\left(\sum_{i=1}^{N-1} CL_i \times DF_i\right) \times CF(N-1)\right]$$

3.4 Plot plan Coincident demand load calculation.

- For Customers' Buildings with LV Meters (from 20 A up to 800 A), calculate their Coincident Demand Load (CDL) on their Public Substation as follows :

$$CDL_{on\ Substation} = \left(\sum_{i=1}^{N}(CBR_i \times DF_i)\right) \times CF(N)$$

Where :

N = Number of all KWH Meters supplied by that Substation.

- For Customers' Buildings designed to be supplied by Private Substation or by MV RMU, calculate their (CDL) according to steps described in MV New Connections section.
- Calculate the Total Coincident Demand Load (CDL) for the (Development Project / Plot Plan) as follows :

$$CDL_{Total} = \sum_{i=1}^{N} CDL_i \times CF_{for\ Substations} \times CF_{for\ MV\ Feeders}$$

Where :

N = Number of all (Public Substations + Private Substations + MV RMUs) which designed to supply all Lots/Buildings within the (Development Project / Plot Plan).

$CDLi$ = Coincident Demand Load in (KVA) for the Individual element (Public Substations + Private Substations + MV RMUs) no. (i).

$CFFor\ Substations$ = Coincident Factor between (Public Substations + Private Substations + MV RMUs) = 0.9

$CFFor\ MV\ Feeders$ = Coincident Factor between (MV Feeders) = 0.9

3.5 Conversion Factor to convert (CDL) from (Amp) to (KVA)

$$CDL_{in\ KVA} = \frac{CDL_{in\ Amp} \times V_{LL} \times \sqrt{3}}{1000}$$

Where:

VLL = Nominal Voltage (line to line) of the LV Network (in volts).

A. This equation can be simplified as follows:

$$CDL_{in\ KVA} = \frac{CDL_{in\ Amp}}{F_{Conversion}}$$

Where: F Conversion =. Its values for different nominal voltages are shown in below.

Conversion factors	Standard nominal voltage		
F Conversion	400	380	220
	1.443	1.519	2.624

Examples for Calculate Coincident Demand Load (CDL) in APPENDIX 2.

4 Procedure for New Connections requests

4.1 Service request

- Electrical service request is submitted through the company's website in accordance with the requirements described in the customer services manual.
- Technical data shall be provided in the service request (load – service voltage – type of request- number of meters…).
- Customers with MV equipment must provide sufficient information such as:-

a. The characteristics of customer's switchgear and Protection data related to the interface Point
b. Single line diagram.
c. Equipment, which produces Harmonic / Fluctuating Loads.
d. Technical study

4.2 Site Visit Procedure

The detailed procedures which are to be followed when visiting or surveying the site of a proposed Project are set out as following.

Before The Site Visit

Before visiting the site, obtain copies of the following, if possible:

- Basic location data from GIS
- Geographic base map if GIS is not available (from Google Earth)
- Existing network drawing(s) from maps or GIS
- Development drawing(s) (*if available*)
- Plot Plan indicating master design and agreed equipment locations

Obtain load information if not available in GIS from

- Feeder load records
- Substation load records
- Customer files

Determine potential source(s) of supply from GIS/ maps if possible.

On The Site Visit

The main tasks for the site surveyor include but are not limited to:

- Determine existing cable, line routes and equipment positions

- Determine potential source(s) of supply.
- Note the position of proposed cables in relation to geographic features including property lines, footpaths, etc.
- Determine proposed equipment locations precisely to facilitate obtaining necessary permissions to site the equipment. In addition, this is necessary for later mapping of facilities and network information. The locations are to be noted on the layout plan.
- Measure the proposed cable and/or line route lengths. Measurement is to be done using any available method such as measuring tape, measuring wheel, laser devices, GPS devices or optical devices. Note the measured lengths on the layout sketch.
- Check the site carefully for potential ROW problems.
- Note the positions and configuration of existing network and details of equipment which is to be retired or replaced.
- It is essential that final levels and road lines be established with the relevant authority/developer to avoid future problems concerning the placing of plant and/or cable laying depths.
- The site visit check list which can be found in the subsequent section gives a more detailed outline of information to be captured during the site visit.

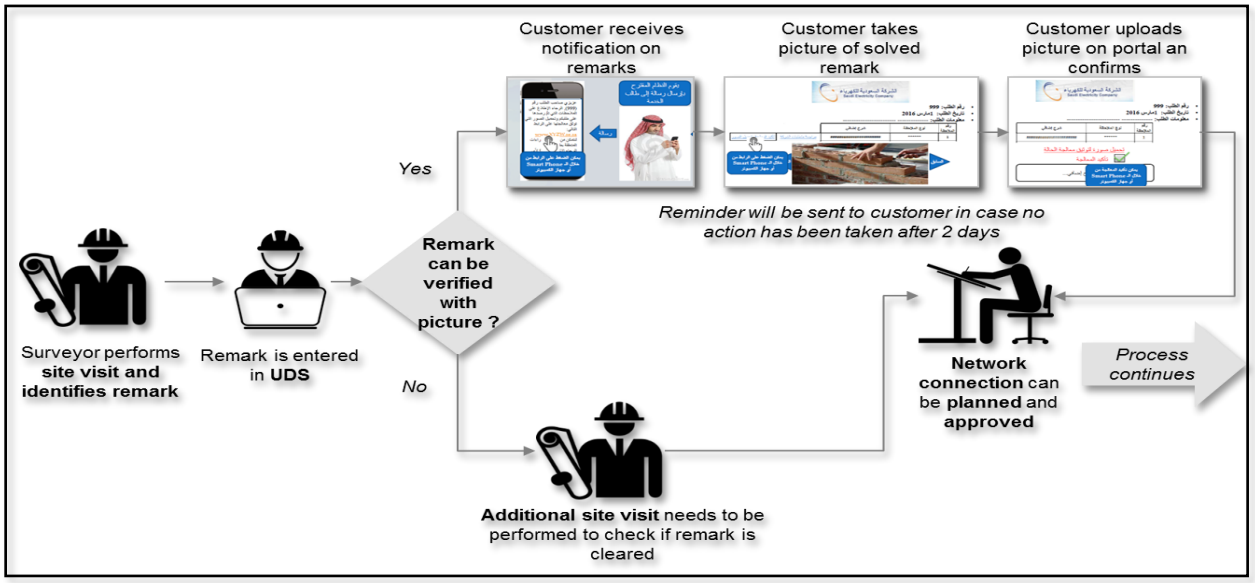
4.3 Customer Remarks

In case the site surveyor finds obstacles that would not allow for a permanent supply connection on the customer premises, he would leave remarks. Such remarks would for example be that the construction of the house it not advanced enough or that there is construction material that would block the connection construction.

In such cases, the site surveyor has to take note of these remarks on site and needs to collect the required material to document (pictures, sketches etc.) these remarks properly. Once back in the office or onsite via FFMS handheld, the site surveyor needs to enter the remarks in the UDS system. He then also needs to decide whether those remarks can be solved via picture, i.e. the customer receives a message to upload a picture of the solved remark or if a second site visit needs to be performed.

This process is to be executed post implementation of all recommendations from new connection process streamlining initiative.

The upcoming figure outlines the customer remark process for both processes.



Customer Remarks Process

Site Visit Check List

The newly developed site visit check list assures that all relevant information is captured at the first time, therewith avoiding delays and extra cost resulting from additionally performed site visits.

Form 3 in Forms provides the most recent version of the site visit check list to be used while inspections.

Substation Check List

The substation check list provides an indication of all relevant information that needs to be captured in addition to the standard site visit check list (Form 4 in Forms) when a substation component is involved in the visit/ the design of the connection.

Exemption of Customers to provide Location for Distribution Substation will be studied if the total contracted load of the entire building of the customer (new, additional, booster) is greater than 166 KVA , For details Exemption of Customers to provide Location for Distribution Substation , refer to SEC Customer Services Manual with latest updates.

4.4 Techincal study

Design Proposal Review

Review of design is conducted after completion of design proposal and cost estimation. The review process must consider the following key points:

- Establish that the most economical solution is provided. In cases of doubt, an alternative design proposal and cost estimate may be requested. The number of these cases should be small since cases requiring alternative proposals to be developed will normally be

identified at an earlier stage (i.e. during design proposal stage) and discussed as part of regular reviews of job progress.

- Establish that the design proposal conforms to SEC standards and design practices.
- Confirm that the cost estimate has been provided correctly and that the design is prepared.to the appropriate level of detail to forward for approval.
- Confirm the project is correctly categorized to ensure costs are correctly allocated during the construction phase.
- Confirm the proposed funding of the project in accordance with the guidelines in the customer services manual.

After Design Review, the Design Engineer must:

Assign equipment numbers and enter the details in the substation register. The identification numbers of substations, distribution pillars and poles are to be shown on the Project scheme(s).

Rights of Way Request

When the design review has been completed, a request for ROW's can be made through customer relations unit using design proposal schemes if necessary.

The date on which the request is sent to customer relations is to be noted in UDS, and a copy of the standard covering letters to be retained in the file.

In some cases, it may be decided that drafting be done before obtaining ROW approval. In such cases, the finished project drawing (or sketch for mini-project) may be used for ROW request. For customer funded cases, ROW request/approval should be carried out prior to the customer undertakings.

A design engineer is to provide assistance as necessary to customer relation in clearing the ROW request. When ROW request have been cleared by customer relations, the ROW drawings/documents are to be returned to the planning section. The drawings/ documents are to be retained in the design file for later use when applying for digging permits.

The date of receipt of ROW is to be noted in UDS again.

DESIGN & COST APPROVAL

- Approval of technical study according to authority matrix
- Approval of of cost Estimation according to authority matrix

Distribution of Documents

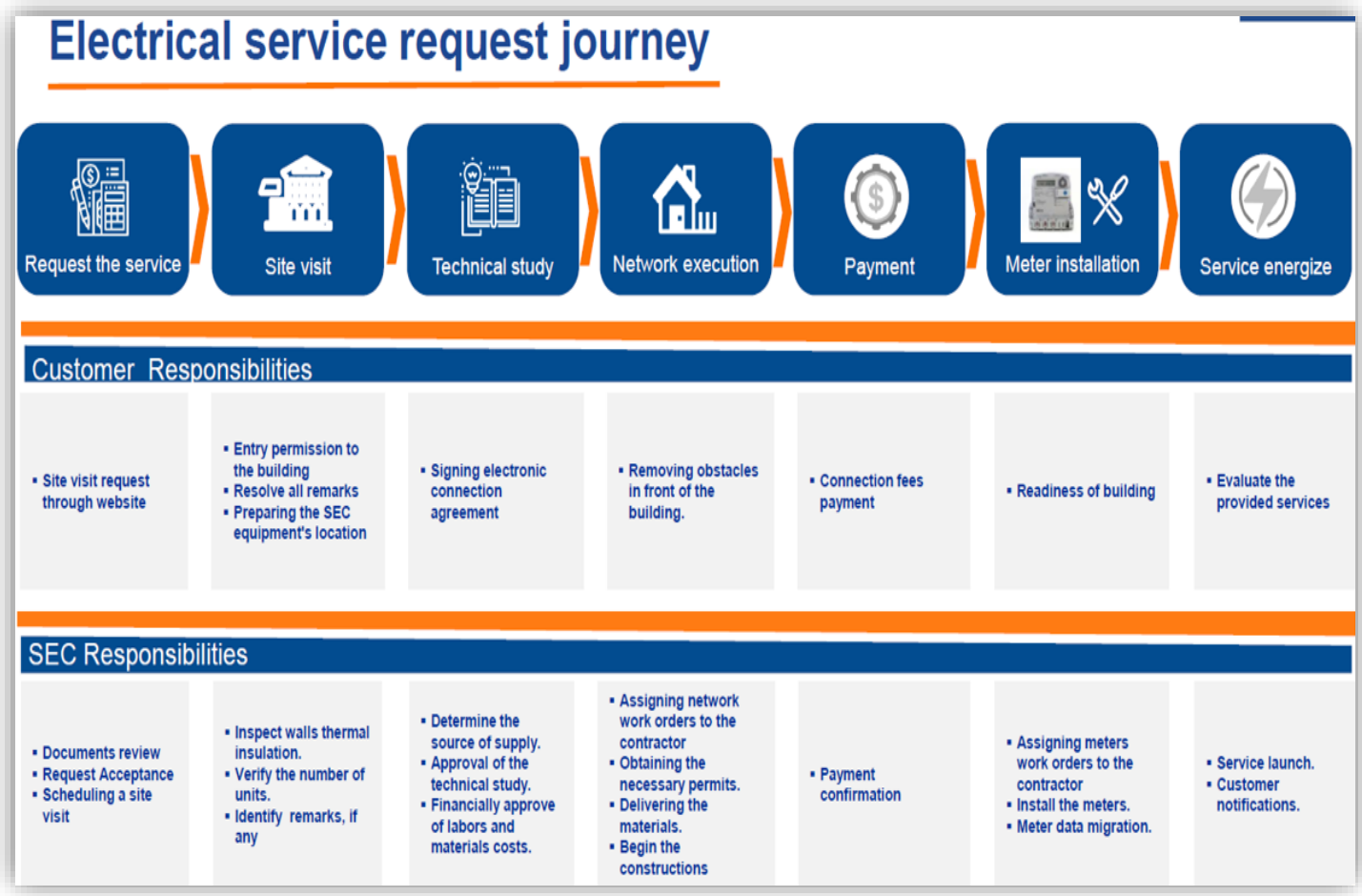
As soon as the connection design and cost calculations are approved by the respective authorities, all relevant design documents need to be shared with related departments. All documents should be stored within UDS and therefore should be able to be transferred electronically only. The main

recipients of the design documents are the construction unit, the O&M section as well as the accounting department.

4.5 Construction Unit

After the project approved by the respective authorities transfered to construction unit for execuation .

The electrical service reguest Journey as shown below



5 Low-Voltage (LV) Connections Planning

5.1 LV Underground Network Planning Process

The following design criteria need to be followed:

- Customer coincident demand load should be satisfied in line with the projected load, The customer coincident demand load should cover all KWH meters as applicable for the customer
- Equipment (LV cables, distribution pillars and distribution substations) shall not be overloaded. In case loading of any equipment exceeds 80%, relevant reinforcement action should be initiated An exception :-

a. Direct Feeder, which fed large, meters (400,500,600,800) A should be not exceeding 100 % of Direct Feeder's rating.

b. Private substation should be not exceeding 100 % of substation rating.

- Voltage drop at customer supply interface points shall not exceed 5% of nominal voltage, i.e. from substation to customer location
- Proposed LV network design should be the most economical for the projected load and layout.
- Optimization first principle – for any network design, existing network elements (substations, feeders, pillars, etc.) should be used as much as possible.

  In other words, the first priority for serving any customer should be through existing equipment (unless the load requirements are high enough to require dedicated equipment, e.g. dedicated substations for customer loads of more than 800A). Only when this is not possible, options involving new equipment (either through reinforcement of existing elements or addition of new elements) should be considered.

- The following connection configurations are available while designing LV underground networks and may be used depending on availability of existing infrastructure and customer coincident demand load requirements:

a. Main LV feeder with 1 distribution pillar – common configuration
b. Direct connection from LV panel to customer meter – heavy load lots
c. 1 main LV feeder with 2 distribution pillars – only suitable for areas with light load density.

- Geographical proximity principles should be used as much as possible:

a. Location of substations and distribution pillars should be as close to center of load area as possible
b. LV feeder / main feeder from SS to customer meter should follow shortest route
c. Street crossings for LV cables should be avoided

- Outgoing circuit breaker ratings for any outgoing supply source (substation, distribution pillar) should be greater than circuit breaker ratings of all KWH meters connected to it and to the coincident customer demand load
- Size of the new substation should take into account:

a. Maximum Demand load of existing customers
b. Existing nearby customers
c. Existing empty lots
d. Buildings under construction nearby

5.2 Network Planning & New Connection Design Procedure

After the site visit, a design proposal is to be prepared for the project.

The drafting of the design consists of, but is not limited to, the following steps:

- Decide on the most appropriate equipment and network configurations and sizes in accordance with SEC network planning standards. A project scheme drawing is to be prepared. The design must specify the outlet number to which LV cables are to be connected at the distribution panel. Pole and equipment numbers are to be added at a later stage. Major steps include

- Examination of contracted load, and circuit breaker (CB) size based on Customer Load Estimation and with reference to the NOC application.
- In case of essential information for engineering design not available in the file, it should be returned with objections to customer service department, e.g. load clarification required from customer. Otherwise, engineering design is commenced.
- Preparation of the connection design based on coincident demand load and on information collected per site visit check list in Forms. KWH meter and size of circuit breaker shall be as per customer's contracted load.

- Prepare a cost estimate for the project.

5.3 Location of LV Distribution Pillars

The following factors should be taken into consideration for installation of distribution pillars:

- Shall be installed at the load center as far as geographically possible to minimize service cable length.
- Shall be installed between two plots to avoid future relocation. as possible.
- May be placed at the inside of a sidewalk closer to customer premises.
- Should be easily accessible from the front of customer's boundary without any obstruction.
- Shall not be located on the top of sewerage system.

Detailed Construction Specifications for Locations of LV Distribution Pillars are referred to SEC distribution construction Standard No. (SDCS-02, PART 4 , Rev.00) with its latest updates.

5.4 Location of Distribution Substation Sites

Distribution substations can be installed at any of the following locations:

- Insets of customer lots
- Municipality land (such as open spaces, schools, mosques, car parking, gardens, public places)

For area electrification SEC will negotiate with the developer or the local Municipality for the land and locations required for substations. When preliminary design of a residential area has been completed and optimum substation sites required have been determined it is essential to indicate the same to the area developer which may be private owner, Municipality, or Ministry of Housing for the provision of the easements for SEC facilities including substations and ring main units.

- The unit substation type and will be installed in all cases except where extensible switch gear is required.
- Shall be installed between two plots to avoid future relocation
- In the services plot, e.g., open spaces, schools, mosques, car parks, etc. if it will be feeding the services area and Municipality land.
- The substation should be located on asphalted or leveled roads so that the medium voltage cables can be laid without any hindrance or difficulty.
- The substation shall be installed at the load center as far as geographically possible to minimize LV cables length.

Size of Inset for Distribution Substation depends on the different rating of this Distribution Substation as shown below

In order to accommodate all the different ratings of Distribution Substations i.e 500, 1000 and 1500 KVA of unit substation, the space 5m x 2.5m

Detailed dimensions and Construction Specifications for Location and Size of Substation Sites should be according to SEC Distribution Construction Standard No. (SDCS-02 -12) with its latest updates.

5.5 Location of energy meters room Sites

The need of meter room for customer depends on the number of units customer may have, for details please refer to customer services manual.

The metering room at customer premises should be adjoining the substation area or as mutually agreed to be the most appropriate as per design.

Detailed Dimensions and Construction Specifications for meters room should be according to SEC Distribution Construction Standard No. (SDCS-02 –part 7) with its latest updates.

5.6 LV Underground Materials Specifications

Distribution substations

Unit substation

A unit substation is commonly used substation in SEC. It combines distribution transformer and LV distribution panel in a single transportable unit. The unit substation is fed from a separate ring main unit. The ring main unit is not an integral part of the unit substation. General characteristics of the unit substation are shown in Table 23. Detailed materials specifications for Unit Substations are referred to SEC Distribution Materials Specification No. (56-SDMS-01, Rev.01) and No. (56-SDMS-03, Rev.00) with its latest updates.

Package substation (Existing at network but non-standard)

The package substation is convenience to install and occupies less space. It consists of Ring Main Unit, Distribution Transformer and Low Voltage Distribution Panel combined in a single unit. General characteristics of the package substation are shown in Table below. Detailed materials specifications for package substations are referred to SEC Distribution Materials Specification No. (56-SDMS-02, Rev.01) and No. (56-SDMS-04, Rev.00) with its latest updates.

Room substations

Separate Transformers and Low Voltage Distribution Boards also are available as well as 13.8 KV Ring Main Unit. These are to be used in indoor substations. As indoor substations usually serves large spot loads, the combinations of transformers and Low Voltage. Distribution Board may differ from those of package unit substations, but the ratings are similar.

LV Distribution panels

Low voltage distribution panels to be used in the distribution substations. The panel contains 400A molded case circuit breakers (MCCB) for out-going circuits. 400A MCCB according to SEC specification No. 37-SDMS-02 latest revision shall be already installed for each outgoing feeder. MCCB outgoing terminals shall be suitable for direct connection of 300mm² Al. cable.

Table 23: LV Distribution panel overview

Sec. voltage	400/230			Dual voltage 220/127,400/230		
Transformer rating (kVA)	500	1000	1500	500	1000	1500
LV Panel Rating (A)	800	1600	2500	1600	3000	4000
Number of outgoing MCCBs	4	8	10	8	12	14
Rating of outgoing MCCBs (A)	400					

Detailed materials specifications for LV Distribution Panels are referred to SEC Distribution Materials Specification No. (31-SDMS-01, Rev.03), No. (31-SDMS-05, Rev.00) and No. (31-SDMS-07) and (31-SDMS-08) with its latest updates.

LV Distribution pillars

Distribution Pillars provide above-ground access for service connections from LV main feeder. Its Bus bars has a rated normal continuous current of 400 Amps. The Distribution Pillar is equipped for seven (7), 3-Phase, 4-Wire, Aluminum Cable circuits. Two (2) circuits for the in-coming and five (5) circuits for the out-going. The two (2) in-coming circuits is located on each side of the Pillar. The five (5) outgoing circuits in the middle are equipped with NH Fuse Ways with rated current of 200 Amps. LV fuse links knife type NH of current rating 200 amps shall be installed.

The incoming circuit terminals are suitable for fixing Aluminum Cable of size 300 mm^2 or 185 mm^2 with the use of cable lugs. The outgoing circuit terminals are suitable for fixing Aluminum Cable of size 185 mm^2 or 70 mm2 with the use of cable lugs. The incoming circuit terminals are used for LV main feeder. The outgoing circuit terminals of distribution pillar are used for service connection to the customers.

The following tables outline the firm capacity of distribution pillars used within SEC

Distribution Pillar	Rating (A)	Rating (kVA) for different voltages (V)	
		400	220
	400 (320)	277 (222)	152 (122)

Detailed materials specifications for LV Distribution Pillars (Mini pillars) are referred to SEC distribution Materials Specification No. (31-SDMS-02, Rev.01) with its latest updates.

LV Cables

The 4 x 300 mm^2 AL/XLPE cable is the standard for LV main Feeder. Two sizes of cable 4 x 185 mm^2 & 4 x 70 mm^2 AL/XLPE shall be used for service connections. Three-phase four wires cable are provided as standard. Detailed materials specifications for LV Cables are referred to SEC Distribution Materials Specification No. (11-SDMS-01, Rev. 02) with its latest updates

The LV cables current ratings and Firm capacity (80%) are given in the table below:

Table 24: LV Cable current ratings

LV Cable size (mm^2) - New	Rating (A)	Firm capacity (80%) (A)
4 x 300 mm^2 Al	310	248
4 x 185 mm^2 Al	230	184
4 x 70 mm^2 Al	135	108

Table 25 Current ratings for Cable 1*800 mm² Al

LV Cable size (mm^2)	Rating (A)	Comment
1 x 800 mm^2 Al	525	Using between transformer and LV distribution panel

Table 26: Current ratings for LV Cable (existing and not standard)

LV Cable size (mm^2) - Old	Rating (A)
1 x 630 mm^2 Cu	525
3 x 185 mm^2 + 95 mm^2 Cu	300
3.5 x 120 mm^2 Cu	280
3.5 x 70 mm^2 Cu	170
3.5 x 35 mm^2 Cu	120
3.5 x 16 mm^2 Cu	75
4 x 500 mm^2 Al	400
4 x 120 mm^2 Al	200
4 x 95 mm^2 Al	160
4 x 50 mm^2 Al	105

Note: Calculation of the Continuous Current Rating of Cables". are based on the cable characteristics. These results are based on data for typical cable types. For more precise data, refer to the specific cable supplier. Correction factors for deviation from these conditions are indicated in (1.2 Standard Conditions) ,Where two or more circuits are installed in proximity, the load rating of all affected cables is reduced.

LV Circuit connections circuit breakers

Molded Case Circuit Breakers (MCCB) for indoor or outdoor installation in an enclosure , intended to be used for Service Connections in the Low Voltage System. The Standard Ratings for the

The incoming terminals shall be suitable for both copper and aluminum conductors of sizes given for the following different ratings as shown in the following table:

Table 27: MCCB ratings and maximum size of conductors

MCCB rating (Amps)	Max size of conductors suitable for the incoming terminals
20, 30, 40, 50, 70, 100, 125, 150	4x70mm² Al
200, 250	2 cables of 4x185mm² Al
300,400	2 cables of 4x300mm² Al
500, 600, 800	2 cables of 4x300mm² Al

Circuit Breakers are 20, 30, 40, 50, 70, 100, 125, 150, 200, 250, 300, 400, 500, 600, and 800 Amps.

Detailed materials specifications for Circuit Breakers referred to SEC Distribution Materials Specification No. 37-SDMS-01, No. 37-SDMS-05 respectively with its latest updates.

SMART meters

Electronic Revenue Whole Current and CT operated smart meters, intended to be used for revenue metering in the system.

The smart meters used by SEC are classified as given in the below table:

Table 28: Meter CB ratings by meter type

Meter Type	Rating (A & V)	CB Rating (A)
Electronic Revenue WC Meter	10(160)A ,20(100)A 400/230/133V	20, 30, 40, 50, 70, 100, 125, 150
Electronic Revenue CT Meter	CT1.5(6)A 133/230/400V,	200, 250, 300, 400, 500, 600, 800 1600,2500

Detailed materials specifications for Electronic Revenue CT and Electronic Revenue Whole Current Meters are referred to SEC Distribution Materials Specification No. 40-SDMS-02A Rev. 0 9.1 , and No. 40-SDMS-02B Rev. 8.1 respectively with its latest updates

Meter boxes

Fiberglass reinforced polyester boxes to be used for Kilo Watt Hour (KWH) smart meters in the distribution system. The meter boxes used by SEC are classified as given below:

Table 29: Overview of meter boxes)

Meter Box Type	Meter Type	Max size of LV Cables suitable for the incoming terminals	Box rating
Single meter box	One whole current meter	Two cables up to 4x70 mm^2 Al	200 Amps
Double meter box	Two whole current meters	Two cables up to 4x185 mm^2 Al	300 Amps
Quadruple meter box	Four whole current meters	Two cables up to 4x300 mm^2 Al	400 Amps
200/250 A CT meter box	One CT meter	Two cables up to one 4x300mm^2 + one 4x185 mm^2	400 Amps
300/400 A CT meter box	One CT meter	Two back to back cables of sizes up to one 4x300 mm^2 + one 4x185 mm^2	400 Amps
500/600 A CT meter box	One CT meter	Two back to back cables of sizes up to 4x300 mm^2	600 Amps
800 A Remote meter box	One CT meter	Two cables of sizes up to 4x300 mm^2	800 Amps

Detailed materials specifications for Meter Boxes are referred to SEC Distribution Materials Specification No. (42-SDMS-01, Rev.06) , No. (42-SDMS-02, Rev.00) and No. (42-SDMS-03, Rev.00) with its latest updates.

5.7 Calculation of Voltage Drop

For a particular supply voltage the voltage drop from the supply point to the customer interface depends on various factors such as customer demand, length and size of cable, and power factor. Formula for voltage drop is provided below:

$$VD_{\%} = \frac{100 \times KVA \times (R \times \cos\varphi + X \times \sin\varphi) \times L}{V^2}$$

Where:

$VD\%$ = Voltage drop percentage on the cable in (%)

KVA = Three phase power in (KVA) = Coincident Demand Load (CDL) on the cable.

R = Resistance of conductor in ohm per kilometer in (Ω/km)

X = Inductive reactance of conductor in ohm per kilometer in (Ω /km)

φ = Power factor angle of the supply

V = Three phase supply nominal voltage in (volts)

L = Length of the cable in (meters)

The formula has reduced to a simple constant K equivalent to the product of KVA and length of cable in meter at power factor of 0.85 lagging. For various values of KVA-meter the voltage drop can be calculated by dividing it with this constant K.

$$K = \frac{V^2}{100 \times (R \times \cos\varphi + X \times \sin\varphi)}$$

Table 30: The value of the K constant are shown in the table below

Cable Size	V	R	X	$cos\varphi$	$sin\varphi$	K
mm2	Volts	Ω/km	Ω/km			V².km/Ω
300	400	0.13	0.09	0.85	0.527	10132
300	220	0.13	0.09	0.85	0.527	3065
185	400	0.211	0.091	0.85	0.527	7040
185	220	0.211	0.091	0.85	0.527	2129
70	400	0.568	0.095	0.85	0.527	3003
70	220	0.568	0.095	0.85	0.527	908

The values of K constant to be used for various standard LV cables are provided below

LV Cable Size	Constant K Standard Nominal Voltages	
	400 V	220 V
4 x 300 mm2 (AL)	10132	3065
4 x 185 mm2 (AL)	7040	2129
4 x 70 mm2 (AL)	3003	908

The simplified formula for voltage drop calculation is:

$$VD_{\%} = \frac{KVA \times L}{K}$$

Where:

$VD\%$ = Voltage drop percentage on the cable in (%)

KVA = Three phase power in (KVA) = Coincident Demand Load (CDL) on the cable

L = Length of the cable in (meters)

K = The constant in (V^2.km/Ω) according to above

Examples of voltage drop calculation in Appendix 2

Voltage Drop calculation form 5 in Forms.

5.8 Underground Low Voltage Network Configuration

There are three standard connection configuration types for customer connections in Low Voltage Underground Network depends on customers demand loads as following.

Connection through Distribution Pillar

This type of connection is shown in Figure 1 and Figure 2. For this condition, cable 300 mm2 AL/XLPE is used from substation to distribution Pillar, and cables 185mm2 AL/XLPE, 70 mm2 AL/XLPE are used from distribution Pillar to the customer meter/meters box. (Common configuration).

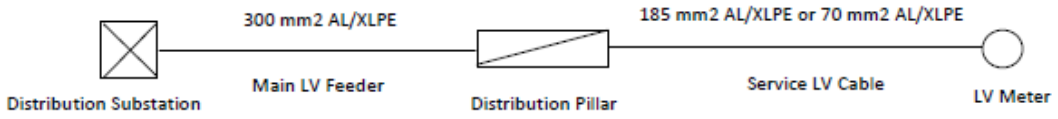


Figure 1: Technical connection through distribution pillar

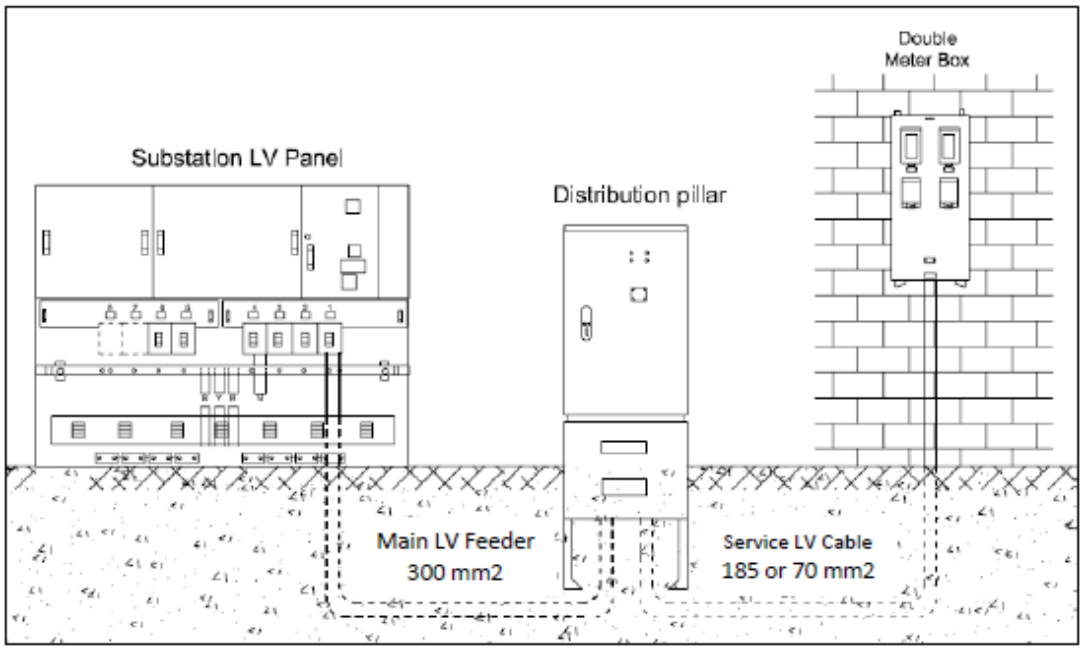


Figure 2: Scheme connection through distribution pillar

Direct Connection

This type of connection is shown in Figure 3 and Figure 4 . For this condition, cable 4x300 mm^2 AL/XLPE or 4x185mm^2 AL/XLPE is used directly from substation to the customer meter/meters box. (Heavy load lots only).

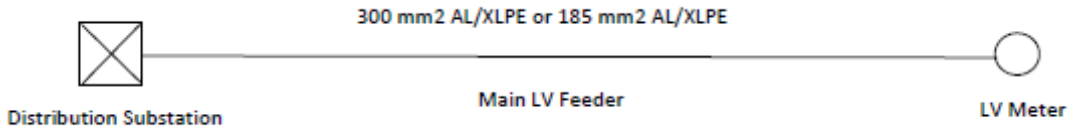


Figure 3: Technical direct connection

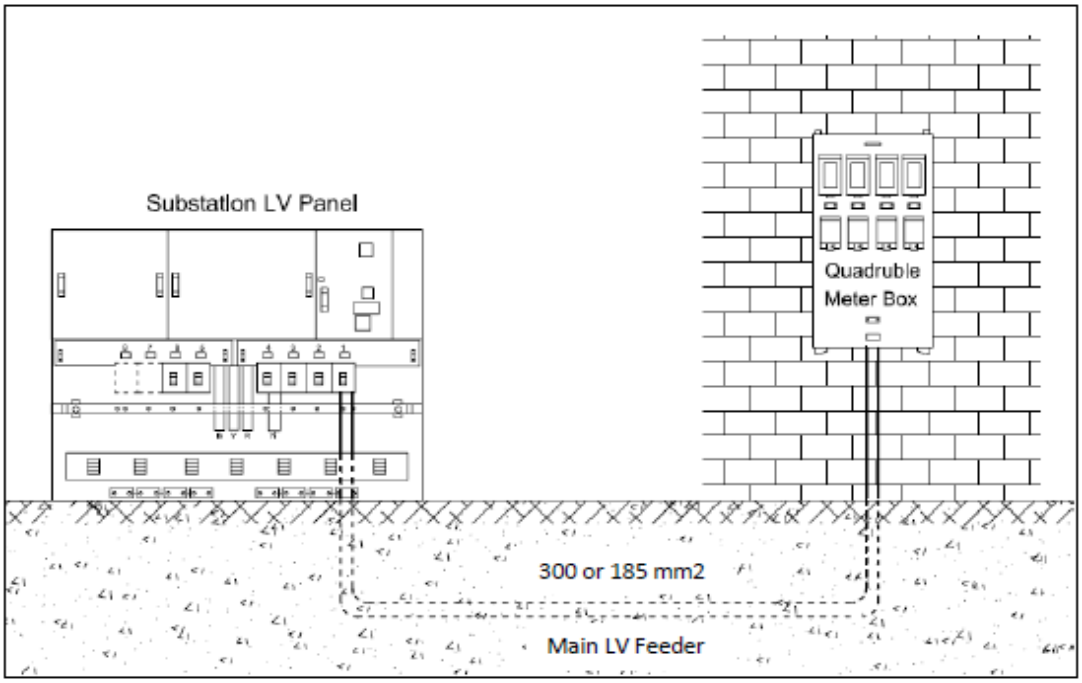


Figure 4: Scheme direct connection

Connection through Two Distribution Pillars

In the areas where load of customers is low, the outgoing of the distribution pillar can be used to feed the second distribution pillar to provide connection to more customers. This type of connection is shown in Figure 5 and Figure 6. For this condition, cable 300 mm2 AL/XLPE is used from substation to the first distribution Pillar, and cables 300 mm2 AL/XLPE or 185 mm2 AL/XLPE are used from the first distribution Pillar to the second distribution Pillar and cables

185mm2 AL/XLPE or 70 mm2 AL/XLPE are used from distribution Pillar to the customer meter/meters box. (Light load lots only).

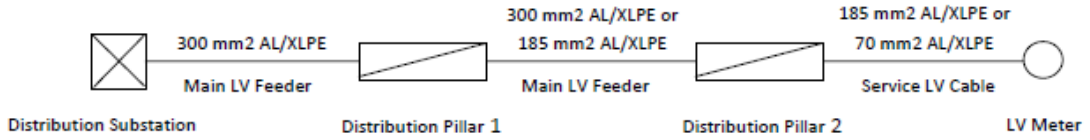


Figure 5: Technical connection through two distribution pillars

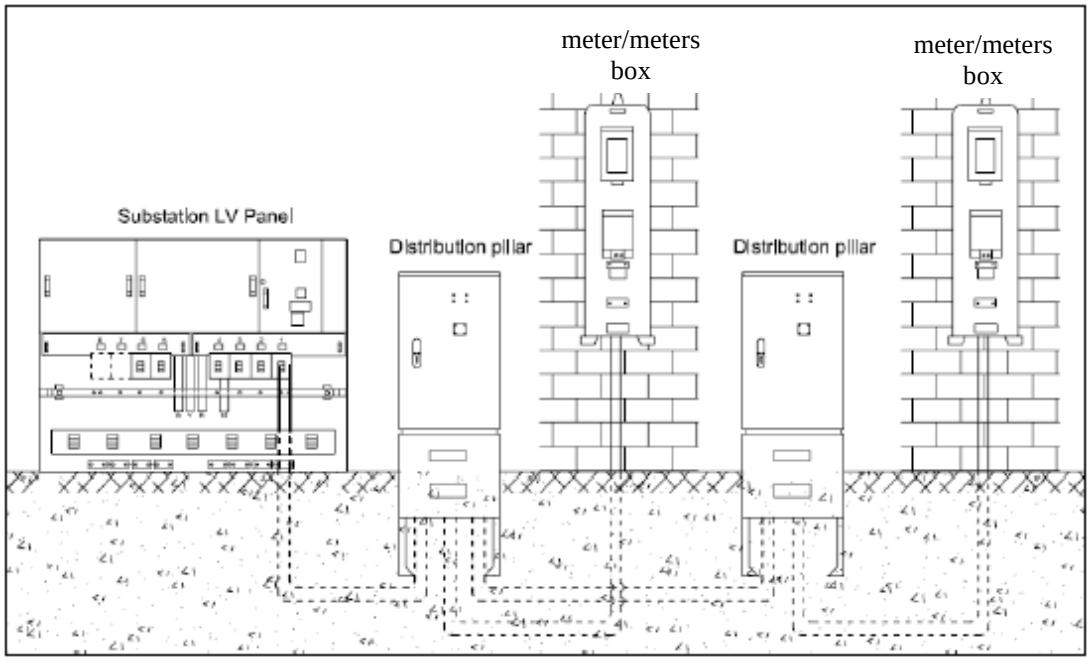


Figure 6: Scheme connection through two distribution pillars

5.9 Additional Planning Design Principles

The general criteria from earlier can be translated into detailed design principles as outlined below:

- Design of any LV network element (Substation, Main LV Feeder, Distribution Pillar , Service Connection Cable) should be based on the Coincident Demand Load (CDL) of all customers KWH Meters supplied from this LV network element.
- To maintain the Loading percentage on any LV network element (Substation, Main LV Feeder , Distribution Pillar , Service Connection Cable) within the Firm Capacity (80 % of Rating) of that LV network element.

- To maintain the Total Voltage Drop percentage on the whole LV network (Main LV Feeder + Service Connection Cable) from the Substation to the customer's location within the Voltage Drop limits (5 % of Nominal Voltage).
- The LV network design should be the most economical (Lowest Cost) as possible to supply the projected customer's load.
- The suitable size of the cable to supply the customer should be selected according to the Coincident Demand Load (CDL) of that customer and should be suitable to satisfy that customer's CDL is not greater than the Firm Capacity (80 % of Rating) of that Cable.
- The suitable connection configuration type to supply the customer should be selected according to the Coincident Demand Load (CDL) of that customer.
- Connection configuration type with Two Distribution Pillars in one main LV feeder can be used only in light load density area.
- CDL on the SS should be not greater than SS's Firm Capacity (i.e. not exceeding 80 % of SS's rating) and is calculated using the following formula

$$CDL\ (KVA)_{on\ SS} = CDL\ for\ all\ N\ Meters\ supplied\ from\ SS$$

$$Loading\ \%_{on\ SS} = \frac{CDL\ (KVA)_{on\ SS}}{Rating_{SS}} \times 100$$

- CDL on the private SS should be not greater than SS's rating (i.e. not exceeding 100 % of SS's rating) and is calculated using the above formula
- CDL on the DP should be not greater than DP's Firm Capacity (i.e. not exceeding 80 % of DP's rating)

$$CDL_{on\ DP} = CDL\ for\ all\ N\ Meters\ supplied\ from\ DP$$

$$Loading\ \%_{on\ DP} = \frac{CDL_{on\ DP}}{Rating_{DP}} \times 100$$

- CDL on the LV Main Feeder should be not greater than LV Main Feeder's Firm Capacity (i.e. not exceeding 80 % of LV Main Feeder's rating)

$$CDL_{on\ Main\ Feeder} = CDL\ for\ all\ N\ Meters\ supplied\ from\ Main\ Feeder$$

$$Loading\ \%_{on\ Main\ Feeder} = \frac{CDL_{on\ Main\ Feeder}}{Rating_{Main\ Feeder}} \times 100$$

- CDL on the Service Cable should be not greater Service Cable's Firm Capacity (i.e. not exceeding 80 % of Service Cable's rating).

$$CDL_{on\ Service\ Cable} = CDL\ for\ all\ N\ Meters\ supplied\ from\ Service\ Cable$$

$$Loading\ \%_{on\ Service\ Cable} = \frac{CDL_{on\ Service\ Cable}}{Rating_{Service\ Cable}} \times 100$$

- CDL on the Direct Feeder should be not greater than Direct Feeder's Firm Capacity (i.e. not exceeding 80 % of Direct Feeder's rating).

$$CDL_{on\ Direct\ Feeder} = CDL\ for\ all\ N\ Meters\ supplied\ from\ Direct\ Feeder$$

$$Loading\ \%_{on\ Direct\ Feeder} = \frac{CDL_{on\ Direct\ Feeder}}{Rating_{Direct\ Feeder}} \times 100$$

- CDL on the Direct Feeder, which fed large meters (400,500,600,800) should be not greater than Direct rating (i.e. not exceeding 100 % of Direct Feeder's rating). and is calculated using the above formula
- Total VD% from SS to customer's location should be not greater than voltage drop limit (i.e. not exceeding 5 %).

$$VD\ \%_{on\ Service\ Cable} = \frac{CDL\ (KVA)_{on\ Service\ Cable} \times L_{Service\ Cable}}{K_{Service\ Cable}} \times 100$$

$$VD\ \%_{on\ Main\ Feeder} = \frac{CDL\ (KVA)_{on\ Main\ Feeder} \times L_{Main\ Feeder}}{K_{Main\ Feeder}}$$

$$VD\ \%_{on\ Direct\ Feeder} = \frac{CDL\ (KVA)_{on\ Direct\ Feeder} \times L_{Direct\ Feeder}}{K_{Direct\ Feeder}}$$

$$VD\ \%_{Total\ from\ SS\ to\ Customer} = VD\ \%_{Main\ Feeder} + VD\ \%_{Service\ Cable}$$

$$VD\ \%_{Total\ from\ SS\ to\ Customer} = VD\ \%_{Direct\ Feeder}$$

- Always try first to supply customer's CDL from any existing nearby DP's (one by one) with priority for the nearest as possible based on the criteria (Loading % , Voltage Drop %) before planning to install new DP.
- Always try first to supply customer's CDL from any existing nearby SS's (one by one) with priority for the nearest as possible based on the criteria (Loading % , Voltage Drop %) before planning to install new SS.
- To supply customer's CDL from any existing DP, first check for availability of any vacant outgoing in that DP.
- To supply customer's CDL from any existing SS, first check for availability of any vacant outgoing in that SS.
- Install the new SS in the center of loads area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, empty lots) as possible.
- Install a new DP near to customers' lots in the center of loads (i.e. in the middle between customers' lots) as possible.

- Select the shortest geographic route for the LV Main Feeder (300 mm2 cable) from SS to the new DP (as possible).
- Select the shortest geographic route for the service cable from the new DP to customer's location (as possible).
- Select the shortest geographic route for the Direct Feeder from SS to the customer's location (as possible).
- Always try to avoid crossing the streets when you design the route of any LV cable as possible as you can.
- It is not allowed to cross any street with width more than 30 meters for any LV cable route.
- CB/Fuse rating of the outgoing from any supply source (Substation , Distribution Pillar) should be not less than the largest CB rating of all KWH Meters supplied from this outgoing CB/Fuse. Same is valid for any two CBs/Fuses outgoings supply customers.
- CB/Fuse rating of the outgoing from any supply source (Substation, Distribution Pillar) should be not less than the Coincident Demand Load (CDL) of all customers KWH Meters supplied from this outgoing CB/Fuse. Same is valid for any two CBs/Fuses outgoings supply customers.
- Size (KVA rating) of the new SS should be selected based on the need of the neighbor area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, existing empty lots) and it should be as minimum as sufficient to meet their total Coincident Demand Load (CDL).
- If multi substations are required to supply a customer , select the no. of the required substations and their ratings from the available SEC standard (500, 1000, 1500 KVA) where the summation of substations ratings should provide minimum sufficient total capacity to meet the calculated Coincident Demand Load (CDL) of the customer with minimum no. of substations.
- For supplying new customers, it is preferred to use the substation with (500 KVA or 1000 KVA) rating. This is to maintain a possibility for reinforcement of these substations (1000 KVA & 500 KVA) by replacing them with 1500 KVA substation without the need to install a new substation.

5.10 Step By Step Design Procedure

- Connected Load (CL) in (KVA) for each Individual unit in customer's building should be estimated, unit by unit, as per Section referring to customer load estimation

- Individual Circuit Breaker Rating (CBR) in (Amp) for the Individual KWH Meter for each Individual unit in customer's building should be determined according to the estimated connected load (CL) of that Individual unit and referring to customer load estimation
- Number of Individual KWH Meters (N) required for the customer's building should be determined according to number of Individual units in customer's building and referring to SEC Customer Services Manual with its latest updates.
- Calculate the Coincident Demand Load (CDL) in (Amp) for the group of all KWH Meters of the customer's building based on coincident demand load (CDL) calculation.
- Based on the calculated Coincident Demand Load (CDL) in (Amp) of the customer's building, select the suitable connection configuration type to supply this Coincident Demand Load (CDL) as shown in Table below. the suitable connection configuration type includes :

a. Size of cable to customer.

b. No. of cables to customer required.

c. Suitable supply source: Direct Feeder from Substation (SS) or Service Connection through Distribution Pillar (DP).

d. No. of outgoing required.

Table 31: Coincident Demand Load (CDL) UG

Coincident Demand Load (A)		Supply Source	No. of Outgoing Fuses / MCCB	Number of LV Cables to Customer	Size of LV Cables to Customer	Main LV Feeder	
From	To					No. of Cables to DP	Cable Size
1	108	DP	1	1	70 mm2	1	300 mm2
109	184	DP	1	1	185 mm2	1	300 mm2
185	216	DP	2	2	70 mm2	1	300 mm2
217	248	DP	2	2	185 mm2	1	300 mm2
185	248	SS	1	1	300 mm2	Direct Feeder	
248	496	SS	2	2	300 mm2	Direct Feeder	

- If the suitable connection configuration type is Service Connection through Distribution Pillar (DP), go to the next step.
- First try to supply customer's CDL from existing nearby DP by using the following steps:

a. Select the nearest existing DP to te customer's location (as possible).

b. Calculate CDL on the DP including of all customers KWH Meters (concerned new customer + existing customers) supplied from this DP.

c. CDL on the DP should be not greater than DP's Firm Capacity (i.e. not exceed 80 % of DP's rating).

d. Calculate CDL on the Main Feeder (300 mm2 cable) from SS to DP including of all customers KWH Meters (concerned new customer + existing customers) supplied from this Main Feeder.

e. CDL on the Main Feeder should be not greater than Main Feeder's Firm Capacity (i.e. not exceed 80 % of Main Feeder's rating)

f. CDL on the Main Feeder which fed large meters (400,500,600,800) should be not greater than Main Feeder's rating (i.e. not exceed 100 % of Main Feeder's rating)

g. Calculate CDL on the SS including of all customers KWH Meters (concerned new customer + existing customers) supplied from this SS.

h. CDL on the SS should be not greater than SS's Firm Capacity (i.e. not exceed 80 % of SS's rating).

i. CDL on the private SS should be not greater than SS's rating (i.e. not exceed 100 % of SS's rating).

j. Select the shortest geographic route for the service cable from DP to customer's location (as possible).

k. Calculate VD% on the Main Feeder (300 mm2 cable) from SS to DP.

l. Calculate VD% on the service cable from DP to customer's location.

m. Calculate the Total VD% from SS to customer's location.

n. Total VD% from SS to customer's location should be not greater than voltage drop limit (i.e. not exceed 5 %).

o. If customer's CDL cannot be supplied from the selected DP because one of the criteria (Loading % , Voltage Drop %) is not satisfied , Try all others nearby existing DP (one by one) with priority for the nearest and by using same steps in above (from "a" to "n").

- If customer's CDL cannot be supplied from all nearby existing DP because one of the criteria (Loading %, Voltage Drop %) is not satisfied, go to the next step.
- Try to supply customer's CDL from existing nearby SS through a new DP by using the following steps :

a. Select the nearest existing SS to the customer's location (as possible).

b. Calculate CDL on the SS as follow:-

c. Option 1 That equal (maximum demand load on the SS + CDL new customer)

d. Option 2 : Calculate CDL on the SS including of all customers KWH Meters (concerned new customer + existing customers) supplied from this SS.(if option 1 is not possible)

e. CDL on the SS should be not greater than SS's Firm Capacity (i.e. not exceed 80 % of SS's rating).

f. CDL on the private SS should be not greater than SS's rating (i.e. not exceed 100 % of SS's rating).

g. Design to install a new DP near to customers' lots in the center of loads (i.e. in the middle between customers' lots) as possible.

h. Select the shortest geographic route for the new Main Feeder (300 mm2 cable) from SS to the new DP (as possible).

i. Select the shortest geographic route for the service cable from the new DP to customer's location (as possible).

j. Calculate VD% on the new Main Feeder (300 mm2 cable) from SS to the new DP.

k. Calculate VD% on the service cable from the new DP to customer's location.

l. Calculate the Total VD% from SS to customer's location.

m. Total VD% from SS to customer's location should be not greater than voltage drop limit (i.e. not exceed 5 %).

n. If customer's CDL cannot be supplied from the selected SS because one of the criteria (Loading % , Voltage Drop %) is not satisfied , Try all others nearby existing SS (one by one) with priority for the nearest and by using same steps in above (from "a" to "m").

- If customer's CDL cannot be supplied from all nearby existing SS because one of the criteria (Loading %, Voltage Drop %) is not satisfied, go to the next step.
- Design to supply customer's CDL from a new SS through a new DP by using the following steps :

  a. Design to install a new SS near to customers lots in the center of loads area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, empty lots) as possible.

  b. Size (KVA rating) of the new SS should be selected based on the need of the neighbor area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, existing empty lots).

  c. Design to install a new DP near to customers' lots in the center of loads (i.e. in the middle between customers' lots) as possible.

d. Select the shortest geographic route for the new Main Feeder (300 mm2 cable) from the new SS to the new DP (as possible).

e. Select the shortest geographic route for the service cable from the new DP to customer's location (as possible).

f. Calculate VD% on the new Main Feeder (300 mm2 cable) from the new SS to the new DP.

g. Calculate VD% on the service cable from the new DP to customer's location.

h. Calculate the Total VD% from the new SS to customer's location.

i. Total VD% from the new SS to customer's location should be not greater than voltage drop limit (i.e. not exceed 5 %).

- If the suitable connection configuration type is Direct Feeder from Substation (SS), go to the next step.
- First try to supply customer's CDL from existing nearby SS by using the following steps:

a. Select the nearest existing SS to the customer's location (as possible).

b. Calculate CDL on the SS including of all customers KWH Meters (concerned new customer + existing customers) supplied from this SS.

c. CDL on the SS should be not greater than SS's Firm Capacity (i.e. not exceed 80 % of SS's rating).

d. CDL on the private SS should be not greater than SS's rating (i.e. not exceed 100 % of SS's rating).

e. Select the shortest geographic route for the Direct Feeder from SS to the customer's location (as possible).

f. Calculate VD% on the Direct Feeder from SS to the customer's location.

g. Total VD% from SS to customer's location should be not greater than voltage drop limit (i.e. not exceed 5 %).

h. If customer's CDL cannot be supplied from the selected SS because one of the criteria (Loading % , Voltage Drop %) is not satisfied , Try all others nearby existing SS (one by one) with priority for the nearest and by using same steps in above (from "a" to "g").

- If customer's CDL cannot be supplied from all nearby existing SS because one of the criteria (Loading %, Voltage Drop %) is not satisfied, go to the next step.
- Design to supply customer's CDL from a new SS by using the following steps:

a. Design to install a new SS near to customers lots in the center of loads area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, empty lots) as possible.

b. Size (KVA rating) of the new SS should be selected based on the need of the neighbor area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, existing empty lots).

c. Select the shortest geographic route for the Direct Feeder from the new SS to the customer's location (as possible).

d. Calculate VD% on the Direct Feeder from the new SS to the customer's location.

e. Total VD% from the new SS to customer's location should be not greater than voltage drop limit (i.e. not exceed 5 %).

5.11 Connection to LV Customers (from 300A to 800A load)

While connecting large LV customers, ratings for LV equipment need to be updated to take into account the additional load requirements.

Table 32 Cable Connections per Out-going Connection Point (Underground)

Connection Point	Cable Connections
Outgoings of the Substation	1 Cable Up to 300 mm^2 per Outgoing
Outgoings of Distribution Pillar	1 Cable Up to 185 mm^2 per Outgoing
Meter Box 300-400 Amp	Two Incomings up to 1 Cable 300 mm^2+1 Cable 185 $mm^{2)}$
Meter Box 400-500-600-800 Amp	Two Incomings Cable 300 mm^2

When encountered with a large customer request, the planning engineer needs to take into account the most economical configuration which takes into account the customer load requirements

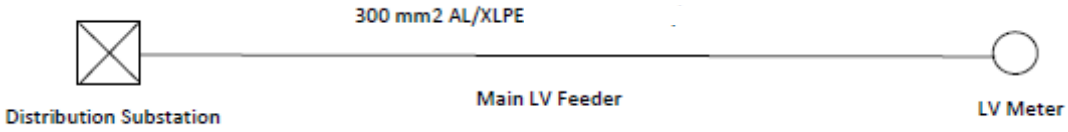
There are two configurations for supplying to customers with large meters

A. Using of 1 Outgoing and 1 Cable.

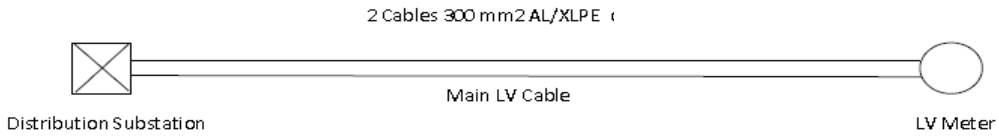
B. Using of 2 Outgoings and 2 Cables.

The planning engineer should study the supply request according to these options and evaluate the cost and the voltage drop for each one then to select the suitable option and the most economical one. The supply method for these types of LV Large Meters need special configurations design as follows:

A. One Outgoings from Substation to Meter Box



B.Two Outgoings from Substation to Meter Box



The following tables outline the supply configuration requirements for s load from 300A to 800A either 1 or 2 outgoing cables

Table 33: Supply Method for LV Customer

CB Rating (A)	Demand Factor	Maximum CDL (A)	Supply Source	No of Outgoing	No. of Cables	Cable Size (mm2)	Comments
400-500-600	0.5	310	SS	1	1	300	One direct cable
400- 500	0.6						
400	0.7						
300-400	0.8						
800	0.5	620	SS	2	2	300	Two direct cable
600-800	0.6						
500-600-800	0.7						
500-600-800*	0.8						

* Can be used private substation 500 KVA

5.12 Connection to LV Customers (from 800A Load and above)

LV Customers with more than 800A connected load will be supplied only through underground configuration. Such customers will require dedicated distribution substations under the following cases:

a. Customer connected load from 800A and the building is considered 1 unit according the SEC Customer Services Manual and municipality permits

b. There is request from the customer for dedicated substation

Under these cases, SEC needs to ensure connection scheme to customer with dedicated substation(s) as required by the customer load. It is the responsibility of the customer to provide location for the substation(s) as per SEC guidelines.

a. Low Voltage Distribution Panel without Outgoing MCCBs is to be used in the distribution substations for this configuration type. The panel shall be supplied with Main Circuit Breaker.

b. The Customer supplied Circuit Breakers (MCCB/ ECB) shall be approved by SEC and be as per specification no. 37-SDMS-04.

c. Outgoing connection to customer from the SEC LV Distribution shall be made by means of connecting single core 800 mm²AL, XLPE or single core 630 mm² CU

d. The main bus bars by using cable lugs. General characteristics of this Panel are shown below along with the connection configuration.

Table 34: LV Distribution Panel Characteristics

Secondary Voltage (V)	400/230			Dual voltage 220/127, 400/230		
Transformer Rating (KVA)	500	1000	1500	500	1000	1500
LV Panel Rating (A)	800	1600	2500	1600	3000	4000
No. of Cable / Phase	2	3	5	3	6	8

There are two potential supply methods for LV customer through private substation:-

- Unit Substation with LV panel with Main Circuit Breaker without Outgoing MCCBs
- RMU ,Transformer and separate LV Panel with Main Circuit Breaker without Outgoing MCCBs.

Figure7: Substation with LV Panel & Main Circuit Breaker

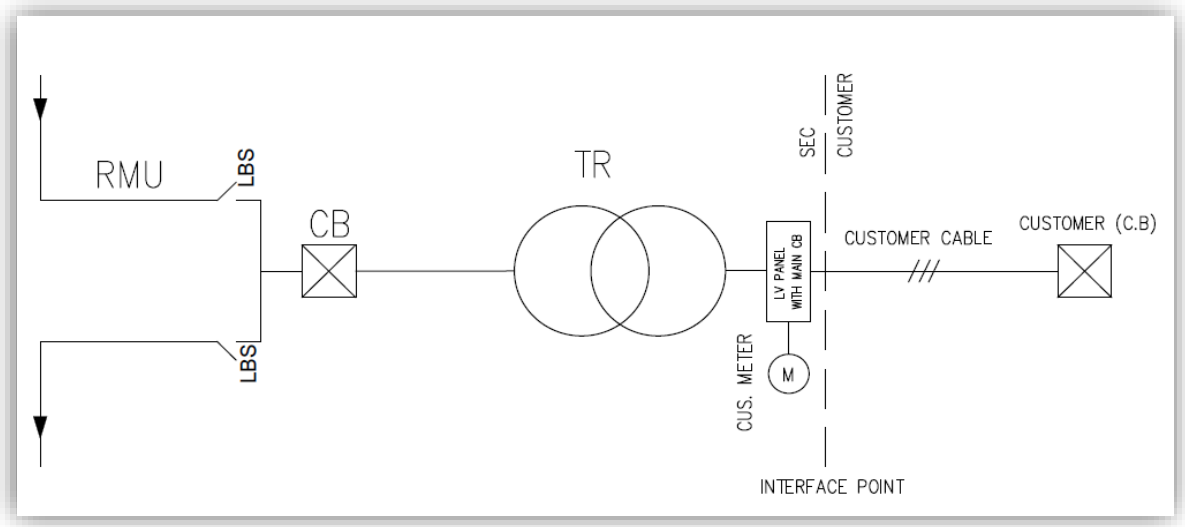
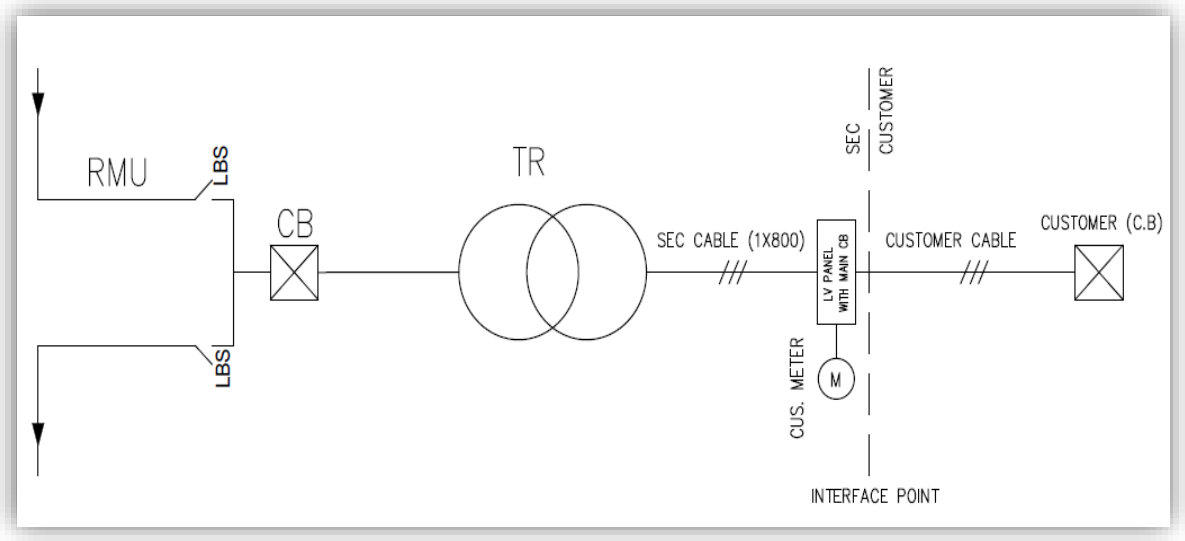


Figure 8: Ring Main Unit, Transformer and separate LV Panel



- The customer CB shall be adjacent to the distribution susbstation which is the interface point.
- Outgoing connection between transformer and SEC separate LV Distribution shall be made by means of connecting single core 800 mm²AL, XLPE as per specification no 11 SDMS 01 with latest updates.

Examples of Underground LV Connection Design

6 LV Overhead Network Planning Process

Network Planning & New Connection Design

Once the site visit has been performed and all customer remarks have been resolved, the network planning section can start to produce the new connection design scheme.

6.1 LV Overhead New Connections Network planning design criteria

The following design criteria need to be followed:

- Customer coincident demand load should be satisfied in line with the projected load, for the upcoming 5 year time period (including current year). The customer coincident demand load should cover all smart meters as applicable for the customer
- Equipment (conductors, cabinets and PMT) shall not be overloaded. In case loading of any equipment exceeds 80%, relevant reinforcement action should be initiated
- Voltage drop at customer supply interface points shall not exceed 5% of nominal voltage, i.e. from substation to customer location

- Proposed LV network design should be the most economical for the projected load (for the 5-year time period) and layout
- Optimization first principle – for any network design, existing network elements (PMTs, feeders, etc.) should be used as much as possible
- The following connection configurations are available while designing LV overhead networks and may be used depending on availability of existing infrastructure and customer coincident demand load requirements:
    a. OH main LV feeder with 50 mm^2 quadruplex conductor as service drop connection – common configuration
    b. OH main LV feeder with 120 mm^2 quadruplex conductor as service drop connection – for heavy load lots
    c. OH main LV feeder with service connection UG cable – exceptional configuration to be used where applicable
- Geographical proximity principles should be used as much as possible:
    a. Location of pole mounted transformers should be as close to center of load area as possible
    b. LV feeder / main feeder from PMT to customer meter should follow shortest route
    c. Street crossings for LV conductors should be avoided
- Outgoing circuit breaker ratings for any outgoing supply source (PMT, LV cabinet) should be greater than circuit breaker ratings of all Smart meters connected to it and to the coincident customer demand load
- Size of the new PMT should take into account:

a. Demand load of existing customers (for this year and for the upcoming 5 year time horizon)
b. Existing nearby customers
c. Existing empty lots
d. Buildings under construction nearby

6.2 LV Overhead Materials Specifications

Pole mounted transformers (PMT)

The standard distribution transformer for overhead system is a pole mounted transformer (PMT) with LV Distribution Cabinet. These are commonly used in rural areas where loads are located in a scattered manner. Sizes and characteristics of the available transformer units are as follow in the tables:

Table 35: Overview of pole mounted transformer sizes and voltage ratios

Transformer		Voltage Ratio
Rated & Firm Capacity	LV Feeder	
100 kVA (80 kVA)	1	13.8KV/(231- 400)V 13.8KV/(231-400,133-220)V Dual 33KV/(231- 400)V 33KV/(231-400,133-220)V Dual
200 kVA (160 kVA)	2	
300 kVA (240 kVA)	3	

Table 36: Overview of pole mounted transformer

Components	TRANSFORMER RATING 400/231 V				
	100 KVA	200 KVA		300 KVA	
	W/ MAIN CB	W/ BRANCHES	W/ MAIN CB	W/ BRANCHES	W/ MAIN CB
Phase B.B Minimum Rating, (A)	200	400	300	600	400
CT Rating on Incoming B.B (A)	200/5	400/5	300/5	600/5	400/5
Number of Outgoing MCCB's	1	2	1	3	1
MCCB's Rating (A)	200	200	300	200	400
NO. of Incoming Feeders	1	1	1	2	2
Size of Incoming Feeders mm²	185	300	300	300	300
NO. of O.H Outgoing Feeders	1	2	-	3	-
NO. of U.G Outgoing Feeders	1	2	1	3	1
Size of O.H Outgoing Feeders mm²	1x120	2x120	-	3x120	-
Size of U.G Outgoing Feeders mm²	1x185	2x185	1x300	3x185	1x300
Pole Type	Single-Pole	H-Pole	H-Pole	H-Pole	H-Pole

Detailed materials specifications for Pole Mounted Transformers are referred to SEC Distribution Materials Specification No. 51-SDMS-01, Rev. 02, No. 51-SDMS-02, Rev. 00, No. 51-SDMS-03, Rev. 00 and No. 51-SDMS-04, Rev. 00 with its latest updates.

100 KVA size transformers shall be installed directly on pole and 200 & 300 KVA transformers shall be mounted on platform using H-pole.

Detailed Construction Specifications for Installation of Pole Mounted Transformers are referred to SEC Distribution Construction Standard No. SDCS-01, SECTION-13, Rev.00 with its latest updates.

Primary MV fuse link ratings for PMT

The standard ratings of type K fuse links for different capacities of the pole mounted transformers which are fed from over head lines and controlled by drop out cutouts are enlisted in the table below:

Table 37: Fuse link ratings

Transformer capacity (kVA)	Fuse link rating (K Type)	
	13.8 kV	33 kV
100 kVA	15 A	6 A
200 kVA	20 A	10 A
300 kVA	30 A	15 A

Detailed materials specifications for Primary MV Fuse Link of PMT are referred to SEC Distribution Materials Specification No. 34-SDMS-02, Rev. 00 with its latest updates.

LV pole

The standard distribution poles used for PMT & LV overhead system are given in the table below

Table 38- Poles used for PMT and LV overhead system

Pole Type	Description	SPAN (meters)
OC10	10 meter Steel Pole, Low Voltage	50
OC10SFS	10 meter Steel Pole, Self-Support, Single Circuit	50

Detailed materials specifications for Poles are referred to SEC Distribution Materials Specification No. 31-SDMS-03A, Rev. 01 with its latest updates.

Detailed construction specifications for Poles are referred to SEC Distribution construction Specification SDCS-01 with its latest updates.

LV Overhead conductors

The LV overhead line conductor shall be a quadruplex cable. The three insulated phase conductors and the bare neutral shall be twisted together to form what is called a quadruplex conductor consisting of three XLPE insulated aluminum conductors laid up around one bare ACSR/AW. The neutral shall act as a messenger for L.V spans up to 50m for main feeder and 30m for service drop.

Two standard sizes of conductors shall be used in the overhead low voltage distribution network as following :

- Quadruplex conductor 3x(1x120 mm^2 XLPE insulated Aluminum Conductor) + 1x120 mm^2 ACSR/AW , for main line/feeder.
- Quadruplex conductor 3x(1x50 mm^2 XLPE insulated Aluminum Conductor) + 1x50 mm^2 ACSR/AW , for service drops as connection to the customer.

Service drop cable is the portion of the system which makes the final connection from the low voltage network to the customer's premises. Detailed materials specifications for LV Overhead Line Conductor are referred to SEC Distribution Materials Specification No. 11-SDMS-02, Rev. 00 with its latest updates.

The conductor current ratings in (A) and equivalent capacities in (KVA) at different low voltages are given in the table below:

Table 39: Conductor current ratings and firm capacities (mentioned within brackets)

LV conductor size - NEW	Rating(A)	Firm Capacities (A)
4x120 mm^2 Al, quadruplex	200	160
4x50 mm^2 Al, quadruplex	110	88

Note: Firm capacity is 80% of rating

Note: These ratings are based on Standard for Calculation of Bare Overhead Conductor Temperature and Ampacity Under Steady-State Conditions. They are based on the standard rating conditions indicated and Correction factors for deviations from these conditions are indicated in (1.2 Standard Conditions)

LV service connections circuit breakers

Molded Case Circuit Breakers (MCCB) for indoor or outdoor installation in an enclosure , intended to be used for Service Connections in the Low Voltage System. The Standard Ratings for the Circuit Breakers are 20, 30, 40, 50, 70, 100, 125, 150, 200, 250, 300 and 400A.

The incoming terminals shall be suitable for aluminum conductors of sizes given for the following different ratings as shown in the table below.

Table 40: MCCB ratings and maximum size of conductors

MCCB rating (Amps)	Max size of conductors suitable for the incoming terminals
20, 30, 40, 50, 70, 100, 125, 150	Quadruplex XLPE insulated 3x50mm² + 1x50mm²
200, 250, 300	Quadruplex XLPE insulated 3x120mm² + 1x120mm²
400-500	Cables of 4x300mm^2 Al

Detailed materials specifications for LV Service Connections Circuit Breakers are referred to SEC Distribution Materials Specification No. 37-SDMS-01, Rev. 03 with its latest updates.

6.3 Calculation of Voltage Drop

For a particular supply voltage the voltage drop from the supply point to the customer interface is provided below:

$$VD_{\%} = \frac{100 \times KVA \times (R \times \cos\varphi + X \times \sin\varphi) \times L}{V^2}$$

The simplified formula for voltage drop is:

$$VD_{\%} = \frac{KVA \times L}{K}$$

Details for the above parameters of equation are explained in section 5.7 of this guideline.

The values of K constant to be used as shown in the table 41 below

Conductor Size	V	R	X	cos ϕ	sin ϕ	K
mm2	Volts	Ω/km	Ω/km			V^2.km/Ω
120	400	0.31037	0.099	0.85	0.527	5064
120	220	0.31037	0.099	0.85	0.527	1532
50	400	0.78353	0.106	0.85	0.527	2217
50	220	0.78353	0.106	0.85	0.527	671

LV Conductor Size	Constant K	
	Standard Nominal Voltages	
	400 V	220 V
4 X 120 mm² AL, Quadruplex	5064	1532
4 X 50 mm² AL, Quadruplex	2217	671

Examples of voltage drop calculation in Appendix 2

Voltage Drop calculation form 5 in Forms.

6.4 Overhead Low Voltage Network Configuration

There are three standard configurations for customer low voltage overhead connections depending on customers demand loads as following.

OH Main Feeder with Service Drop 50 mm2 Quadruplex Conductor

This type of configuration is shown in Figure . For this condition, Quadruplex Conductor 120 mm2 shall be used as main OH LV feeder from PMT LV cabinet to customer location, and Quadruplex Conductor 50 mm2 shall be used as service drop connection from that location to the customer meter/meters box. (Common configuration).

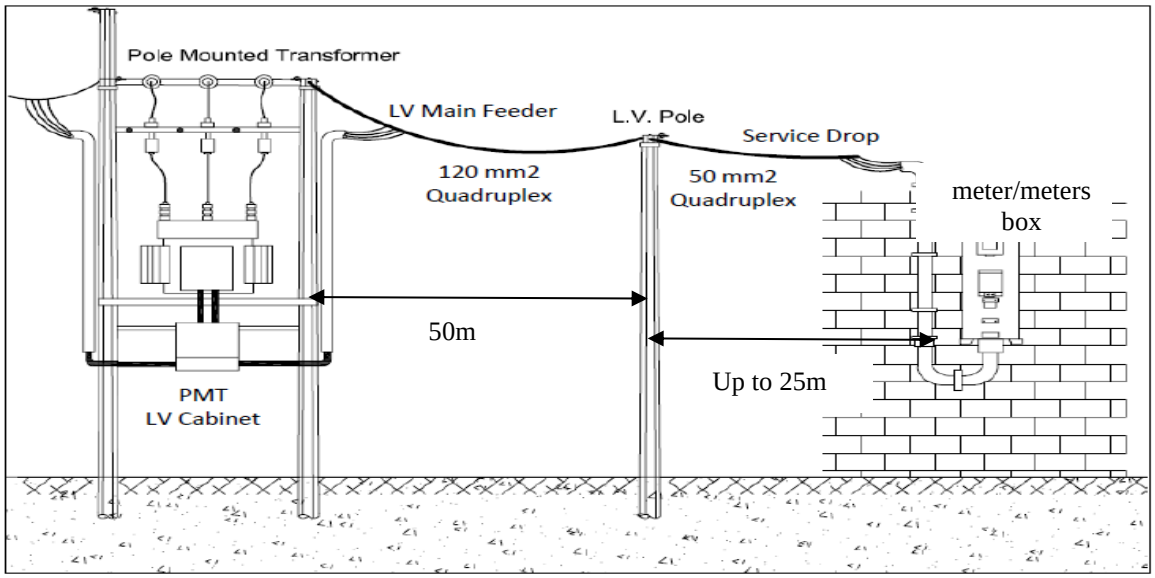


Figure 9: OH Main Feeder with Service Drop 50 mm2 Quadruplex Conductor

OH Main Feeder with Service Drop 120 mm2 Quadruplex Conductor

This type of configuration can be used only when Quadruplex Conductor 50 mm2 is not sufficient to supply the customer demand load. It is shown in Figure 10. For this condition, Quadruplex Conductor 120 mm2 shall be used as main OH LV feeder from PMT LV cabinet to customer location, and Quadruplex Conductor 120 mm2 can also be used as service drop connection from that location to the customer meter/meters box. (Heavy load lots only).

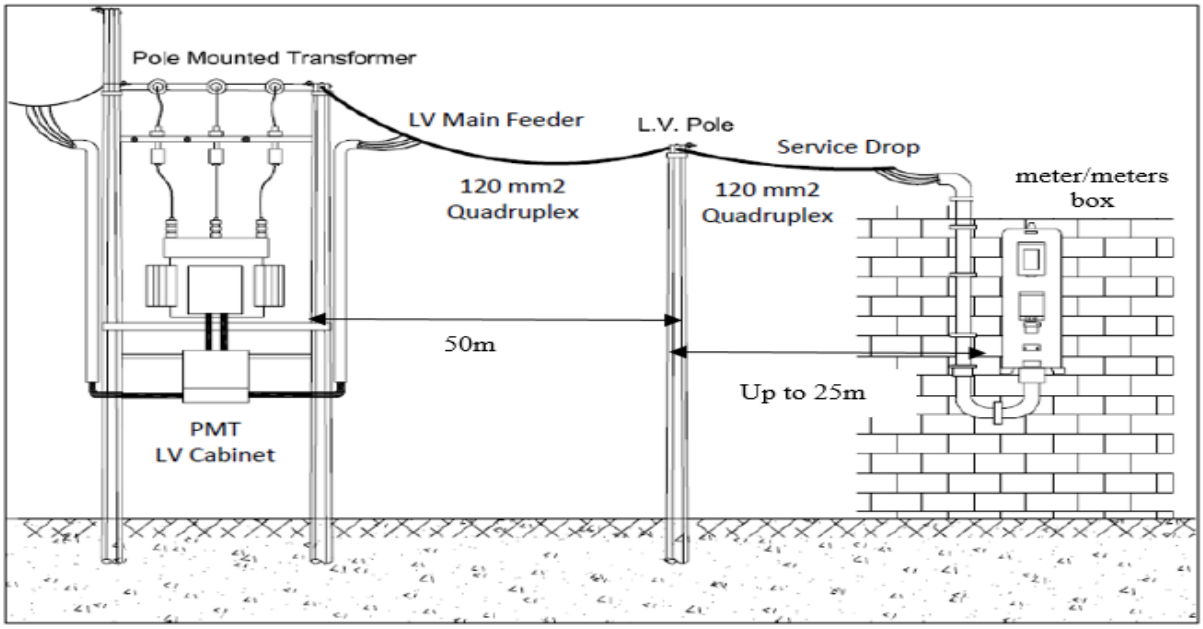


Figure 10: OH Main Feeder with Service Drop 120 mm2 Quadruplex Conductor

OH Main Feeder with Service Connection UG Cable

This type of configuration can be used only when there is a physical hindrance to use Quadruplex Conductor as service drop connection to the customer. It is shown in Figure . For this condition, Quadruplex Conductor 120 mm2 shall be used as main OH LV feeder from PMT LV cabinet to the nearest pole to customer location, and UG Cable 70 mm2 or 185 mm2 (depending on customer load) can be used as service connection from the nearest pole to the customer meter/meters box. (Exceptional configuration).

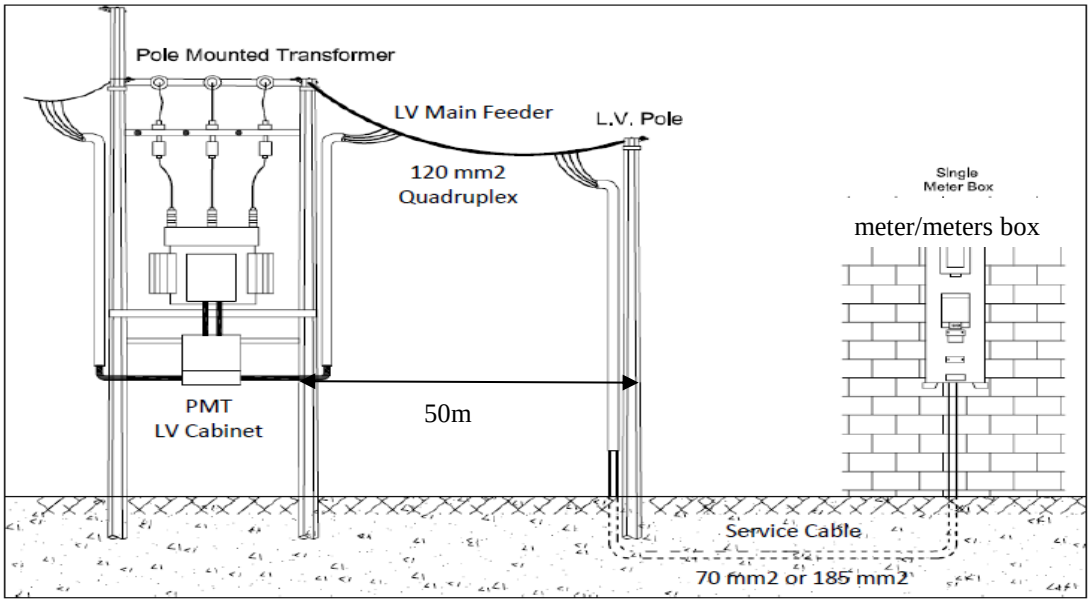


Figure 11: OH Main Feeder with Service Connection UG Cable

Additional Planning Design Principles

The general criteria from earlier can be translated into detailed design principles as outlined below:

- Design of any LV network element (PMT , LV Main Feeder , Service Drop Connection) should be based on the Coincident Demand Load (CDL) of all customers KWH Meters supplied from this element LV network element.
- To maintain the Loading percentage on any LV network element (PMT , LV Main Feeder, Service Drop Connection) within the Firm Capacity (80 % of Rating) of that LV network element.
- To maintain the Total Voltage Drop percentage on the whole LV network (LV Main Feeder + Service Drop Connection) from the PMT to the customer's location within the Voltage Drop limits (5 % of Nominal Voltage).

- The LV network design should be the most economical (Lowest Cost) as possible to supply the projected customer's load.
- The suitable size of the conductor to supply the customer should be selected according to the Coincident Demand Load (CDL) of that customer and should be suitable to satisfy that customer's CDL is not greater than the Firm Capacity (80 % of Rating) of that conductor.
- The suitable connection configuration type to supply the customer should be selected according to the Coincident Demand Load (CDL) of that customer.
- CDL on the PMT should be not greater than PMT's Firm Capacity (i.e. not exceeding 80 % of PMT's rating).

$$CDL\ (KVA)_{on\ PMT} = CDL\ for\ all\ N\ Meters\ supplied\ from\ PMT$$

$$Loading\ \%_{on\ PMT} = \frac{CDL\ (KVA)_{on\ PMT}}{Rating_{PMT}} \times 100$$

- CDL on the private PMT should be not greater than PMT's rating (i.e. not exceeding 100 % of PMT's rating) and is calculated using the above formula
- CDL on the LV Main Feeder should be not greater than LV Main Feeder's Firm Capacity i.e. not exceeding 80 % of LV Main Feeder's rating).

$$CDL_{on\ Main\ Feeder} = CDL\ for\ all\ N\ Meters\ supplied\ from\ Main\ Feeder$$

$$Loading\ \%_{on\ Main\ Feeder} = \frac{CDL_{on\ Main\ Feeder}}{Rating_{Main\ Feeder}} \times 100$$

- CDL on the Service Drop should be not greater than Service Drop's Firm Capacity (i.e. not exceeding 80 % of Service Drop's rating).

$$CDL_{on\ Service\ Drop} = CDL\ for\ all\ N\ Meters\ supplied\ from\ Service\ Drop$$

$$Loading\ \%_{on\ Service\ Drop} = \frac{CDL_{on\ Service\ Drop}}{Rating_{Service\ Drop}} \times 100$$

- CDL on the Direct Feeder should be not greater than Direct Feeder's Firm Capacity (i.e. not exceeding 80 % of Direct Feeder's rating).

$$CDL_{on\ Direct\ Feeder} = CDL\ for\ all\ N\ Meters\ supplied\ from\ Direct\ Feeder$$

$$Loading\ \%_{on\ Direct\ Feeder} = \frac{CDL_{on\ Direct\ Feeder}}{Rating_{Direct\ Feeder}} \times 100$$

- CDL on the Direct Feeder, which fed large meters (300, 400, 500) should be not greater than Direct Feeder's rating (i.e. not exceeding 100 % of Direct Feeder's rating).

$$CDL_{on\ Direct\ Feeder} = CDL\ Meter\ supplied\ from\ Direct\ Feeder$$

$$Loading\ \%_{on\ Direct\ Feeder} = \frac{CDL_{on\ Direct\ Feeder}}{Rating_{Direct\ Feeder}} \times 100$$

- Total VD% from PMT to customer's location should be not greater than voltage drop limit (i.e. not exceeding 5 %).

$$VD\ \%_{on\ Service\ Drop} = \frac{CDL\ (KVA)_{on\ Service\ Drop} \times L_{Service\ Drop}}{K_{Service\ Drop}} \times 100$$

$$VD\ \%_{on\ Main\ Feeder} = \frac{CDL\ (KVA)_{on\ Main\ Feeder} \times L_{Main\ Feeder}}{K_{Main\ Feeder}}$$

$$VD\ \%_{on\ Direct\ Feeder} = \frac{CDL\ (KVA)_{on\ Direct\ Feeder} \times L_{Direct\ Feeder}}{K_{Direct\ Feeder}}$$

$$VD\ \%_{Total\ from\ SS\ to\ Customer} = VD\ \%_{Main\ Feeder} + VD\ \%_{Service\ Drop}$$

$$VD\ \%_{Total\ from\ PMT\ to\ Customer} = VD\ \%_{Direct\ Feeder}$$

- Always try first to supply customer's CDL from any existing nearby LV Main Feeders (one by one) with priority for the nearest as possible based on the criteria (Loading % , Voltage Drop %) before planning to install new LV Main Feeder.
- Always try first to supply customer's CDL from any existing nearby PMT's (one by one) with priority for the nearest as possible based on the criteria (Loading % , Voltage Drop %) before planning to install new PMT.
- To supply customer's CDL from any existing LV Main Feeder, first check for capability of supply from that LV Main Feeder.
- To supply customer's CDL from any existing PMT , first check for availability of any vacant outgoing in that PMT.
- Install the new PMT in the center of loads area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, empty lots) as possible.
- Select the shortest geographic route for the LV Main Feeder (120 mm² cable) from PMT to the customer's location (as possible).
- Select the shortest geographic route for the service drop from the LV Main Feeder to customer's location (as possible).
- Select the shortest geographic route for the Direct Feeder from PMT to the customer's location (as possible).
- Avoid crossing the streets when you design the route of any LV conductor as possible as you can.

- It is not allowed to cross any street with width more than 30 meters for any LV conductor route.
- CB rating of the outgoing from PMT LV Cabinet should be not less than the largest CB rating of all KWH Meters supplied from this outgoing CB. Same is valid for any two CBs outgoings supply customers.
- CB rating of the outgoing from PMT LV Cabinet should be not less than the Coincident Demand Load (CDL) of all customers KWH Meters supplied from this outgoing CB. Same is valid for any two CBs outgoings supply customers.
- Size (KVA rating) of the new PMT should be selected based on the need of the neighbor area (including : concerned new customer , existing others nearby supply requests , nearby under constructions buildings , existing empty lots) and it should be as minimum as sufficient to meet their total Coincident Demand Load (CDL).
- If multi PMTs are required to supply a customer , select the no. of the required PMTs and their ratings from the available SEC standard (100, 200, 300 KVA) where the summation of PMTs ratings should provide minimum sufficient total capacity to meet the calculated Coincident Demand Load (CDL) of the customer with minimum no. of PMTs.
- For supplying new customers , It is preferred to avoid using the PMT with 300 KVA rating as possible and it is preferred to use the PMT with (100 KVA or 200 KVA) rating instead of that. This is to maintain a possibility for reinforcement of these PMTs (100 KVA & 200 KVA) by replacing them with 300 KVA PMTs without the need to install a new PMT.
- No. of Meter Boxes and their sizes required to handle the KWH Meters required to supply a customer should be as minimum as sufficient with minimum no. of Meter Boxes. i.e. always use larger size of Meter Box to handle more possible KWH Meters in one box instead to use multi smaller size of Meter Boxes for same no. of KWH Meters.

6.5 Step By Step Design Procedure

- Connected Load (CL) in (KVA) for each Individual unit in customer's building should be estimated, unit by unit, as per Section referring to (Customer Load Estimation)
- Individual Circuit Breaker Rating (CBR) in (Amp) for the Individual KWH Meter for each Individual unit in customer's building should be determined according to the estimated connected load (CL) of that Individual unit and referring to (Customer Load Estimation)
- Number of Individual KWH Meters (N) required for the customer's building should be determined according to number of Individual units in customer's building and referring to SEC Customer Services Manual with its latest updates.
- Calculate the Coincident Demand Load (CDL) in (Amp) for the group of all KWH Meters of the customer's building based on (coincident demand load (CDL) calculation).

- Based on the calculated Coincident Demand Load (CDL) in (Amp) of the customer's building, select the suitable connection configuration type to supply this Coincident Demand Load (CDL) as shown in table 44 Hereunder. the suitable connection configuration type includes :

a. Size of conductor to customer.

b. No. of conductors to customer required.

c. Suitable supply source: Direct Feeder from PMT or Service Drop through LV Main Feeder.

d. No. of outgoing required.

Table 42: Coincident Demand Load (CDL)

Coincident Demand Load (A)		Supply Source	No. of Outgoing MCCB	Number of LV Conductors to Customer	Size of LV Conductors to Customer	Main OH LV Feeder	
From	To					No. of Conductors to OH	Conductor Size
1	88	PMT	1	1	50mm2	1	120mm2
89	160	PMT	1	1	120mm2	1	120mm2
1	108	PMT	1	1	70mm2	1	120mm2
109	184	PMT	1	1	185 mm2	Direct UG Feeder	
185	370	PMT	2	2	Up to 300 mm2	Direct UG Feeder	

- If the suitable connection configuration type is Service Drop through Main Feeder, go to the next step.
- First try to supply customer's CDL from existing nearby Main Feeder by using the following steps:

a. Select the nearest existing Main Feeder to the customer's location (as possible).

b. Calculate CDL on the Main Feeder (120 mm2 cable) including of all customers KWH Meters (concerned new customer + existing customers) supplied from this Main Feeder.

c. CDL on the Main Feeder should be not greater than Main Feeder's Firm Capacity (i.e. not exceed 80 % of Main Feeder's rating).

d. CDL on the Direct Feeder, which fed large meters (300, 400, 500) should be not greater than Direct Feeder's rating (i.e. not exceeding 100 % of Direct Feeder's rating)

e. Calculate CDL on the PMT including of all customers KWH Meters (concerned new customer + existing customers) supplied from this PMT.

f. CDL on the PMT should be not greater than PMT's Firm Capacity (i.e. not exceed 80 % of PMT's rating). CDL on the private PMT should be not greater than PMT's rating (i.e. not exceeding 100 % of PMT's rating)

g. Select the shortest geographic route for the Service Drop from Main Feeder to customer's location (as possible).

h. Calculate VD% on the Main Feeder (120 mm2 cable) from PMT to customer's location.

i. Calculate VD% on the Service Drop from Main Feeder to customer's location.

j. Calculate the Total VD% from PMT to customer's location.

k. Total VD% from PMT to customer's location should be not greater than voltage drop limit (i.e. not exceed 5 %).

l. If customer's CDL cannot be supplied from the selected Main Feeder because one of the criteria (Loading % , Voltage Drop %) is not satisfied , Try all others nearby existing Main Feeders (one by one) with priority for the nearest and by using same steps in above (from "a" to "k").

- If customer's CDL cannot be supplied from all nearby existing Main Feeders because one of the criteria (Loading %, Voltage Drop %) is not satisfied, go to the next step.
- Try to supply customer's CDL from existing nearby PMT through a new Main Feeder by using the following steps:

a. Select the nearest existing PMT to the customer's location (as possible).

b. Calculate CDL on the PMT including of all customers KWH Meters (concerned new customer + existing customers) supplied from this PMT.

c. CDL on the PMT should be not greater than PMT's Firm Capacity (i.e. not exceed 80 % of PMT's rating).

d. Select the shortest geographic route for the new Main Feeder (120 mm^2 cable) from PMT to the customer's location (as possible).

e. Select the shortest geographic route for the Service Drop from the new Main Feeder to customer's location (as possible).

f. Calculate VD% on the new Main Feeder (120 mm^2 cable) from PMT to customer's location.

g. Calculate VD% on the Service Drop from the new Main Feeder to customer's location.

h. Calculate the Total VD% from PMT to customer's location.

i. Total VD% from PMT to customer's location should be not greater than voltage drop limit (i.e. not exceed 5 %).

j. If customer's CDL cannot be supplied from the selected PMT because one of the criteria (Loading % , Voltage Drop %) is not satisfied , Try all others nearby existing

PMT (one by one) with priority for the nearest and by using same steps in above (from "a" to "i").

- If customer's CDL cannot be supplied from all nearby existing PMT because one of the criteria (Loading %, Voltage Drop %) is not satisfied, go to the next step.
- Design to supply customer's CDL from a new PMT through a new Main Feeder by using the following steps:
  a. Design to install a new PMT near to customers lots in the center of loads area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, empty lots) as possible.
  b. Size (KVA rating) of the new PMT should be selected based on the need of the neighbor area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, existing empty lots).
  c. Select the shortest geographic route for the new Main Feeder (120 mm2 cable) from the new PMT to the customer's location (as possible).
  d. Select the shortest geographic route for the Service Drop from the new Main Feeder to customer's location (as possible).
  e. Calculate VD% on the new Main Feeder (120 mm2 cable) from the new PMT to customer's location.
  f. Calculate VD% on the Service Drop from the new Main Feeder to customer's location.
  g. Calculate the Total VD% from the new PMT to customer's location.
  h. Total VD% from the new PMT to customer's location should be not greater than voltage drop limit (i.e. not exceed 5 %).
- If the suitable connection configuration type is Direct Feeder from PMT, go to the next step.
- First try to supply customer's CDL from existing nearby PMT by using the following steps:
  a. Select the nearest existing PMT to the customer's location (as possible).
  b. Calculate CDL on the PMT including of all customers KWH Meters (concerned new customer + existing customers) supplied from this PMT.
  c. CDL on the PMT should be not greater than PMT's Firm Capacity (i.e. not exceed 80 % of PMT's rating).CDL on the PMT should be not greater than PMT's rating (i.e. not exceed 100 % of PMT's rating).
  d. Select the shortest geographic route for the Direct Feeder from PMT to the customer's location (as possible).
  e. Calculate VD% on the Direct Feeder from PMT to the customer's location.

f. Total VD% from PMT to customer's location should be not greater than voltage drop limit (i.e. not exceed 5 %).

g. If customer's CDL cannot be supplied from the selected PMT because one of the criteria (Loading % , Voltage Drop %) is not satisfied , Try all others nearby existing PMTs (one by one) with priority for the nearest and by using same steps in above (from "a" to "f").

- If customer's CDL cannot be supplied from all nearby existing PMTs because one of the criteria (Loading %, Voltage Drop %) is not satisfied, go to the next step.
- Design to supply customer's CDL from a new PMT by using the following steps:

a. Design to install a new PMT near to customers lots in the center of loads area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, empty lots) as possible.

b. Size (KVA rating) of the new PMT should be selected based on the need of the neighbor area (including: concerned new customer, existing others nearby supply requests, nearby under constructions buildings, existing empty lots).

c. Select the shortest geographic route for the Direct Feeder from the new PMT to the customer's location (as possible).

d. Calculate VD% on the Direct Feeder from the new PMT to the customer's location.

e. Total VD% from the new PMT to customer's location should be not greater than voltage drop limit (i.e. not exceed 5 %).

6.6 Connection to LV Customers (from 300A to 500A load)

The process for connecting bulk customers at LV side is the same as the process for LV new connections (which is detailed in Section….) However, while connecting large customers, ratings for LV equipment need to be updated to take into account the additional load requirements

Meters of more than 500A (600A, 800A and more than 800A) cannot be supplied through overhead PMT cabinet configuration since maximum capacity of supply of PMT LV cabinet is 400A

Table 43: Supply Method for LV Customer (Overhead) FROM 300A TO 500A LOAD)

CB Rating (A)	Demand Factor	CDL (A)	Supply Source	No of Outgoing	No. of Cables	Cable Size (mm2)	Comments
300	0.5	150	PMT	1	1	120 /185	One direct cable
	0.6	180	PMT	1	1	120 /185	One direct cable
	0.7	210	PMT	1	1	185	One direct cable
	0.8	240	PMT	1	1	300	One direct cable
400	0.5	200	PMT	1	1	120/185	One direct cable
	0.6	240	PMT	1	1	300	One direct cable
	0.7	280	PMT	1	1	300	One direct cable
	0.8	320	PMT	1	1	300	One direct cable
500	0.5	250	PMT	1	1	300	One direct cable
	0.6	300	PMT	1	1	300	One direct cable
	0.7	350	PMT	2	2	300	Two direct cable
	0.8	400	PMT	2	2	300	Two direct cable

Note: Meters of more than 500A (600A, 800A and more than 800A) cannot be supplied through overhead PMT cabinet configuration since maximum capacity of supply from 2 outgoing MCCBs of PMT LV cabinet is 400A. Hence, for loads of more than 500A, underground configuration needs to be used

Examples of Overhead LV Connection Design

7 Medium Voltage (MV) Connections Planning

The rules and guidelines in this section will be applicable for all network connections at medium voltage .This will include the following types of connection requests:

a. New connection requests.

b. Temporary MV connection to customer

c. Reinforcement, replacement and integration of new grid station or MDN by MV networks.

7.1 Voltage drop calculation

The formula for voltage drop is provided below:

$$\%\,V.D = \frac{kVA\,(R\cos\emptyset + X\sin\emptyset)\,L}{10kV^2}$$

Where:

kVA = Three phase load in kVA.

R = Resistance of conductor in ohms per kilometer

X = Inductive resistance of conductor in ohms per kilometer

kV = Three phase supply voltage in kilovolts at sending end

L = Length of cable in kilometers

Ø = Angle of supply

This formula can be modified to $\% \text{ V. D} = \frac{\text{kVA x L}}{\text{K}}$

Where K is a constant $K = \frac{10kV^2}{\text{Rcos}\emptyset + \text{Xsin}\emptyset}$

Table 44: Voltage Drop Calculations for MV Cables and Conductors

Cable / Conductor size	R (Ω/km)	X (Ω/km)	Voltage	At Power Factor 0.85		
				cos ϕ	sin ϕ	K
3 x 500 mm2 Al	0.0818	0.105	13.8	0.85	0.527	15252
3 x 300 mm² Cu	0.0808	0.108	13.8	0.85	0.527	15163
3 x 185 mm² Cu	0.129	0.116	13.8	0.85	0.527	11151
3 x 300 mm² Al	0.13	0.108	13.8	0.85	0.527	11375
170 mm² ACSR	0.210	0.391	13.8	0.85	0.527	4952
70 mm² ACSR	0.529	0.422	13.8	0.85	0.527	2834
240 mm² ACSR	0.150	0.374	13.8	0.85	0.527	5867
3 x1 x 500 mm2 Cu	0.0597	0.13	33	0.85	0.527	91317
3 x 240 mm2 Cu	0.0987	0.131	33	0.85	0.527	71208
3 x 400 mm2 Al	0.1023	0.124	33	0.85	0.527	71502
3 x 185 mm² Cu	0.129	0.116	33	0.85	0.527	63766
170 mm² ACSR	0.210	0.404	33	0.85	0.527	27823
70 mm² ACSR	0.529	0.436	33	0.85	0.527	16028
240 mm² ACSR	0.150	0.388	33	0.85	0.527	32804

*Existing but non standard

Examples for calculation voltage drop (MV network) in appendix 2

Voltage Drop calculation form 5 in Forms.

7.2 Processes & Procedures for Connection Design

MV Design Criteria & Principles

The following criteria should be taken into account while designing MV network for new connections:

- Optimization first principle
- Grid station criteria
- Transformer loading criteria
- Feeders loading criteria

- Contingency planning
- Normal open point location
- Voltage drop
- Length of feeder
- Feeders configuration criteria

Additionally, the "Distribution Security Standard" issued by WERA, with latest updates should be followed

1- Optimization first principle

The order of priority for solutions to handle any network request should be as follows:

- Network optimization
- Reinforcement
- Expansion

2- Grid stations criteria

The grid station is the interface point between transmission level and distribution level. In planning stage, the following should be considered:

- The study for new connection application of a customer from an existing grid station will depend upon the forecasted load of that grid station.
- The load should not exceed the firm capacity of the grid station based on N-1 criteria.
- The location of the new grid station should be in load center
- The length of feeders should be limited to avoid extra ordinary long lengths
- The capacity of the new grid station should be appropriate for the forecasted load.
- The time required to build the grid station should be in line with the timeline of demand realization
- The number / rating of outgoing MV feeders from each grid station should be reasonable
- For the interface details between transmission level and distribution level, refer to "Operation Interface Agreement" signed between SEC' Distribution business unit and National Grid.
- For details about developing the new grid station for MV customer, refer to SEC Customer services Manual with latest updates.
- There are specific guidelines and conditions to guide whether network planner should request for grid station or for main distribution network (MDN) substation (if available in the area). These are mentioned below:
- The first priority would be to check if the new demand load can be met through adding feeder(s) in existing MDN substations or grid stations, using the conditions mentioned below.

If and only if the above cannot be done, request for new MDN substations should be made3. If and only if the demand load cannot be met through addition of new MDN substations, new grid stations may be requested. However, exceptions may be made based on the rules for requested load and depending on the specific site scenario

- An MDN substation should be requested only if any of the following conditions are met:
  a. A customer connection with very high demand load which requires dedicated MDN substation or it is specified based on rules for requested load or there are specific site scenarios
  b. The new demand load (over the 5-year time horizon) cannot be met through new feeders from existing MDN substations or grid stations, i.e. adding such feeders will lead to load exceeding firm capacities of existing MDN substations and / or grid stations
  c. Catering to the new demand load (over the 5-year time horizon) from existing MDN substations / grid stations will lead to voltage drop in excess of 5%. This will be relevant for demand in areas that are geographically far away from existing MDN substations / grid stations
  d. Existing interties cannot sufficiently take care of contingency situations. For example, load transfer from existing feeders leading to overloading across other feeders and equipment and no new feeders can be added. Under this situation, the n-1 reliability of the system is compromised (although the situation can be handled through use of mobile equipment). A permanent solution would mean creation of new interties through new MDN substations
  e. There are geographical constraints for new feeders from existing MDN substations
- Request for grid station, which is the interface point between National Grid and Distribution Business Unit, should be submitted to National Grid
- Such a request should be made only after assessing the need for grid station through 5-year network plan .
- A new grid station request can only be made if addition of new MDN substation is not possible:
  a. Addition of new MDN substation is not possible due to geographical constraints (either remoteness of area or congestion in area)
  b. Addition of new MDN substation cannot sufficiently cater to new demand load (over the upcoming 5-year time period)

[3] There will be a need to align this with the overall strategic objectives of SEC with respect to MDN substations. For example, if SEC plans to phase-out MDN substations across all areas, requests for new MDN substations will not be accepted. Under such circumstances, only requests for new grid stations should be made

c. All existing grid stations catering to the area are fully loaded and new MDN substation will lead to grid station exceeding its firm capacity

d. There is strong economic rationale for new grid station, i.e. new grid station will be economically more feasible than new MDN substation

- However, exceptions to the above may be made based on the rules for requested load and depending on the specific site scenario. This should be verified with operating areas

3- Transformer loading

Maximum load for grid station / MDN substation shall not exceed 100% of the installed capacity of substation when load reaches 80%, reinforcement should be planned according to load forecasting Guidelines

4- Feeder loading

The table below describes the maximum loading of different types of feeders.

Table 45: Maximum loading of different types of feeders

Feeder type	Maximum loading with relation to de-rated capacity	Comment
Radial feeder	100%	For public feeder : Reinforcement should be commenced when load reaches 80%
Single loop	50%	N-1 criterion must be maintained. This takes higher precedence than loading of individual feeder
Tee loop	66%	
Multi loop	66%	N-1 criterion must be maintained

For N-1 loop. Offline criterion, in case of power failure in one of the feeders, the other feeder should be capable to meet the whole demand until the repair work is completed.

5- Contingency planning

Distribution network plans shall be developed to meet the first level contingency conditions and not for multiple contingencies as per distribution planning criteria. Abnormal (though rarely) multiple contingencies may arise in the network resulting in loss of supply to customers.

Contingencies to be considered in Distribution Network Planning

The following first level contingencies shall be considered in the development of a 5-year network plan:

- Failure of any one of the power transformers in any substation having single, double or multiple power transformers.

- Failure of any one of the bus sections in any substation having single, double or multiple bus sections, which normally involves interruption to all the loads associated with the bus section. but the first contingency criteria require that the power supply shall be restored within reasonable time through available standby/alternate supply
- Failure of any one-feeder segment in any feeder network configuration.

Grid station overloading

Load shifting in the distribution network should be proposed to avoid the expected overloading, if load transfer capability is available in the system at distribution level. The interties between different grid stations can facilitate load transfer from one grid station to the other neighboring grid station(s) by shifting the normal open point in the loop. The first step of the contingency plan shall be the identification of the interties and the available relief through each intertie depending upon the overload on the grid station and the spare capacity available in the neighboring grid stations. Factors such as auto change over switches; operational inconvenience, important loads and geographical location restrict or limit the load transfer through an intertie and therefore are required to be thoroughly examined. Accordingly, proposals shall be made to suitably shift the normal open points to efficiently utilize the system spare capacity to relieve an overloaded facility.

6- Normal open point location

Based on the following factors, normal open points in the distribution network should be decided:

- Distribution of load on each feeder
- Distribution of load on the grid station
- Continuity performance
- VIP customers
- Voltage drop
- Auto-change over switches
- Easy accessibility
- Equipment operational flexibility
- Optimal energy loss

All temporary shifting shall be by operating officials as per operational requirements.

The most desirable design condition for a normally open point in any loop is to have equal loading on the individual circuits of the loop and to have each circuit supplied from separate grid stations as possible as it can be (to achieve maximum load transfer capability between grid stations). As an alternative to supply from separate grid stations, if other grid station is not available in the same zone, the circuits may be looped onto different bus sections at the same grid station (with this arrangement, station capability will not be achieved).

7- Voltage Drop

The voltage drop on any feeder should not exceed 5%. Any new plan needs to be assessed-(load flow software can be used) to determine the extent of voltage drop for any customer from the source, in both peak load and low load scenarios. If any over-voltage or under-voltage is observed, appropriate measures need to be taken, which can include use of voltage regulators, capacitor banks or other solutions, These are outlined in the subsequent chapter on MV Network Performance Improvement.

8- Maximum feeder length

The length of feeders to be controlled by the following:-

- Optimal utilization of the rated capacity of the feeder.
- Voltage drop shall be within +/- 5% limit and voltage regulators can be used for overhead network
- Operating circumstances.
- Number of customers

9- Feeder configuration

Single loop is preferred under normal circumstances but due to customer location and feeder loading, other configurations may be used:

- For feeders with high load, tee loop (Option 1 – equal sharing of load) is preferred
- Tee loop (Option 2 – 3rd feeder used for emergencies with 2 loaded feeders) is not preferred. When the combined feeder load in a single loop exceeds normal rating of the cable depending on the size and construction of the line, tee loop arrangement shall be considered. There are two options for use of tee loops:
  a. Option 1: three feeders sharing approximately equal load connected together
  b. Option 2: two feeders each loaded to full capacity and one feeder as express circuit to provide back up to either of the two feeders in case of emergency
- Radial configuration is to be used for remote area customers. This is the most economical type of supply but offers minimum reliability (circuit-out conditions)
- Multi loop configuration consists of more than three feeders and used to increase reliability between feeders if more than different gird station is available in the same zone, but it should not be used in new project or new area.

Single loop consists of two radial feeders. Such radial feeders should be looped between two neighbouring grid stations. Alternatively, they may be looped between different MV buses of the same grid station.

Wherever practical and economical, loop supply should be provided with diversified sources. The network shall be operated radially and the total load of loop shall not exceed the normal rating of the conductor / cable. This type of feeder arrangement offers an acceptable degree of reliability but at a higher initial cost.

Figure 12: Single Loop

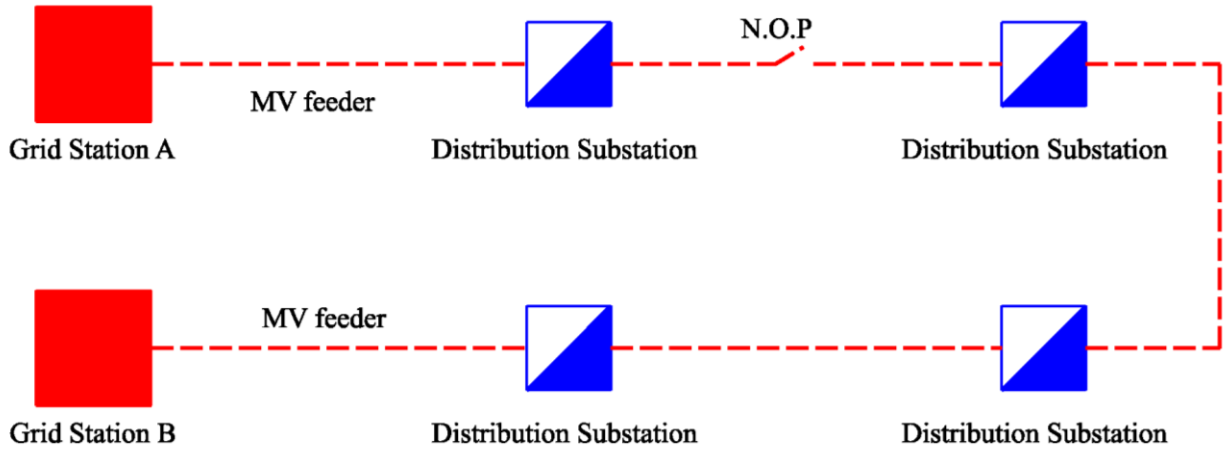


Figure 13: Radial System

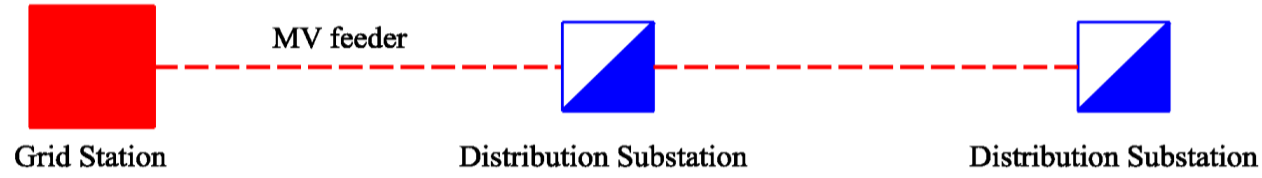


Figure 14: Tee Loop (Option 1)

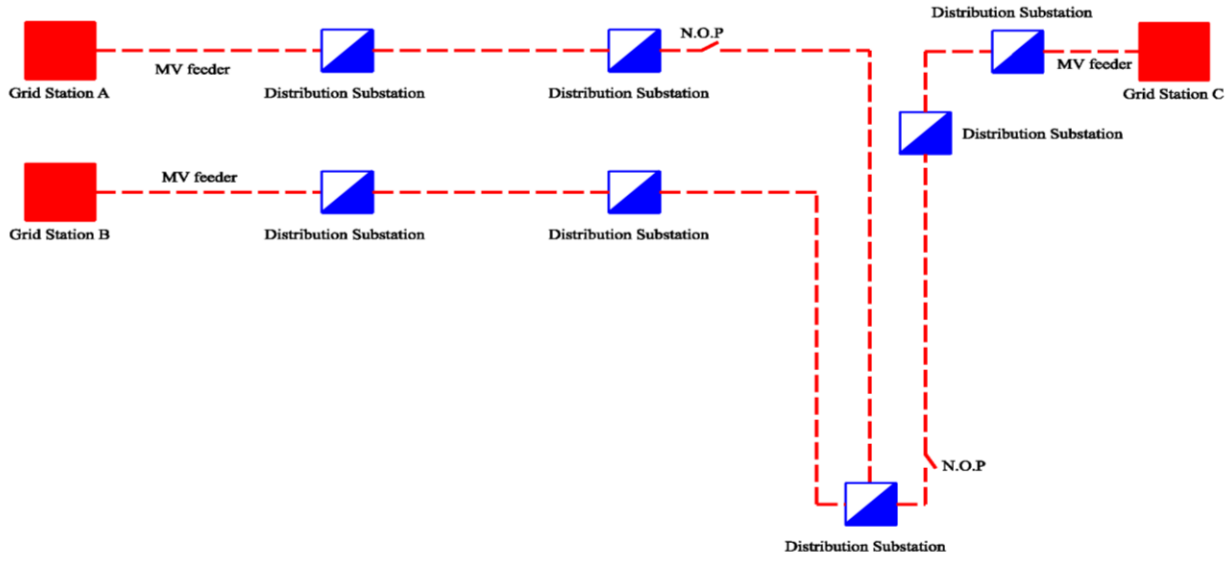


Figure 14: Tee Loop (Option 2)

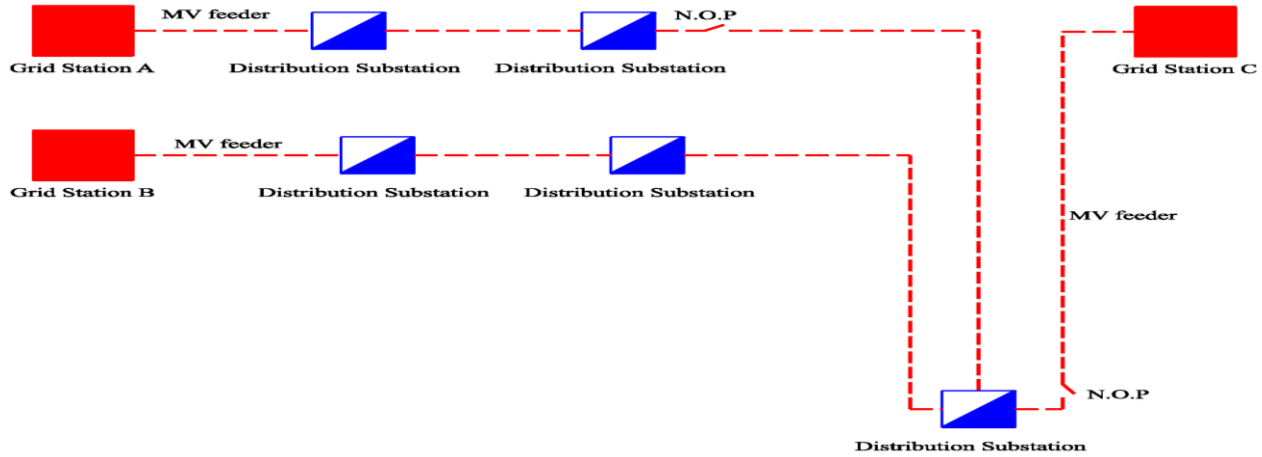
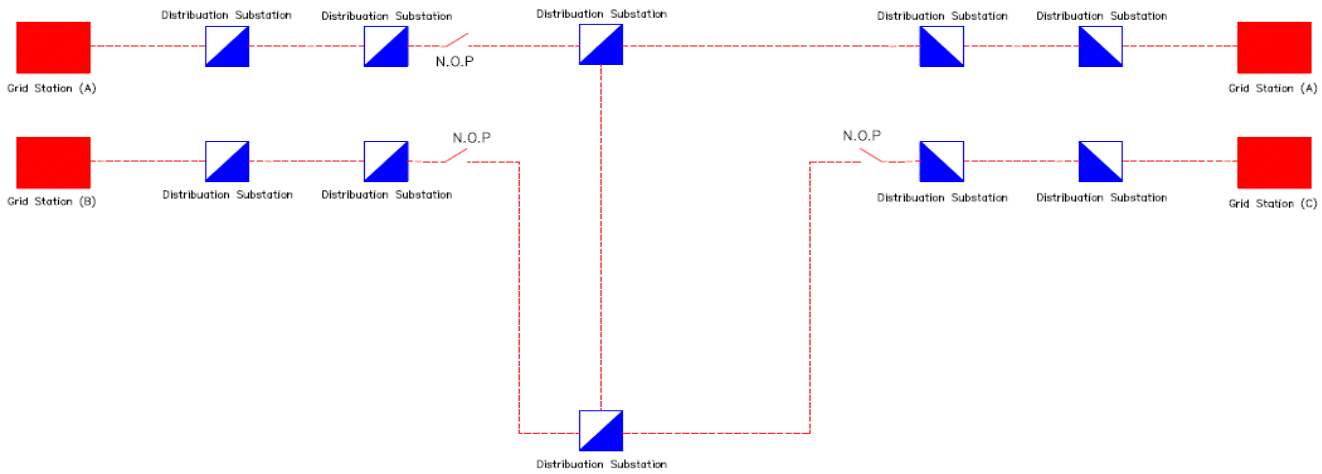


Figure 15 multi Loop



Segmentation of long direct feeders

For long underground / overhead direct MV feeders can using RMU / LBS to divide feeders which contributes to improving reliability and repair of outages,

The table 46 below describes divided of long feeders:-

Feeder type	Voltage KV	Distance km	Equipment used
underground	13.8	7	RMU
	33	12	
Over head	13.8	5	LBS
	33	10	

7.3 Materials Specifications for MV network (Underground & Overhead).

1- Cables and conductors

Frequently used cables and conductors include the following:

Table 47: MV Underground cable

Size	Voltage level	Direct buried cable rating	
		Rating (A)	Rating (MVA)
3x500 mm² Al	13.8 kV	380	9
3x400 mm² AL	33 kV	350	20
3 x240 mm² Cu	33 kV	350	20
3 x(1x500) mm² Cu	33 kV	500	29

Table 48: Overhead conductors

Size	Voltage level	Rating (A)	Rating (MVA)
170 mm² ACSR	33 kV	361	20.6
70 mm² ACSR		207	11.8
170 mm² ACSR	13.8 kV	361	8.6
70 mm² ACSR		207	4.9

Table 49 Old MV cables& conductors are existing but currently non-standard as the following:

Cable size	Voltage Level	Rating (A)	Rating MVA
3 x 185 mm² Cu	33 kV	290	17
240 mm² ACSR		450	25.7
3 x 300 mm² Cu	13.8 kV	390	9
3 x 300 mm² Al		300	7
3 x 185 mm² Cu		290	7
240 mm² ACSR		450	10.8

Standard cable ratings are presented as guidelines only and are based on the indicated assumptions. Variations from standard conditions and the general suitability of the ratings method shall be checked before using the ratings. Special surveys to define environmental and operating conditions should be carried out prior to major engineering works. These load ratings are based solely on the thermal rating of the equipment. For details please referred to 1.2 Standard conditions.

2- RMU

The non-extensible ring main unit (RMU consist of load break switches 400 A and circuit breakers (200A tee off):

Type	Voltage Level	Configuration
3 WAY RMU	13.8- 33 KV	(LBS-CB-LBS)
	13.8- 33 KV	(LBS-LBS-LBS)
4 WAY RMU	13.8- 33 KV	(LBS-LBS-CB- LBS)
	13.8-33 KV	(LBS-CB-CB-LBS)
	13.8-33 KV	(LBS-LBS-LBS-LBS)

For materials specification details, refer to 32-SDMS-01, 32-SDMS-04, 32-SDMS-07, 32-SDMS-11 with latest updates. For constructions specification details, refer to SDCS 02 part (11-12) with latest updates.

3- MRMU

The metered ring main unit (MRMU) at customer end rating 400A/630A are given in the table 50 below:

Type	Rating A	Voltage	(LBS)	Outgoing (CB)	Current transformer for CB
MRMU	400	13.8/33 KV	2 or 3	1	200/400
	630		2 or 3	1	300/600

For materials specification details, refer to 32-SDMS-02, 32-SDMS-05, 32-SDMS-06 ,32-SDMS-12 with latest updates. For constructions specification details, refer to SDCS 02 part 10 with latest updates

4- MV OVERHEAD POLES

The standard distribution poles used for MV overhead system are given in the table 51 below:

Pole Type	Description	SPAN (meters) Single Circuit	SPAN (meters) Double Circuit	comment
OC (12-13-14) S	(12-13-14) meter Steel Pole, Medium Voltage, Single Circuit	100		
OC14D	14 meter Steel Pole, Medium Voltage, Double Circuit	100	100	
OC15S/D	15 meter Steel Pole, Medium Voltage, Single & Double Circuit	100	100	
OC (12-13-14-15) SFS	(12-13-14-15) meter Steel Pole, Self-Support, Single Circuit	100		
OC18S/D	18 meter special Steel Pole, Medium Voltage, Single & Double Circuit	200	150	For Crossing wide streets or valleys
OC23S/D	23 meter special Steel Pole, Medium Voltage, Single & Double Circuit	250	200	
OC29S/D	29 meter special Steel Pole, Medium Voltage, Single & Double Circuit	300	250	

Detailed materials specifications for Poles are referred to No. 20-SDMS-01, 20-SDMS-03 with latest updates. For constructions specification details, refer to SDCS with latest updates

5- Load Break Switch (LBS)

Load break switches are used in the overhead distribution system. Load break switch operates manually only.
Load break switches are added where necessary, if the number of existing switches is not considered as being appropriate.

Voltage level	Current Rating
13.8 kV	400 A & 600 A
33 kV	400 A & 600 A

For details, refer to 30-SDMS-01 and DOM 01-20 with latest updates.

6- Fuse

Dropout fuse cutouts are used in the overhead distribution system. Installation of fuses on shorter or lightly loaded laterals is recommended. These are considered as a low cost, yet efficient solution for line sectionalization and to protect equipment against short circuits.
Since the fuse does not have reclosing capability, faults whether temporary or permanent by nature, will cause a sustained outage.
Selection of fuse should be done based on short circuit study in consultation with Protection Engineering function.
The basic technical parameters of the fuse cutout are:

- rated current of the fuse holder
- rated voltage
- short-circuit current interruption rating
- nominal current of the fuse-link. The load current should not exceed this magnitude.
- the time-current curve.

Voltage level	Current Rating
13.8 kV	100 A & 200 A
33 kV	100 A

For details, refer to 34-SDMS-01 with latest updates.
Note: fuse rating has to account for in-rush currents and cold load pick-up

7- Energy smart meter

For revenue metering, the CT & VT operated smart energy meter(s) are given in the table 57 below

Type meter	Rated Current	Voltage	Elements	Starting current (CL = 0.5)
Electronic revenue CT & VT	(In) 1.5 (A)/ (Imax) 6 A)	110 V (phase to phase)	2 or 3 elements	0.001 In

For materials specification details, refer to 40-SDMS-02A Rev 9.1 with latest updates.

7.4 Additional Design Principles for MV Connections

In addition to the MV network design criteria, the following additional guidelines need to be taken into account for connected at MV (13.8kV and 33kV):

- All equipment should be utilized in an optimal manner (for example, shifting of load from overloaded grid station to lightly loaded grid station) and design should take into account availability of key equipment such as switchgear panels
- System reliability and power quality should be maximized
- Expansion of network should be systematic and economical
- Design should take into account SEC safety guidelines
- Design should ensure standardization of system
- Zone-based planning should be used
- For rural areas, MV system can be of 13.8kV or 33kV as per availability, but preferable by of 33kV (as detailed in Section 7.6.2)
- The MV connection line from SEC should not pass through the customer's premises
- Parallel operation of SEC MV feeders or standby generators operating in parallel to SEC network are not allowed for bulk customers.
- Customers with sensitive supply requirements may be provided additional supply sources by SEC.
- Any backup supply to customer should either be from another MV bus bar within the same grid station or from another grid station (the second is preferred particularly for customers with sensitive supply requirements)
- The preferred configuration of supply is single loop, if loop system is available in the area
- If the customer requests for single loop system and the existing network in the area is radial, the customer needs to pay for the additional cost
- The backup feeder either should be from another MV bus bar within the same grid station OR can be from another grid station/ MDN substation if feasible OR can be from another extra high voltage grid station if feasible
- The backup feeder for sensitive nature customers (e.g. big hospitals, military / government head offices etc.), may be preferably from another grid station, if feasible.
- The customer is responsible for safety and reliable protection of its plant.
- Single line diagram illustrating schemes along with relay setting shall be submitted for SEC comments and approval at design stage.
- The plug settings of relays should be according to the contracted load.

- For details on charging of costs (e.g. of cables, conductors, backup supply, switchgear, sharing of network etc.), refer to SEC Customer Service Manual with its latest updates.
- The customer shall comply with all relevant Saudi Arabian' Codes, Regulations & Standards. All other relevant government and statutory requirements shall be adhered to. The customer is required to comply with all relevant good electricity industry practices.
- All designed MV networks can be validated through use of load flow software (e.g. CYME)

MV Design Process

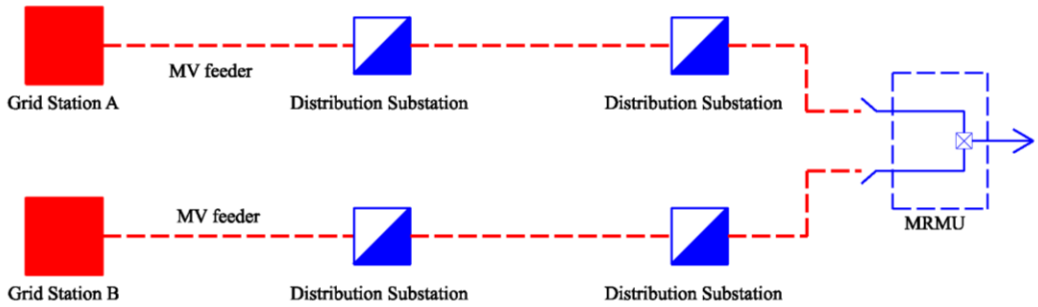
The process for handling MV connection requests is outlined below:

- Receive connection request for bulk customer at MV network (for loads between 4 MVA and 25 MVA), which will include the following information:
  a. Plan area, location and ownership along with relevant approval forms from other government entities, like Baladiya and Ministry of Commerce
  b. Type of facility (new, extension, re-connection, reduction)
  c. Filled out load declaration form outlining the customer load details in Forms
  d. Time schedule of construction and date by when connection is required
  e. Single line diagram for the connection, outlining backup requirements as well as presence of backup generators, if relevant
  f. Distribution voltage level
  g. MV customer switch gear
  h. Number and Location of interface point
  i. The customers which are likely to create disturbance / distortion / fluctuation in SEC' network (e.g. steel furnaces etc.) are required to perform proper studies and implement remedial measures (e.g. current limiting reactors, harmonic filters etc.), so as to ensure compliance with the SEC' power quality standards for harmonics, voltage dips etc.
  j. Also refer to SEC Customer Service Manual with its latest updates. SEC will verify the submitted load detail, according to its Rules
- Verify the provided details through physical inspection and / or discussions with the customer
- Ensure receipt of study detailing impact and remedial measures to prevent disturbance / distortion / fluctuation in SEC network due to equipment in customer premises (such as furnaces), if relevant
- Estimate impact of customer connection on existing SEC MV assets. In cases of interference with existing assets, the customer is to finance project(s) to eliminate interference
- Define configuration of connection depending on the nature of system in the area (loop vs. radial) and requirement for back-up connection by the customer (for customer with sensitive load requirements)

- Ensure availability of MV switching / metering room at the boundary of the customer premises (at ground level, adjacent to its boundary wall, with door on the outer side). This room should be constructed in coordination with SEC. This room shall always be kept locked. For details, refer to SDCS-02 part 10 with latest updates.

7.5 MV Network Configuration Schemes

- The customers can be supplied from existing MV network if it is technically feasible (feeder load permits, voltage drop is within permissible limits etc.), otherwise SEC will have the right to ask for new feeder(s) as per the prevailing standards / policies.

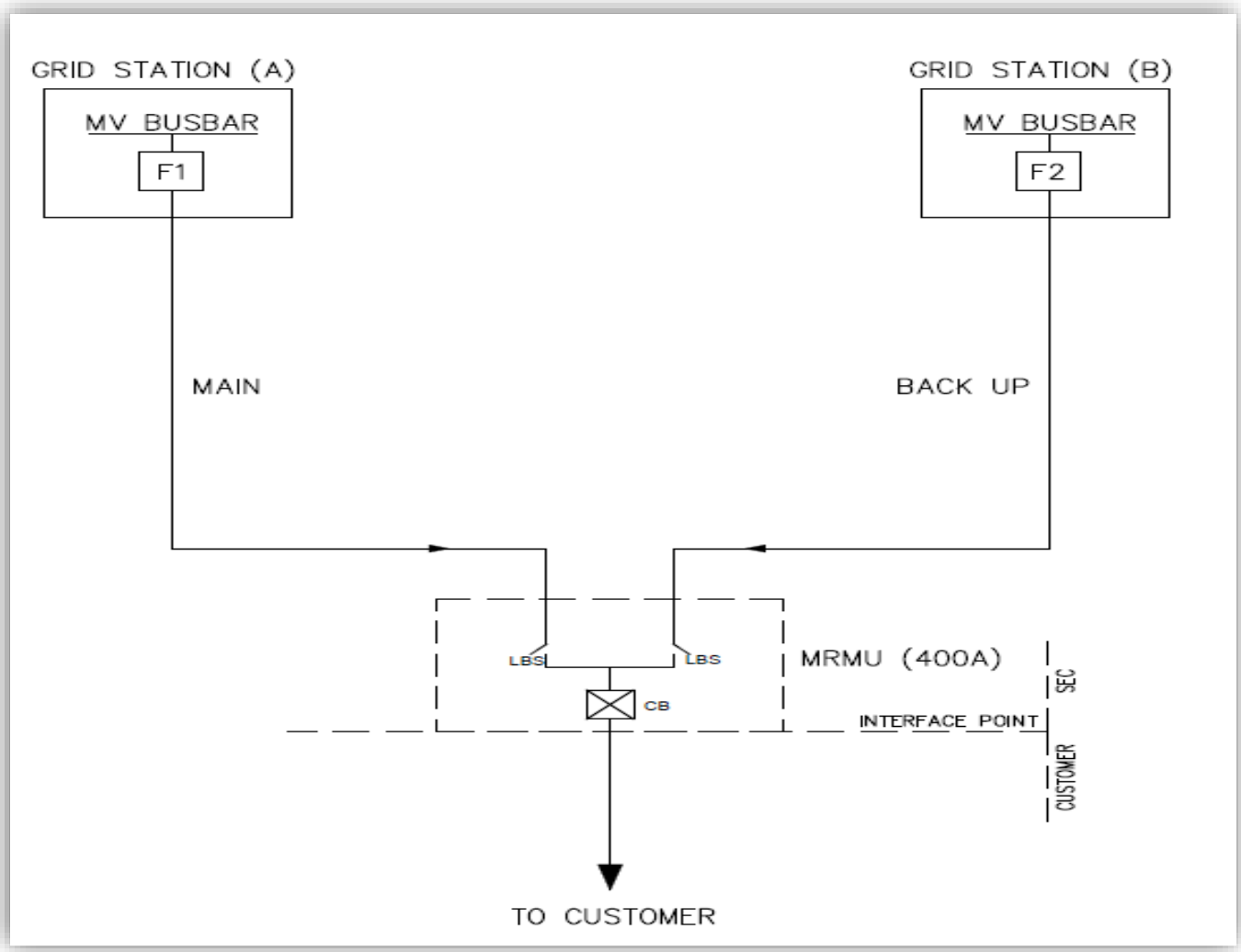


- The maximum load for MV feeder, which feed public customers, shall be 80% from the rating capacity of cables /conductors
- The maximum load for MV feeder, which feed dedicated customers, shall be 100% from the rating capacity of cables /conductors
- The customers can be supplied by creating new feeders from the existing grid station if it is technically feasible (grid station load permits, spare switchgear is available, voltage drop is within allowed limits etc.), otherwise SEC will have the right to ask for new grid station as per the prevailing standards / policies.

The following underground connection schemes

1- Load ≤ 9 MVA at 13.8 kV

Figure 16

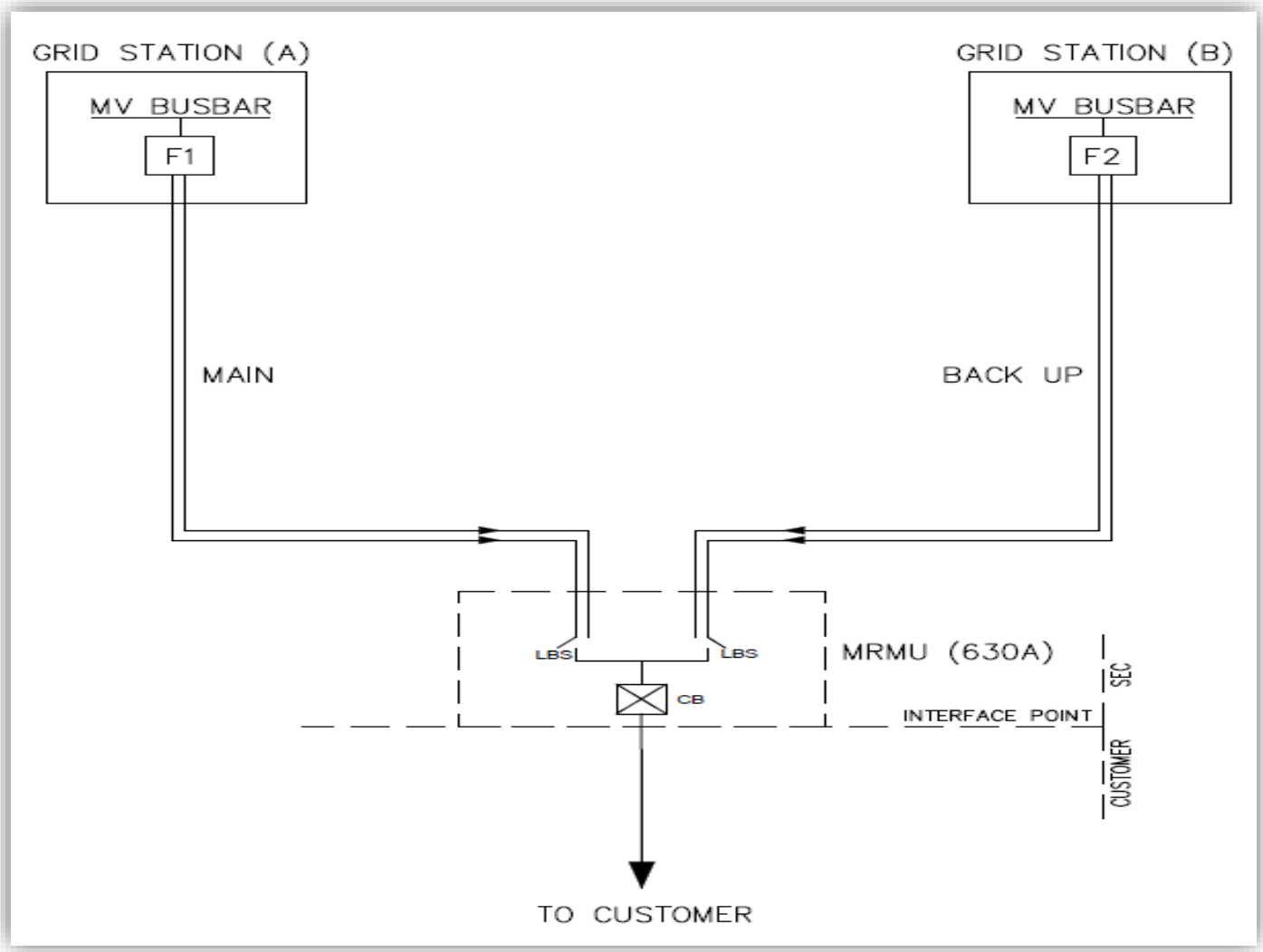


Note:

- There is no need for special cable for such customers and connection from existing networks should be used while maintaining n-1 condition, If this is not possible, the above configuration can be used
- All cables are 3x500 mm² Al

2- 9 MVA < Load ≤ 14 MVA at 13.8 kV

Figure 17

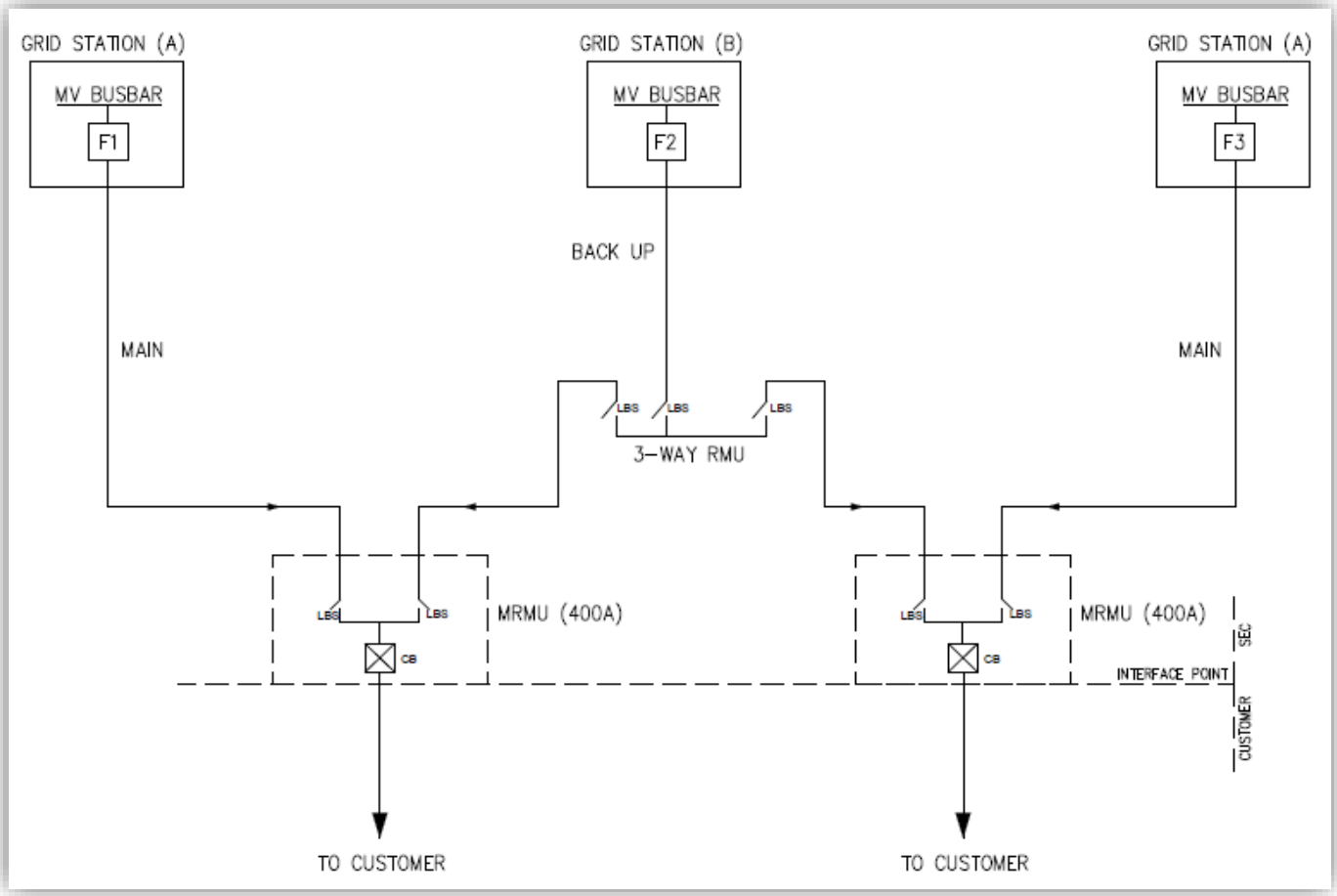


Note:

- This configuration is to be implemented if circuit breakers in grid stations are not available or there is requirement for more sources.
- All cables are 3x500 mm² Al

3- 9 MVA < Load ≤ 18 MVA at 13.8 kV

Figure 18

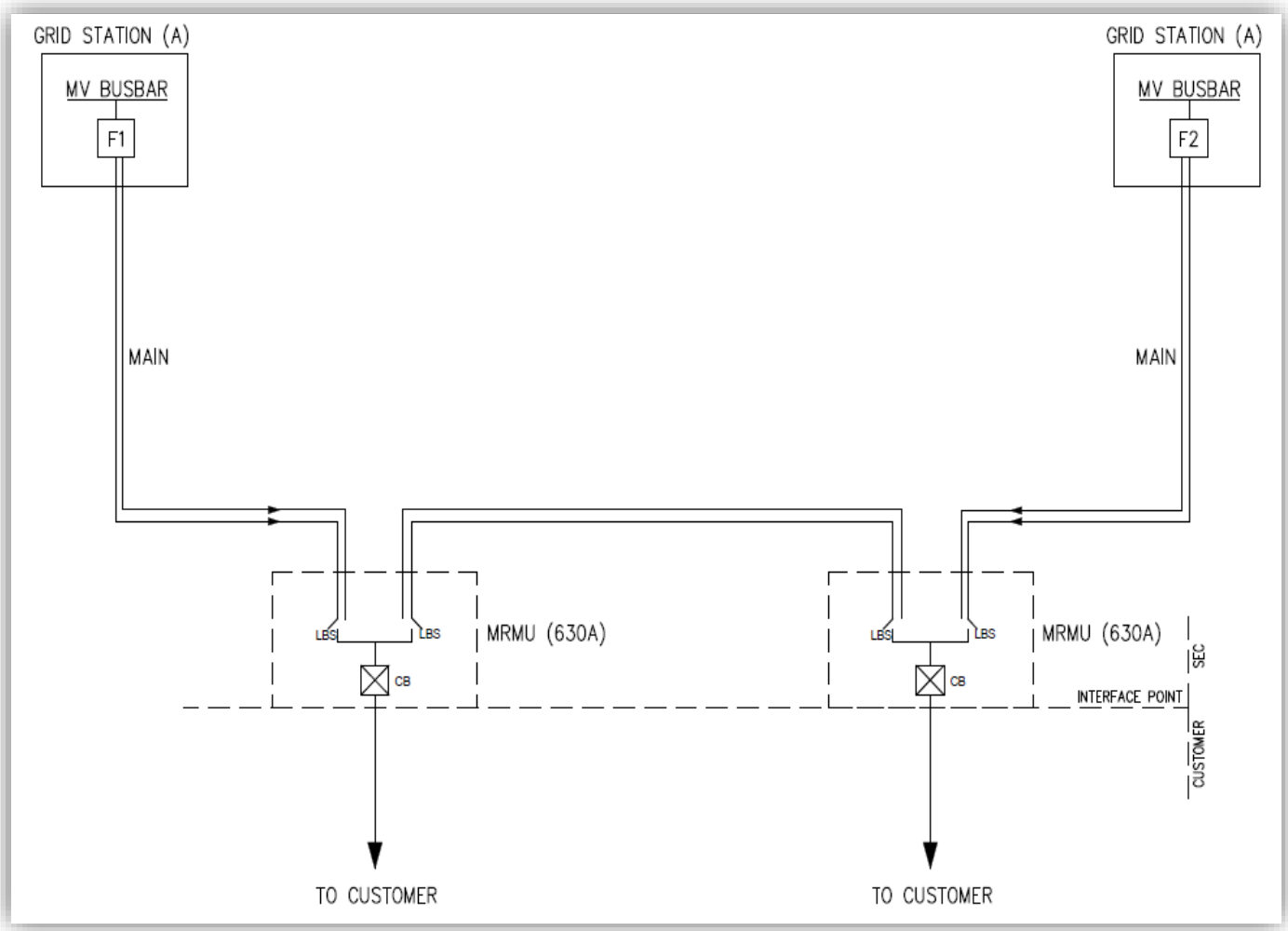


Note:

- All cables are 3x500 mm² Al

4- 9 MVA < Load ≤ 18 MVA at 13.8 kV

Figure 19

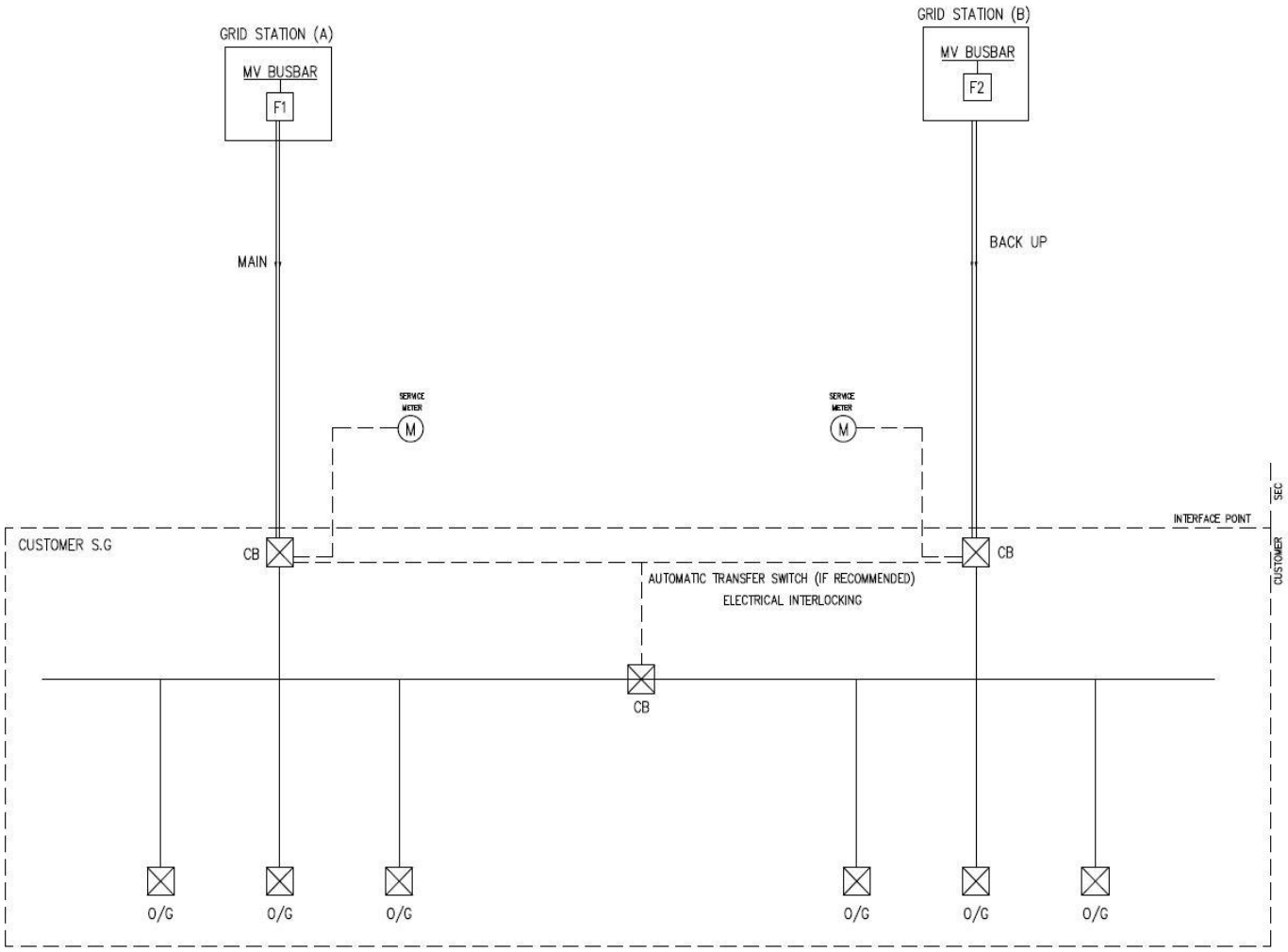


Note:

- This configuration is to be implemented if circuit breakers in grid stations are not available or there is requirement for more sources.
- All cables are 3x500 mm² Al

5- 9 MVA < Load ≤ 18 MVA at 13.8 kV

Figure 20

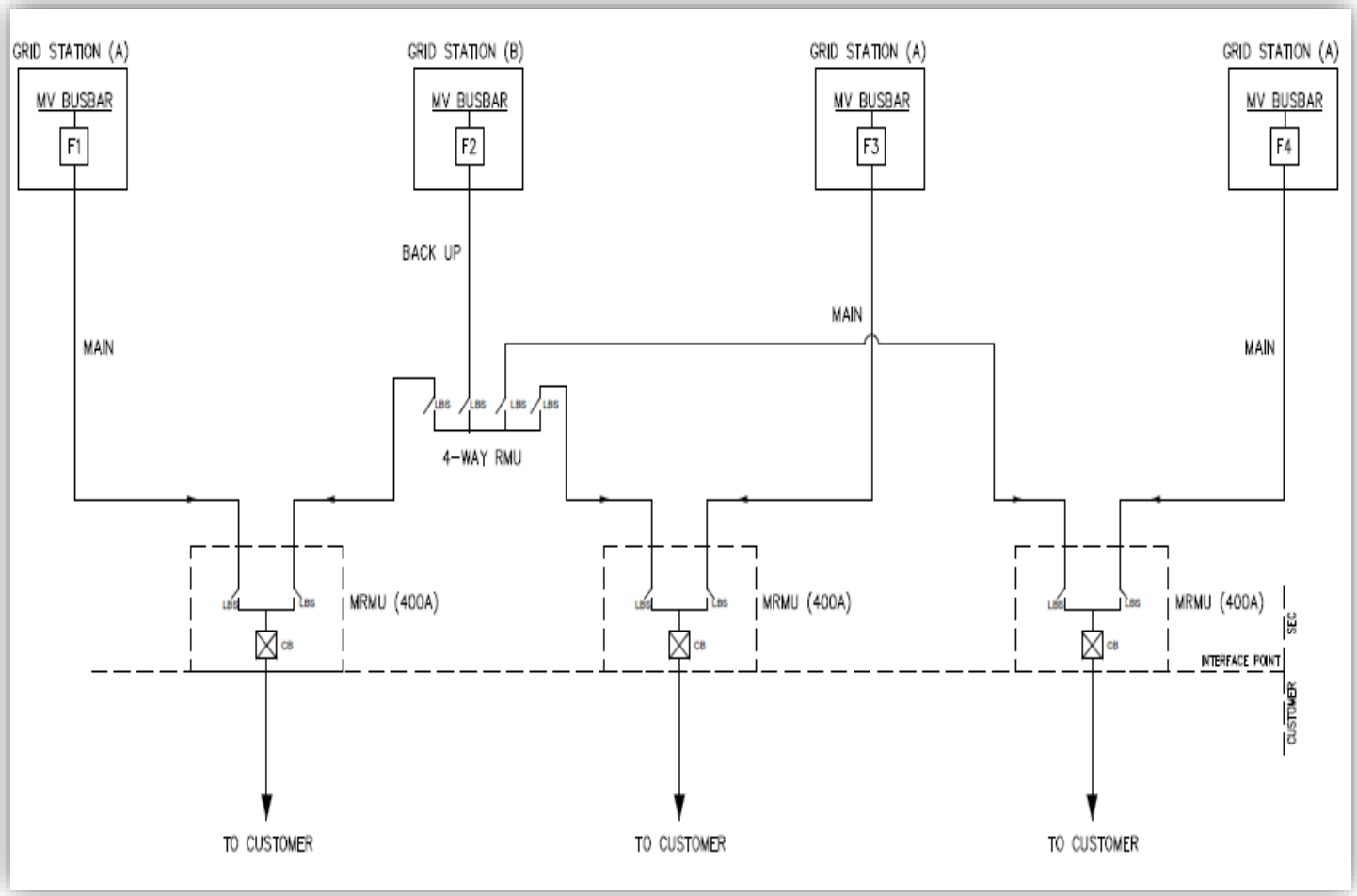


Note:

- This configuration is applied if circuit breakers in the grid stations are not available, or in case the customer requested his loads to be fed in one point.
- Switchgear will be supplied by the customer.
- All cables are 3x500 mm² Al

6- 18 MVA < Load ≤ 25 MVA at 13.8 kV

Figure 21

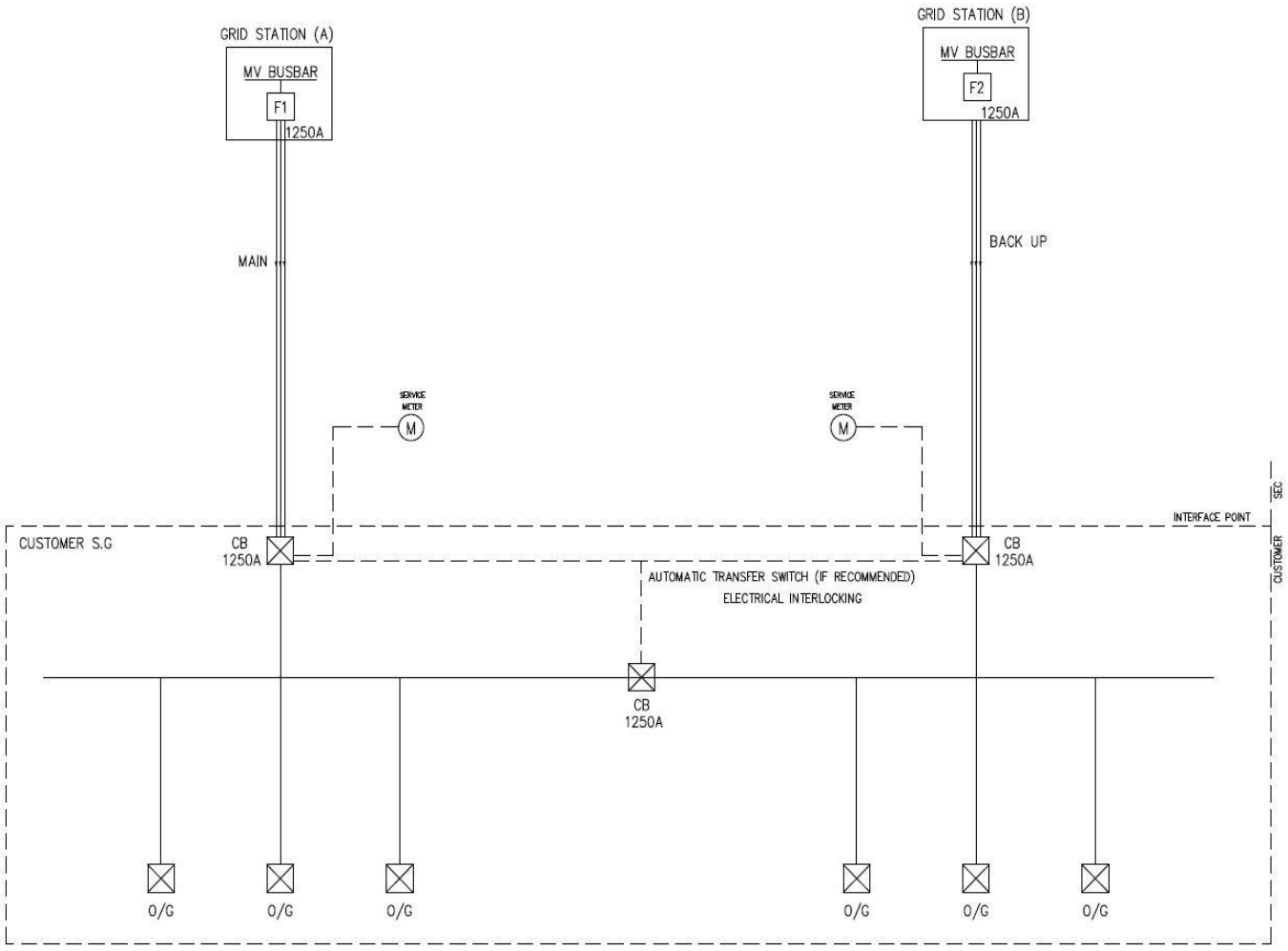


Note:

- All cables are 3x500 mm² Al

7- 18 MVA < Load ≤ 25 MVA at 13.8 kV

Figure 22

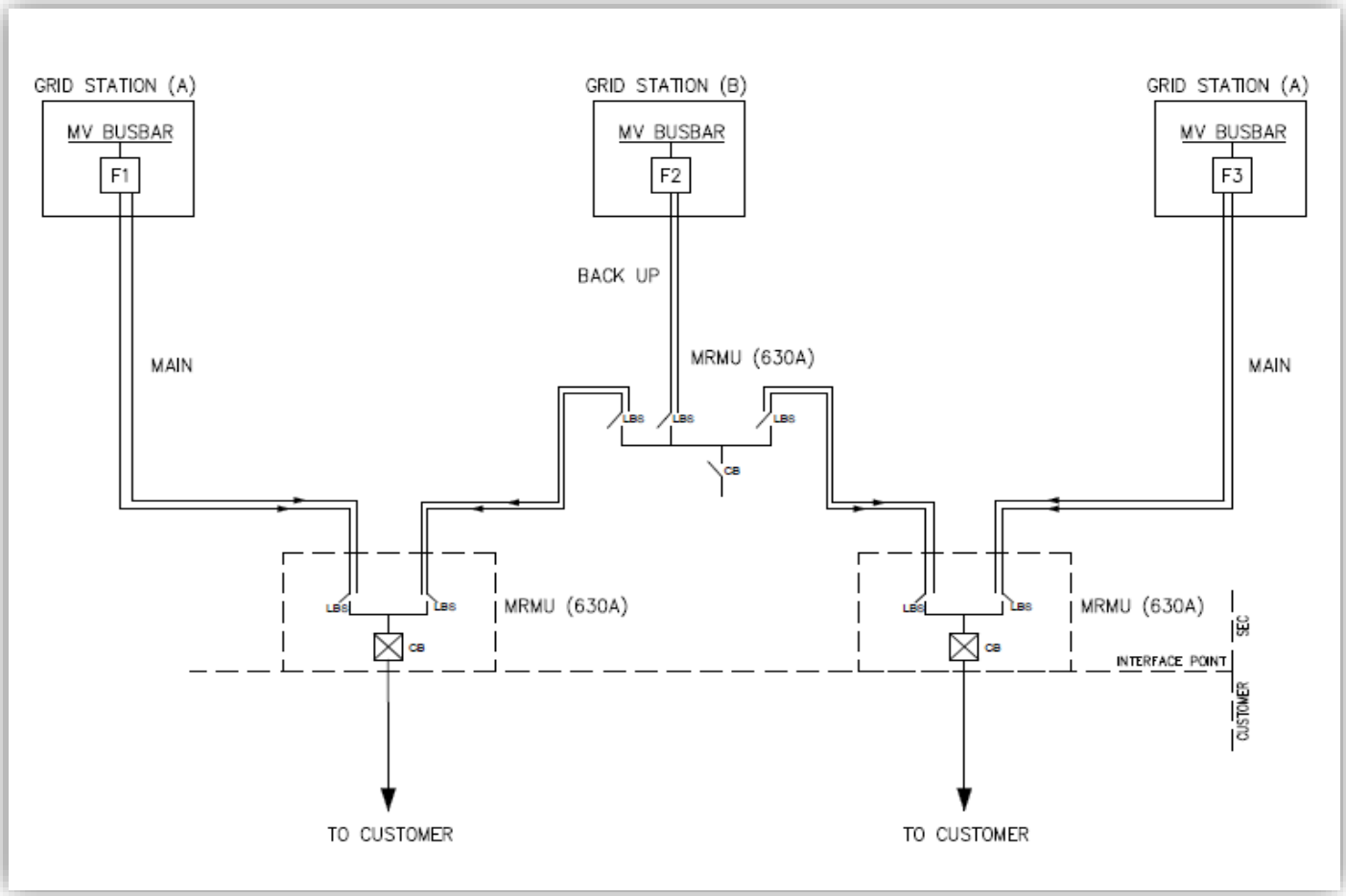


Note:

- This configuration is applied if circuit breakers in the grid stations are not available, or in case the customer requested his loads to be fed in one point.
- Switchgear will be supplied by the customer.
- All cables are 3x500 mm² Al

8- 18 MVA < Load ≤ 25 MVA at 13.8 kV

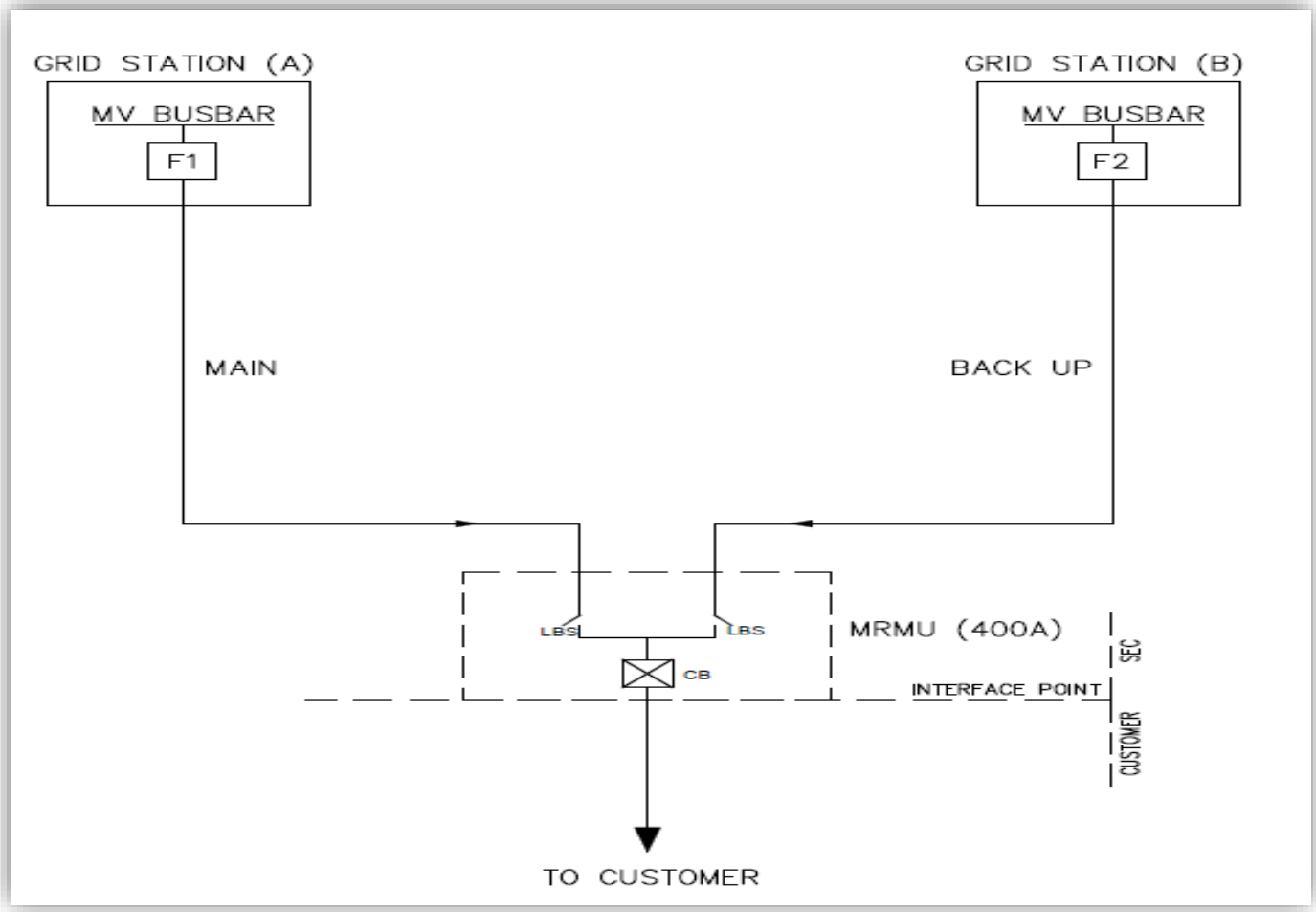
Figure 23



- This configuration is to be implemented if circuit breakers in grid stations are not available or there is requirement for more sources.
- All cables are 3x500 mm² Al

1- Load ≤ 20 MVA at 33 kV

Figure 24

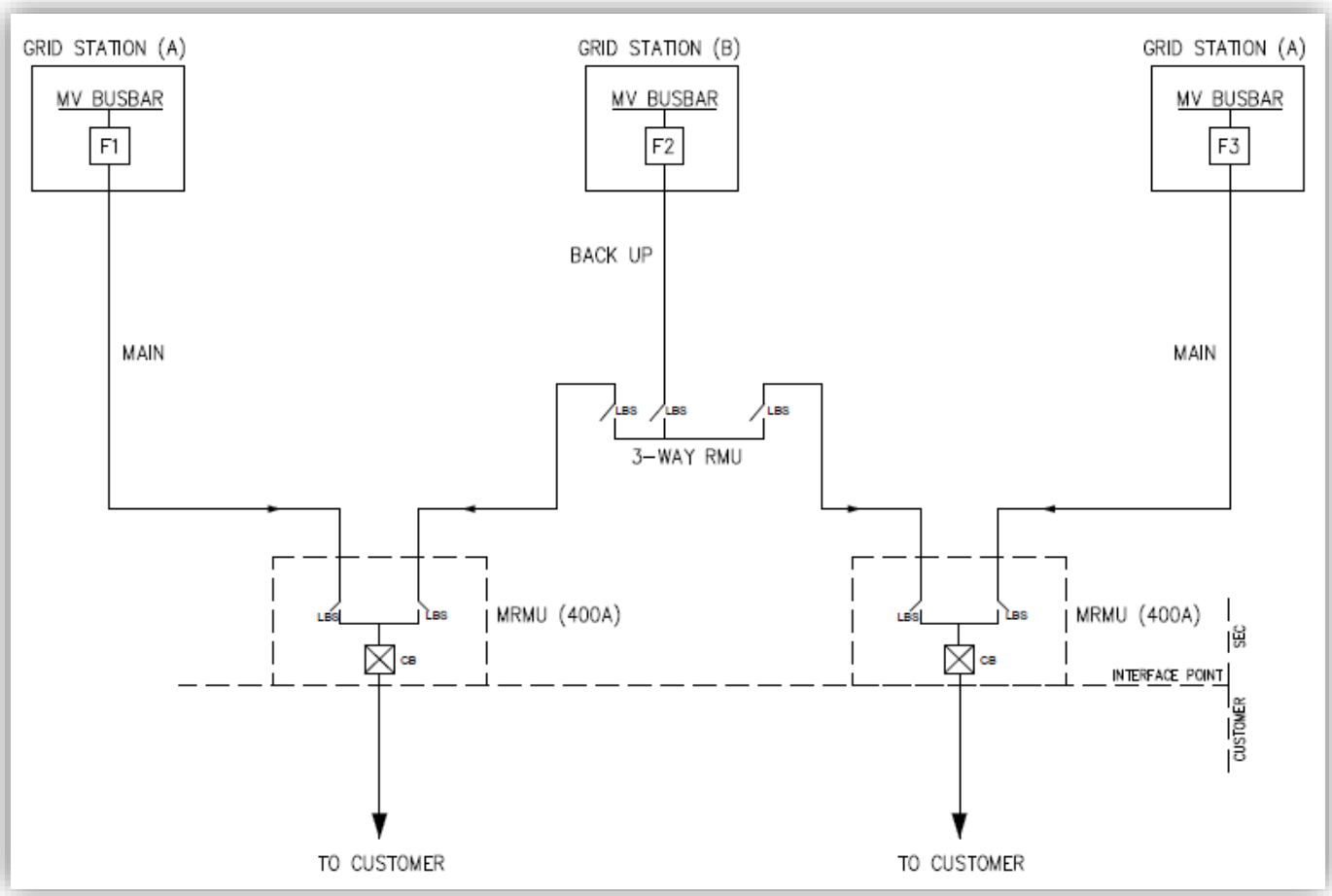


Note:

- There is no need for special cable for such customers and connection from existing networks should be used while maintaining n-1 condition
- If this is not possible, the above configuration can be used
- All cables are 3x400 mm² Al

2- 20 MVA < Load ≤ 25 MVA at 33 kV

Figure 25



Note:

All cables are 3x400 mm² Al

Rural supply

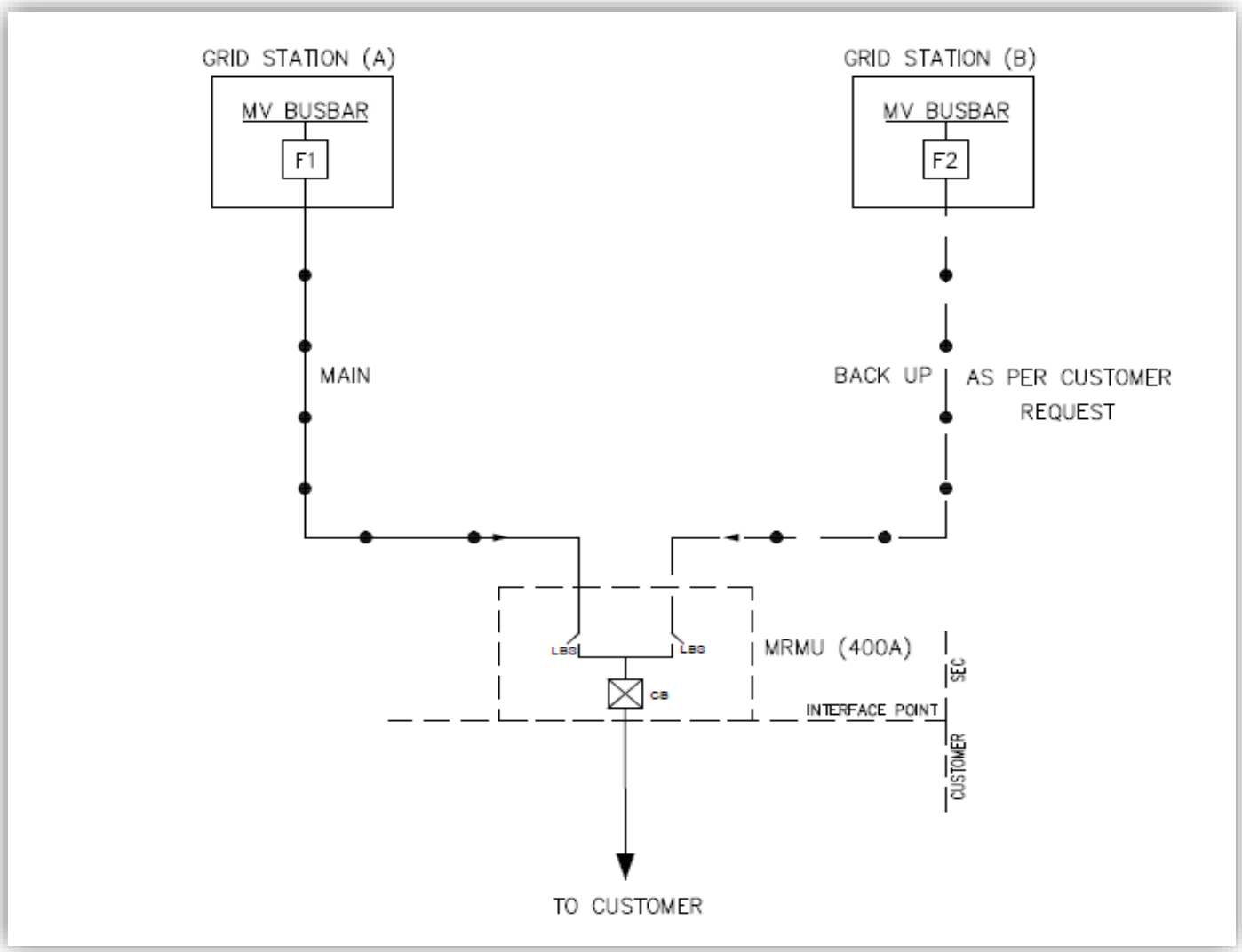
Given the long distances and low loads involved, the following guidelines need to be followed:

- It is preferred to supply rural areas using 33kV systems. However, other voltages such as 13.8kV may be considered depending on availability.
- Preferred feeder configuration is radial and overhead. Interconnection between MV feeders may be used depending on availability.
- Interconnection between MV feeders must be used. When the maximum load for radial feeder more than 2 MVA.
- The preferred overhead configuration of supply is single circuit / pole and it can also be double circuit / pole, In case necessity such as (reducing project costs – unavailability of root to achieve the clearance of overhead networks, no source available nearby).
- Normally 170 mm² ACSR conductor is used for main / branches OH section which has a high impedance & it can restrict the optimal utilization of the conductor capacity.
- Voltage drop considerations must be given high importance. Voltage regulators and Capacitor Banks can be used to mitigate potential situations of voltage drop.
- Minimizing outages and outage duration on such feeders should also be given high importance, In light of this, use of auto-reclosers and sectionalizers should be explored on such feeders

The following overhead mv connection schemes as below:

1- Load ≤ 8.6 MVA at 13.8 kV

Figure 26

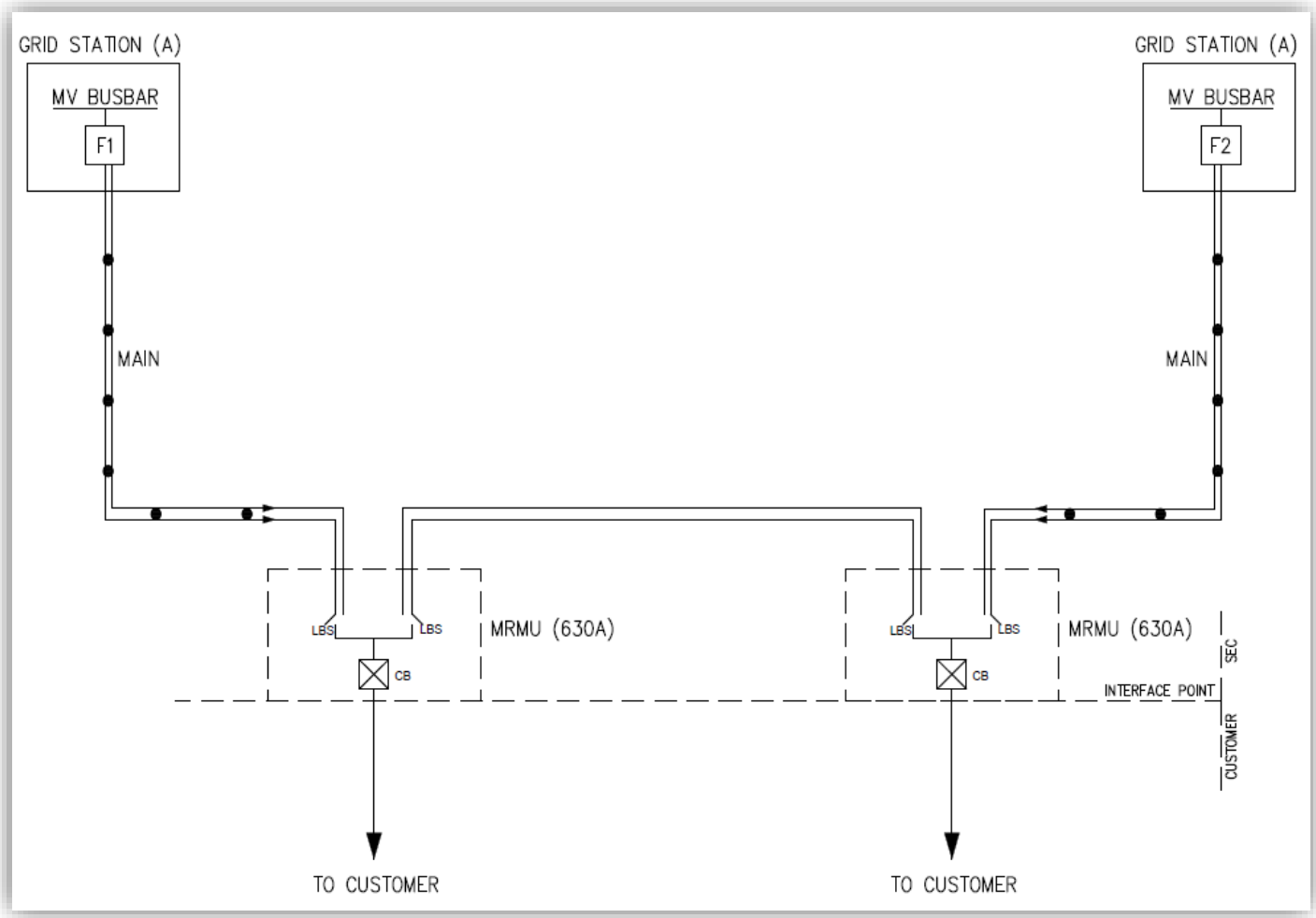


Note:

- There is no need for dedicated feeder for customers and connection from existing networks if it is not possible, the above configuration shall be used instead.
- All conductors 170 mm² Al.

2- 8.6 MVA < Load ≤ 17.2 MVA at 13.8 kV

Figure 27

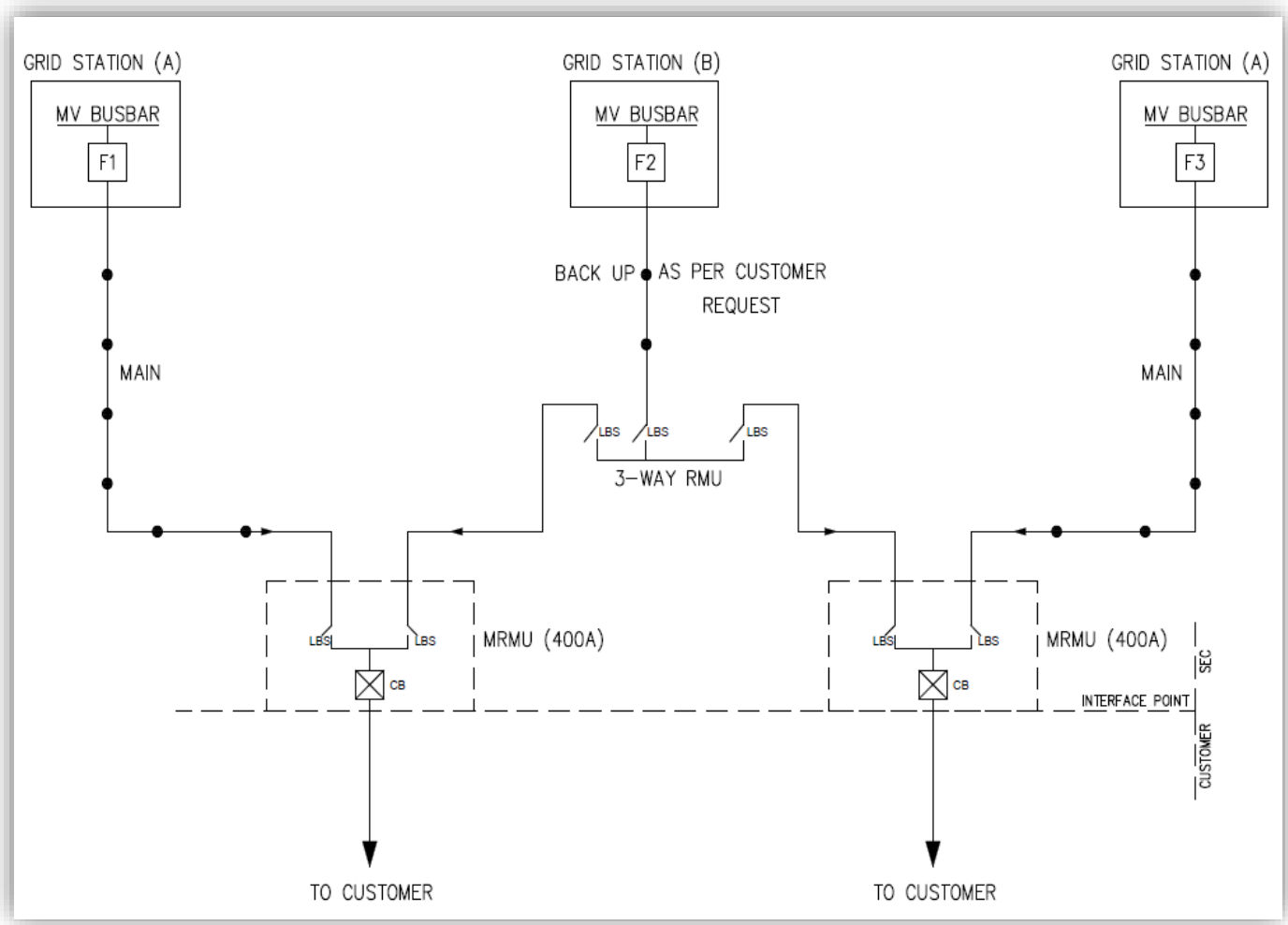


Note

- Above shown configuration is to be implemented double circuit at the same pole.
- All conductors 170 mm².

3- 8.6 MVA < Load ≤ 17.2 MVA at 13.8 KV

Figure 28

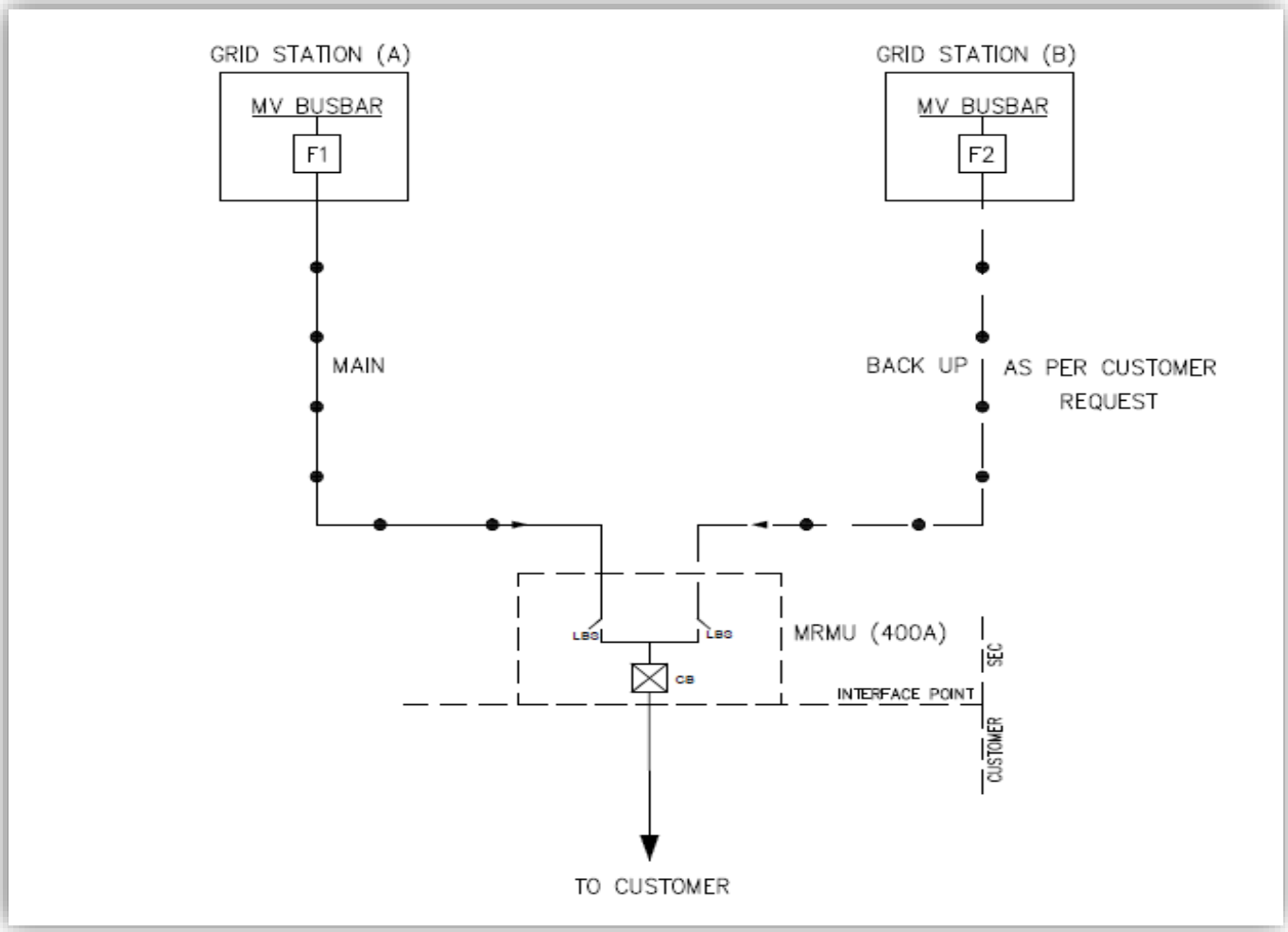


Note

- All conductors 170 mm².

4- Load ≤ 20 MVA at 33 kV.

Figure 29

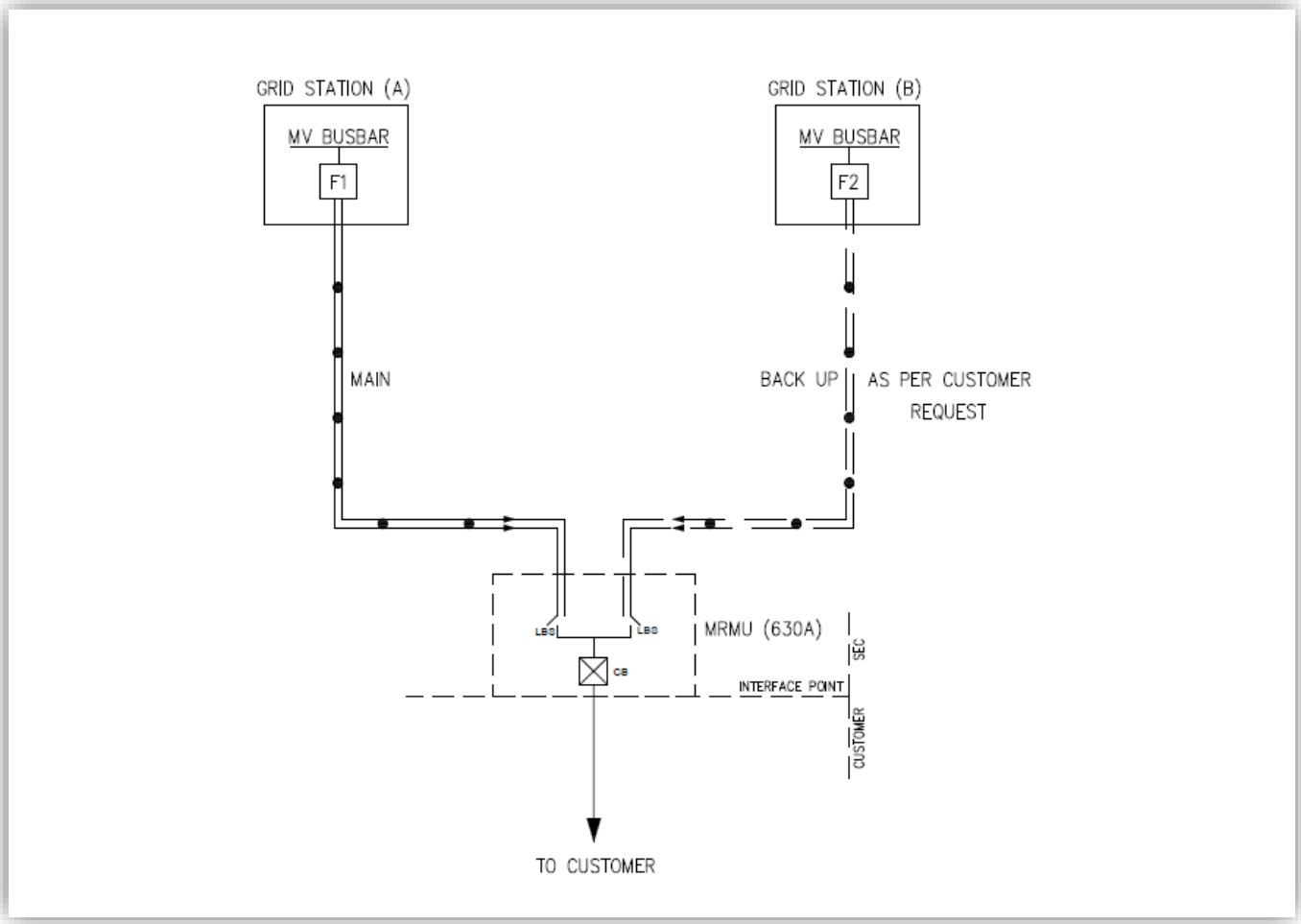


Note:

- There is no need for dedicated feeder for customers and connection from existing networks if it is not possible, the above configuration shall be used instead.
- All conductors 170 mm² Al.

5- 20 MVA < Load ≤ 25 MVA at 33 kV

Figure 30

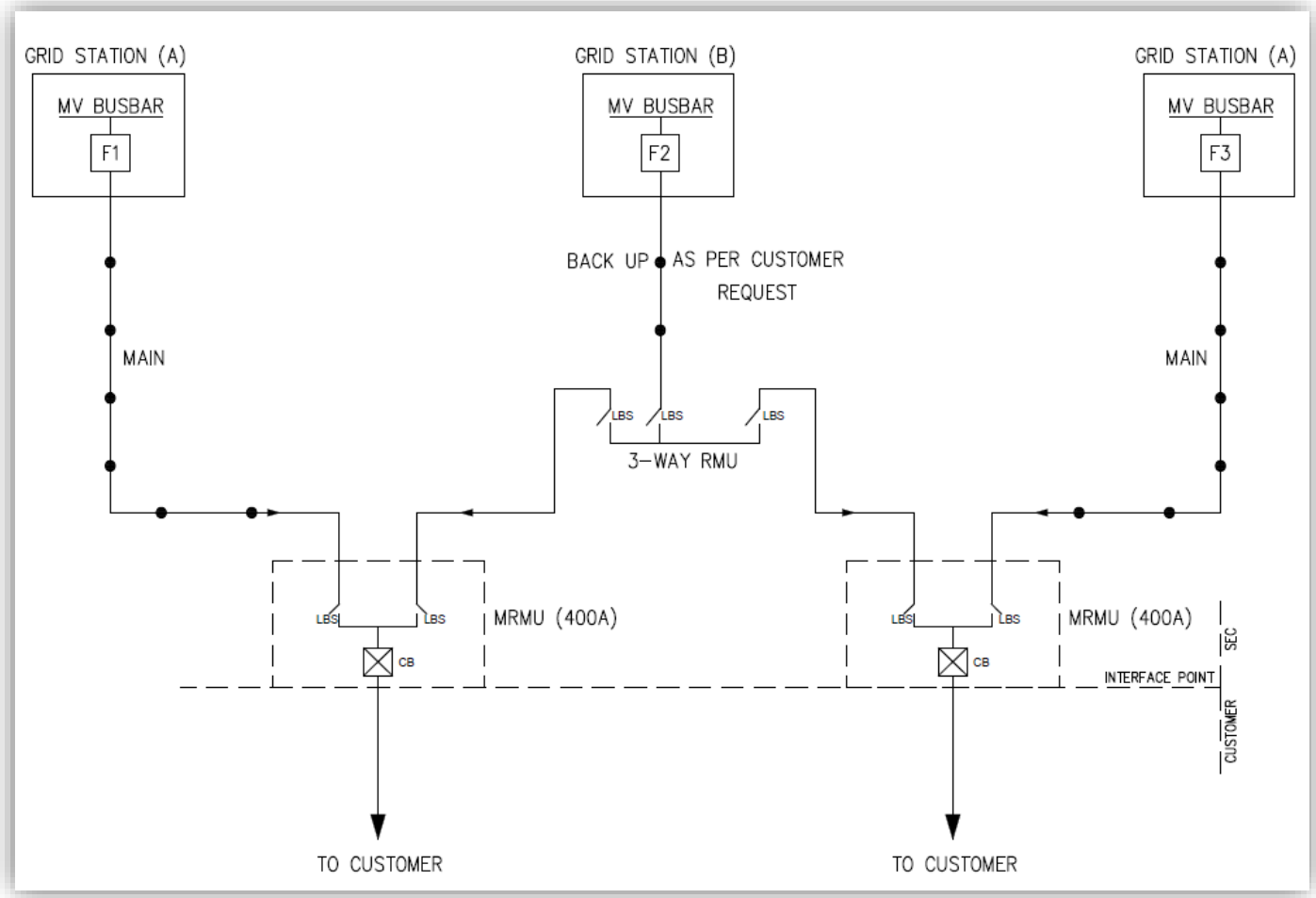


Note

- Above shown configuration is to be implemented double circuit at the same pole
- All conductors 170 mm².

6- 20 MVA < Load ≤ 25 MVA at 33 kV

Figure 31



Note

- All conductors 170 mm² Al.

8 System Improvement

A key objective of network planning is to identify system improvements projects over 1-year to 5-year plan. This would enable the management and network planning team to have a clear, robust and forward-looking view of required changes to the network thereby ensuring an integrated view of CAPEX spending and network development.

8.1 Reinforcement

Reinforcement of network elements is undertaken to relieve overloaded equipment or to improve performance of the network, the different types of reinforcement activities that can be undertaken include:

- Underground reinforcement
- Overhead reinforcement

Network reinforcement shall be covered the following:

- New feeder for reducing Network load including (MV or LV).
- Establish a new feeder link between existing substations for reducing loads.
- Voltage drop (MV or LV).
- Creating a new feeder to re-distribute existing customers.
- Division of feeders.
- Capital Emergency works related to network capacity increase
- Installation of network improvement devices.
- Increasing capacity of transformers not related to any new connection request or other projects.
- Reinforcement of main feeders 33KV (MDN) substation.
- Improve power factor.
- Company funded projects to remove hazardous conditions.
- Modifying / simplifying network design.
- Transfer of equipment from a site to the other site with the purpose of network improvement
- Conversion of Overhead Network to Underground.

- **Criteria for selection of network elements for reinforcement**

The feeders should be selected for reinforcement on the basis of certain criteria e.g. peak load, voltage drop, equipment loading, benefit ~ cost, length, technical/geographical aspects etc.. Specific geographical constraints should be taken into account so that the objectives of reinforcement are met.

- Priority should be given to those MV feeders, which are heavily loaded and contribute high technical losses to the system.
- No doubt, the load and the losses on a particular feeder are the main criteria for bifurcation of a feeder but in some cases, a lightly loaded but lengthy feeder also requires bifurcation to reduce the line losses, improve the voltage drop at the tail end and reliability of supply.
- Each feeder involved in the particular proposal should be evaluated technically based on latest data collected from field formation and voltage drop. Thus as per existing condition of the network, those proposals should be executed which give maximum technical as well as financial benefits.
- Any proposal can be evaluated using load flow software (such as CYME) to ensure robustness of any proposal.

- **Design principles for MV Reinforcement**

The following principles should be taken into consideration for any MV reinforcement exercise:

- All customer coincident demand load for the system is satisfied for the upcoming 5-year time period.
- Voltage drop does not exceed 5%.
- The capacity of any MV equipment should not exceed its firm capacity.
- All relevant contingencies are planned and accounted for (such as feeder overloading & outage, transformer overloading & outage and grid station outage).
- The proposed action is the most economical option to achieve the objective for reinforcement.

- **Area planning**

The heavily loaded feeders are selected and their load can be partially shifted to nearby lightly loaded feeders to balance the load amongst them. This may involve re-conductoring and / or creation of MV links.
Sometimes, one or more new feeders are proposed to share load of the overloaded feeders. Sometimes, due to overloading of grid stations, area planning of MV feeders is exercised for shifting of load from one grid station to another grid station by making MV links between the grids. This will provide relief to the grid station.

- **Re-conductoring**

The undersized / deteriorated conductor or cable should be replaced with that of higher capacity wherever required.

- **Installation of MV capacitor banks**

Installation of capacitor banks on MV lines at proper places results in loss reduction and improvement in the voltage drop conditions.
In case of change in network configuration, re-location of existing capacitor may also be required.

- **Replacement of undersized cables at grid end**

At grid end, the undersized cable should be replaced with cable of higher capacity wherever required.

- **The Reinforcement form 6 is given in Forms**

8.2 Integration

When a new source of supply is planned for an area / system (a new source of supply can be grid station, MDN substation, new feeders are planned along with it which can either:

- Load transfer between new gird station / MDN and the existing gird station / MDN.
- Increase reliability of MV network requirements (such as industrial & commercial customers and critical customers).

- **Design principles for Integration Projects**

- Voltage drop does not exceed 5% in any part of the feeder.
- The capacity of any MV equipment should not exceed its firm capacity for feeders.
- All relevant contingencies are planned and accounted for (such as feeder overloading & outage, transformer overloading & outage and grid station outage).
- The proposed action is the most economical option to achieve objective for reinforcement.
- Prioritization of customers for integration projects.
- Load flow software (such as CYME) is used to ensure robustness of design.

- Connection projects for new (grid station – MDN substation) should be executed in sufficient time frame before the (grid station – MDN substation) is operational
- **The Integration form 7 is given in Forms**

8.3 Network Replacement

Replacement project execuation will be applied as per network type either the project is Underground (UG) or Overhead (OH) without jeopardizing the nework type e.g UG will remain UG and OH will remain OH with same capacity.. Replacement will hahppen on the following reasons:

- Network change due to usage or equipment age
- Change in Network, Due to Network malfunctioning associated with abrupt and abnormal operating conditions (such as short circuit, leakage, faults).

For example, if there is a transformer blow-out due to overloading beyond the firm capacity, it will need to be replaced with another transformer of higher capacity. This should be treated as reinforcement. However, the same capacity, this will be treated as replacement.

Replacement projects will entail close collaboration between maintenance and planning functions. This will be particularly true for MV networks where cost of failure and equipment costs are higher. The lifecycle of MV network replacement projects will comprise of the following activities:

1. **Planning:**
   - The maintenance function in each department will be responsible for developing the Maintenance Roadmap for the department, outlining preventive maintenance requirements for MV network for each year over the time horizon.
   - This will be developed in line with Value Based Maintenance principles
   - Once defined, the roadmap will need to be aligned with the network plan, which is defined by the operating Area network planning function.
   - This alignment will need to ensure alignment on equipment for reinforcement and replacement.
   - The alignment is particularly critical for time horizon due to CAPEX planning requirements (which are outlined below).
   - Once the maintenance and network planning roadmaps are aligned, the CAPEX requirements need to be estimated.
   - The CAPEX for all replacement projects (as well as other projects) will be allocated under the correct project category (new connection vs. reinforcement vs. replacement vs. integration) and will be allocated to network planning function of the Sectors, who will be responsible for monitoring and controlling CAPEX spending for the Sectors.

2. **Design**
   - Design of replacement project (whether preventive or corrective) will need to be undertaken before it is executed.
   - The ED maintenance function will need to develop the design plan for the project which should include:
     a. Details of equipment to be replaced.
     b. Material / equipment requirements for the project.
     c. Layout (current and changes, if any).
     d. Geographical coordinates.
   - The ED maintenance function will also define the inspection schedule.

3. **Approva**l
   a. Every replacement project needs to be approved by network planning team.
   b. The first check for approval is compliance with the defined maintenance plan roadmap.
   c. In case the replacement project is a deviation from roadmap, the network planning team has the option of seeking justification from the ED maintenance team.
   d. The second approval is from technical perspective – whether the proposed design and equipment requirements are in line with standards and specifications. The network planning team may proposed changes to design to ensure compliance and discuss with the maintenance planning team.
   e. The second approval ~~is~~ from Feasibility Studies perspective – within stage Gate processes.

- **The Replacement form 8 is given in** Forms

8.4 Conversion of Overhead Network to Underground

Conversion of existing MV network from overhead to underground can be undertaken under special circumstances:

- If an existing overhead, MV line is a risk to public safety.
- If there is a special request from the government.
- If there is no other option available to improve network.
- Request from maintenance due to usage or equipment age.
- According to SEC plans when needed for the purpose of improving reliability or Re-planning the network according to urban expansion.

9 Medium Voltage (MV) Network Performance Improvement

This section describes the optimal placement of voltage regulator, capacitor, auto-recloser, sectionalizer, load break switch, fuse & fault indicator on medium voltage distribution network and short circuit performance & losses evaluation. It is intended to assist SEC engineers to achieve standardization in planning & design and to ensure a satisfactory and economical level of service.

SEC' Distribution Planning Standards (DPS), Distribution Material Specifications (SDMS), Distribution Construction Standards (SDCS), Distribution Operation Manual (DOM) (including any updated amendments) are indispensable for the application of this document: this document should be read in conjunction with SEC' DPS, SDMS, SDCS and DOM, unless otherwise specified.

- The overall objective of improving MV network performance is to enhance the security of supply in light of Distribution Security of Supply by WERA please refer to the table 17.
- However, it should be noted that full compliance to the above standards of security cannot be ensured only through use of auto-reclosers and sectionalizers. This would require projects on distribution automation, distribution management system and smart grids, which are being pursued separately. Furthermore, this would also require firm capacity and interconnectivity.
- One of the objectives behind installation of protection devices is to ensure compliance to WERA standards. However, these decisions need to be aligned with distribution protection analysis team before final installation to account for various factors such as:

a. Coordination between devices for robust protection
b. Ensuring protection against correct level of fault current
c. Presence of other protection devices such as switches and circuit breakers

This is typically achieved through Sectionalizing studies using established load flow software tools like CYME

9.1 Voltage Regulator (VR)

The voltage problems of existing overhead MV distribution network can be solved by utilization of line voltage regulators. This may be necessary, in many instances, for rural areas. Voltage regulator provides continuous voltage regulating capacity by either increasing or decreasing the voltage.

Furthermore, voltage rise (or over-voltage) is also observed in certain systems. This condition can potentially exist on long overhead lines in low loading conditions. The phenomenon, known as Ferranti effect in transmission lines, can also occur in distribution lines, especially those of 33 kV if they are long and lightly loaded. It is due to the leading power factor. In this case, instead of voltage drop, a voltage rise is produced. A particular concern in use of line voltage regulator is that the voltage during both peak and light load periods should comply with the voltage criteria.

Voltage regulator will cause a voltage rise at the point of application. The voltage rise at the voltage regulator will vary depending upon line current and the output voltage settings.

When installed on a MV feeder, the voltage increase at the voltage regulator location will vary for both light load and high load feeder voltage profiles. The voltage at light load must be calculated to ensure that customers are not over voltage, especially when capacitor banks are located on the same feeder.

The secondary voltages are to be maintained according to SEC standard distribution voltage ranges.

It should be noted that there are potentially multiple options to handle voltage issues in MV systems, such as:

- Placement of voltage regulator.
- Placement of capacitor banks.
- Changes in transformer tap-changer settings
- Reinforcement of feeder (addition of second conductor)
- New transformer / substation
- When a voltage drop / over-voltage situation is encountered, the above options should be evaluated simultaneously before any recommendation is made.
- Both technical and economic factors need to be considered while selecting the optimal option to handle voltage drop issues.

Technical factors will include:

- Voltage drop for heavy-load and light-load conditions before and after the option is exercised, for current load and future load for five years.
- No overvoltage at any demand point.
- Reduction in technical losses in the system before and after the option.

Economic factors will include:

- Cost of installation of new equipment(s).
- Incremental operating costs of new equipment(s).
- Cost associated with technical losses, all if applicable.

- *For details, refer to 43-SDMS-02 with latest updates.*

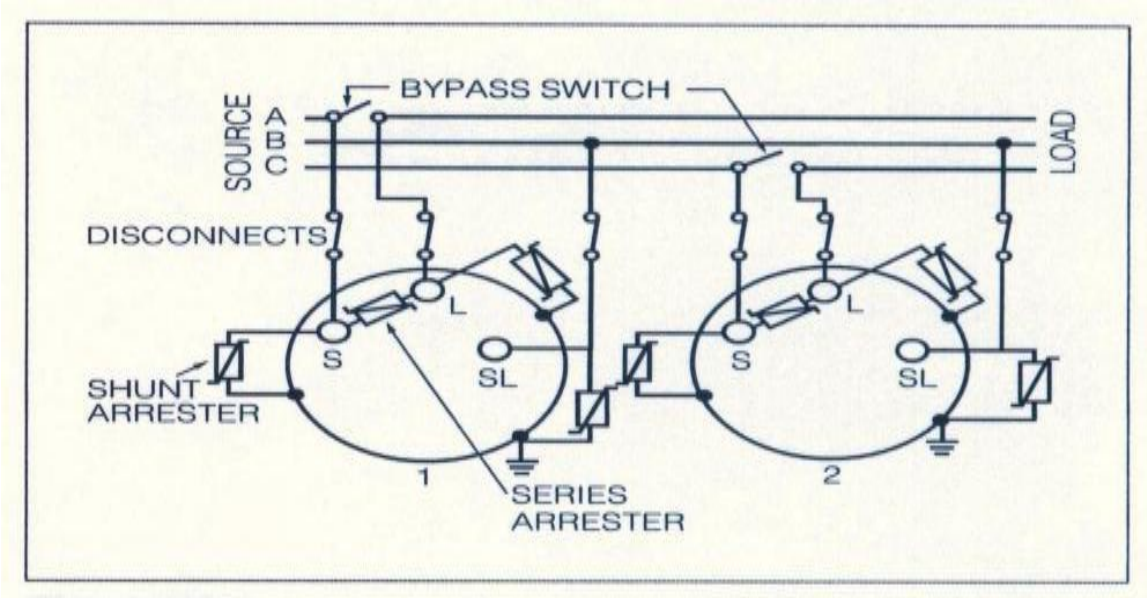
Benefits:

- Provides more uniform voltage levels, more satisfactory to customers.
- By better utilization of existing SEC network, investments in creation of new feeders or additional grid station capacity can often be cancelled or delayed.
- Often provide a small reduction in energy and demand losses and release line and substation capacity due to a small reduction in line currents. This depends upon the situation, system parameters and voltage profile.

System Connection

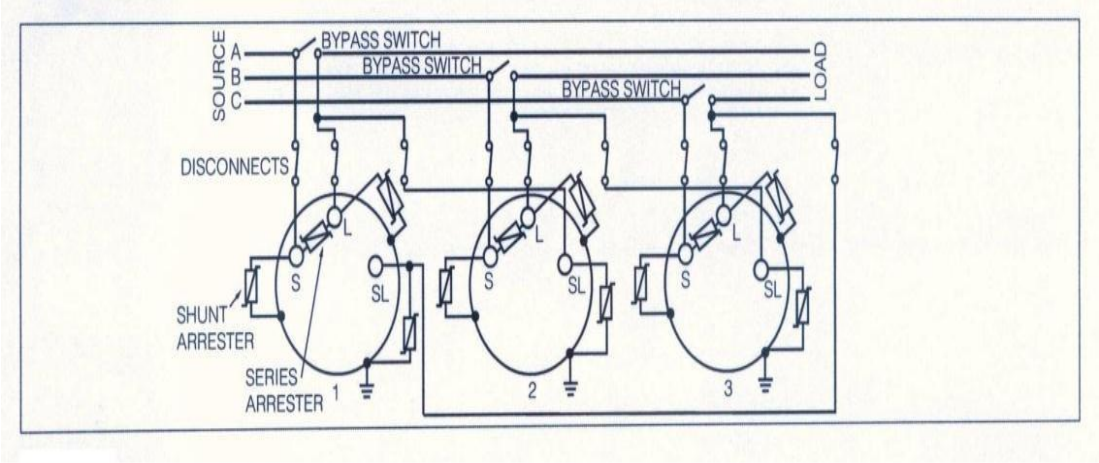
- Two or three single-phase regulators banked together, regulate the voltage of a three phase, three-wire system when connected according to these configurations:
- Open delta connection: Maximum regulation is ± 10% of input. Two single-phase regulators are used.

Figure 31: Regulating a three-phase, three-wire circuit with two regulators



- Closed delta connection: Maximum regulation is ± 15% of input. Three single-phase regulators are used.

Figure 32: Regulating a three-phase, three-wire circuit with three regulators



The regulators should be properly located in harmony with the current load forecast and planned feeder configurations. Economic placement of voltage regulator is preferred, if possible.

Any operational problems, such as over voltages, or special compensation setting requirements with voltage regulator installations should be reported.

Whenever there is change in network configuration or addition of new bulk customer or reinforcement, the re-location of existing voltage regulator(s) should also be done accordingly.

Software (CYME) is used to ensure robustness of design for assessment of need, size and location of Voltage Regulator

- Determine size of voltage regulator using the following formula:

$$KVA\ (3\emptyset) = Range\ x\ KV_{feeder}\ x\ \sqrt{3}\ x\ I_{Peak}$$

Where: $KVA\ (3\emptyset)$ = Size of voltage regulator in KVA (3-phase. For a single phase voltage regulator, the square root 3 factor should be removed from the formula)
$Range$ = Range of regulation needed (+/- 10% or +/- 15%)
KV_{feeder} = Voltage rating of feeder in KV
I_{Peak} = Peak load of feeder

Example illustrates the manual calculation method for voltage regulator in Appendix 2

9.2 Capacitor Banks

Shunt capacitors are installed along the overhead MV feeders to correct poor power factor, reduce losses, and, as a side effect, improve the voltage. Capacitors can be used in two possible configurations:

- Fixed: The network operator can switch on capacitor bank whenever required and it can be permanently switched on.
- Switched: where the capacitor bank can be switched on automatically based on control settings (VAR, V, date.).

A particular concern in using fixed capacitor banks is that the voltage during both peak and light load periods should comply with the voltage criteria. Whereas, during periods of light load, customers should not be over-voltaged

Shunt capacitor will cause a voltage rise from the capacitor bank location back to the source. Fixed capacitor banks will not appreciably improve voltage regulation, but will provide a constant increase in the voltage level. When installed on a MV feeder, the voltage increase at the capacitor location is the same for both light load and high load feeder voltage profiles. The voltage at light load must be calculated to ensure that customers are not over voltaged.
For details, refer to 43-SDMS-01 with latest updates.

- Improvement in efficiency of the power system
- Power factor improvement
- Voltage improvement

- Increased power flow capability
- Reduction in energy and power losses
- Release in capacity of line and substation

Fixed and Switched Capacitor Banks

In order to safely select the total kVAR per feeder that can be compensated by fixed capacitors, it is necessary to know the exact loading pattern throughout the year and possess information about the lightest load condition power factor. Based on this, a safe determination of the kVAR output of the fixed capacitor banks can be affected.

The fixed capacitor banks should be sized such as to compensate the minimum reactive kVARs that are constantly required in the network.

The application of fixed capacitor banks that actually exceed the reactive kVAR requirements of the network at a given point in time can result in an over compensated network where the power factor changes from lagging to leading. In this case the total current flowing though the feeder increases and the losses are again high. Moreover, the feeder voltage increases significantly due to the capacitive voltage increase phenomenon.

Use an established load flow software (such as CYME). These programs typically include dedicated functionalities that recommend the optimal location for capacitors on a feeder to correct voltage drop and/or power factor issues.

For use of capacitor banks in light of observed voltage drop in a feeder, the minimum size of capacitor required to correct the voltage drop can be estimated using the following formula:

$$C_{KVA} = \frac{(95\% - Volt_{Obs})\ x\ 10\ x\ kV^2}{\sum(x_i.\ d_i)}$$

Where: C_{KVA} = Size of capacitor in KVA

$Volt_{Obs}$ = Voltage observed at point of unacceptable voltage drop, with the assumption that 95% is the minimum acceptable voltage

kV = Voltage of line in kV

x_i = Reactance of conductor of each segment between capacitor & source

d_i = length of conductor of each segment between capacitor & source in km

Various capacity options for capacitor banks available within SEC include: 600 KVAR (3x200 KVAR), 900 KVAR (3x300 KVAR) and 1200 KVAR

Example illustrating manual calculation method for capacitor in Appendix 2

9.3 Auto-Recloser (AR) & Sectionalizer

To improve supply standards & customer services on overhead MV feeders, installation of auto recloser on the overhead MV line has been considered as one of the most effective and practical solution.

Automatic line sectionalization offers substantial improvement in reliability of the overhead MV system. Sectionalizer opens automatically in case of permanent faults on their load side, thus isolating the faulted section of the line rapidly and automatically.
Improving customer services both in urban and the rural areas is one of the important goals, which is affected by following:

a. Very long length of feeders.
b. The frequency and the average outage duration of faults is higher.
c. Temporary nature of most of the fault outages.

Principle:

Auto recloser is a protective device with the ability to detect phase and phase-to-earth overcurrent conditions, to interrupt the circuit if the overcurrent persists after a predetermined time, and then to automatically reclose to re-energize the line. If the fault that originated the operation still exists, then the recloser will stay open after preset number of operations, thus isolating the faulted section from the rest of the system. In an overhead distribution system, about 80-90% of the faults are of a temporary nature and last, at the most, for a few cycles or seconds. Coordination with other protection devices is important in order to ensure that when a fault occurs, the smallest section of the circuit is disconnected to minimize disruption of supplies to customers. Generally, the time characteristic and the sequence of operation of the recloser are selected to coordinate with mechanisms upstream towards the source. After selecting the size and sequence of operation of the recloser, the devices downstream are to be adjusted in order to achieve correct co-ordination.
Auto recloser operates in a similar manner to that of the MV feeder breaker. Its main function is to trip on a fault and reclose successfully in case of either a transient fault, or a fault cleared by a downstream protective device, or trip and reclose till it reaches lockout in case of permanent fault.

For details, refer to 33-SDMS-01,33 SDMS-03 and Distribution Operation Manual (DOM) 01-20 with latest updates.

Sectionalizer does not have fault interrupting capabilities. It does not interrupt short circuit current and it is not used for protection, but just for isolation of the faulted section of the line. The fault current interruptions are performed by the backup device (such as an auto-recloser or circuit breaker) that is used for the protection of the line. Sectionalizers count the operations of the backup device during fault conditions. The following conditions should be fulfilled in order to initiate operation of the Sectionalizer:

- Sufficient overcurrent to activate the Sectionalizer.
- Interruption of the overcurrent by a recloser within a specific time.

The Sectionalize counts the number of the fault current flow interruptions and after a pre-selected number, it opens to isolate the faulted section of the line. This always takes place when the backup recloser is in open position.

Sectionalizers can be set to open after the first, the second or the third interruption of the short circuit current. After the sectionalizer opens, the backup device automatically recloses to restore to service that portion of the line up to the sectionalizer location. It will reset counts that do not reach the counts-to-open setting within the selected reset time due to clearing temporary faults.

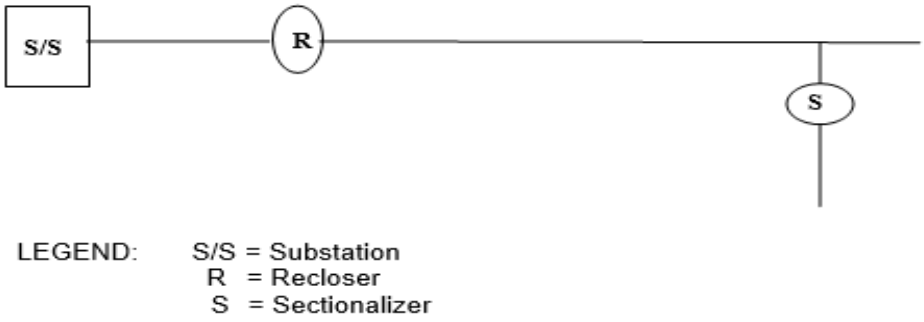
For details, refer to 33-SDMS-02,33 SDMS-04 and DOM 01-20 with latest updates.

Benefits:

Use of auto-reclosers and sectionalizers will lead to the following benefits:

- Reduction in the outage duration.
- Auto clearing of outages caused by temporary faults. It will save field staff from unnecessary patrolling.
- Isolating faulty sections from the healthy sections in case of permanent fault. It will reduce post fault line patrolling effort by field staff.

Figure 33: Location Diagram of Sectionalizer in Overhead Networks



Installation Criteria, Number and Location

The installation of auto-reclosers and sectionalizers need to ensure compliance to WERA standards. However, decisions on auto-reclosers and sectionalizers need to be aligned with distribution protection analysis team before final installation to account for various factors such as:

- Coordination between devices for robust protection
- Ensuring protection against correct level of fault current
- Presence of other protection devices such as switches and circuit breakers

This is typically achieved through Sectionalizing studies using established load flow software tools like CYME

All overhead feeders may not require auto-reclosers and sectionalizers. Selection of feeders will depend on the historical incidence of faults for the feeder and its load, as outlined the figure below.

Figure 34: Identification of Feeders for Installation of Auto-reclosers and Sectionalizers

		Feeder load (current and / or 5-year forecast)		
		<= 2 MVA	> 2 MVA to 12 MVA	> 12 MVA
Fault incidence	>= 4 faults per year or historical average annual fault duration >= 2 hours	No use	Priority 3 selection	**Highest priority (Priority 1) → immediate selection for use**
	< 4 faults per year or historical average annual fault duration< 2 hours	No use	Priority 4 selection (need-based)	Priority 2 selection

Identification of feeders as per above figure will facilitate compliance with WERA Security of Supply standards. As per this, feeders where the load is greater than 12 MVA and historical average number of faults per year (permanent or transient) is more than 4 or maximum historical fault duration is more than 2 hours (for the entire year) will have the highest priority for use of auto-reclosers and sectionalizers – these feeders do not comply with the current WERA security of supply standards.

Additional factors need to be considered as mentioned below:

Auto-reclosers:

- Line auto-reclosers (if not present), will follow the prioritization set in the earlier framework
- Line auto-reclosers (if not present on the feeder) will be applicable for feeders with overall length > 30 kms for 13.8kV feeders and 60 kms for 33 kV feeders (for feeders that are less than 30 kms in length for 13.8kV voltage, other protection devices such as load break switches, fuses, etc. may be applicable)
- Single phase to ground and 3-phase short circuit values at the location on which auto-recloser / Sectionalizer is proposed should be considered
- The short circuit level should not be exceeded at location i.e. auto-recloser / Sectionalizer maximum interrupting capacity
- Cable terminations and joints should not be exposed to high short circuit currents since faults

Sectionalizers:

- Sectionalizers are to used downline from line auto-reclosers or circuit breakers
- If radial feeder with high priority as per earlier framework:

  a. Sectionalizer to be used on each branch with more than 2 MVA load

b. After each section of 2 MVA load on the main feeder, when other sectionalizing devices such as circuit breakers, load break switches or switchgear are not present

- If loop feeder with high priority as per earlier framework:
  a. No devices needed if automatic switching is present
  b. If automatic switching is not present, the time required manual switching needs to be taken into account. If time required for above is more than 2 hours, sectionalizers will be required

The following table outlines the criteria regarding number and location of auto-reclosers to be used on feeders:

Table 52: Criteria for Number & Location of Auto-Reclosers on Feeder

	Feeder's total distance in km	Number of line auto-reclosers required	Install line recloser at a distance of about
1	> 30 km and ≤ 60 km	1	Half way between the source and the farthest point on the line
2	> 60 km and ≤ 90 km	2	1/3rd and 2/3rd respectively from the source point.
3	> 90 km and ≤ 120 km	3	1/4th and 1/2 and 3/4th respectively from the source point.

Additionally, auto-reclosers should be installed after key load points (such as industrial parks, bulk customers, etc.) on the main feeder.

The sizing of auto-reclosers and sectionalizers should match the load and maximum fault current requirements of the feeders as well as match the voltage level of the feeder

Process for Evaluation of Auto-reclosers and Sectionalizers

- Identify priority for each feeders for each ED:

a. From the outage register for the last 3 years, identify number of outages, average duration of fault and maximum duration of fault for each overhead feeder within the ED
b. For each overhead feeder, identify current peak load (in MVA) and forecast peak load for 5 years (in MVA)
c. Categorize feeder as 'priority 1 feeder' if:
d. Average number of outages per year for the feeder > 4; AND
e. Average annual duration of fault for the feeder > 2 hours; AND
f. Current peak load OR forecast peak load for any of the upcoming 5 years > 12 MVA (for all MV voltages)
g. Similarly, assign priorities for other feeders based on the earlier defined framework

- For each priority 1 feeder, assess need for auto-reclosers, sectionalizers, load-break switches and / or fault indicators as per earlier defined criteria (the need for such devices will be recommended by Network Planning).
- For each priority 1 feeder, confirm equipment requirements after aligning with protection engineering team to take into account short-circuit study and sectionalizing study conducted by them.
- Conduct economic assessment to determine present value of investments due to equipment installation, while estimating customer minutes lost (CML) before and after equipment to showcase as benefit.

NOTE: Priority 1 feeders are the most suited to showcase benefits of distribution automation

9.4 Distribution Automation criteria and SMART switches/equipment:

- Distribution Automation & smart switches help in quickly isolate the affected parts of the distribution feeders due to faults and restore electrical service to the unaffected parts, which in the following :

- Improve reliability indicators (SAIDI / SAIFI)
- Reduces the effort and time expended by the field operator to reach to for the affected area and restore service.
- Raising occupational safety and health standards.

Criteria for determining the optimal rate of automation for individual feeder:

A. Underground feeders:

- In overall, the optimal rate for automation (SMART switch or equipment) is 40% of total number of switches in entire distribution network at any scale, whether narrow or broad scale (office / department / city).
- To determine the optimal rate of automation for individual feeder, there are two factors shall be taken into account since they are directly related to reliability indexes (SAIDI & SAIFI) . In the following the two factors are: -

1- Ration of Number of Customers on the feeder, comparing to the average number of total Customer in the network.
2- Ration of the length of the feeder, comparing to the average length of the network.

Moreover , Studying the operational needs of the feeders also contributes to determining the optimal rate of automation for individual feeder and the number of switches required to be automated by coordinating between planning and operation as follows: -

- The geographical nature of the area through which the feeder passes.
- Population distribution.
- Feeder loads.
- Historical record of faults.

B. Overhead lines:

The optimal automation rate for OHL is 100%.

Determine the location of the smart equipment in the feeder:

After determining the optimal rate of automation for individual feeder, the most appropriate location for installing the smart switches on the feeders shall be determined in coordination between planning and operation. Below are the sites that need to be taken into consideration:

1. MV customers with (MRMU).
2. VIP Customers.
3. Sites that require entry/access permits.
4. Sites are difficult to access/reaches.
5. NOP.
6. Extensible RMU.
7. any RMU with special specifications pertaining to the customer`s premises where its responsibility for maintenance and operation will be transferred to SEC.
8. It should not be used for temporary purposes.

Selection the type of the smart equipment:

A. Overhead lines:

All of OHL switches (LBS/ Auto-RECLOSER / SECTIONELIZER) must be as per specifications (33-SDMS-04/ 33-SDMS-03/ 30-SDMS-05) shown in table below:

#	Item No	Overhead lines	Remarks
1	908305011	OHD LBS,13.8 KV,3P,600A, 60Hz,110 KV BIL,SF6,PM	The SMART OHL switches (LBS/ Auto-RECLOSER / SECTIONELIZER) must be as per 33-SDMS-04/ 33-SDMS-03/ 30-SDMS-05
2	908305012	OHD LBS,13.8 KV,3P,400A, 60Hz,110 KV BIL,SF6,PM	
3	908305009	OHD LBS, 33 KV,3P,600A, 60Hz,200 KV BIL,SF6,PM	
4	908305010	OHD LBS, 33 KV,3P,400A, 60Hz,200 KV BIL,SF6,PM	
5	908331011 908333011	Auto-RECLOSER,33KV,560A,SF6,170KV BIL,825MM CRP RECLOSER:CIRCUIT,33 KV,3 P,60 HZ,170 KV	
6	908331007	Auto-RECLOSER,33KV,560A,SF6,170KV BIL,1320MM CRP	
7	908331010 908333007	Auto-RECLOSER,33KV,560A,SF6, 200KV BIL,1320MM CRP RECLOSER:CIRCUIT,33 KV,3 P,60 HZ,200 KV	
8	908331008 908333008	Auto-RECLOSER,13.8KV,560A,SF6, 95KV BIL,345MM CRP RECLOSER:CIRCUIT,13.8 KV,3 P,60 HZ,95 KV	
9	908331009 908333009	Auto-RECLOSER,13.8KV,560A,SF6,110KV BIL,552MM CRP RECLOSER:CIRCUIT,13.8 KV,3 P,60 HZ,110 K	
10	908334004	SECTIONELIZER,33KV,400A,SF6, 200KV BIL,1320MM CRP	
11	908334005	SECTIONELIZER,13.8KV,400A,SF6,110KV BIL,552MM CRP	

B. Underground network :

All of UG Switches (RMU and MRMU) must be as per specification (32-SDMS-11/ 32-SDMS-12) except RMU (3way: 2LBS+1CB) only which it should remain within limits to achieves 40% of automation of entire distribution network as shown in table below:

#	Item No	RMU 13.8 KV	Remarks
1	908321106	SWGR, RMU, ODTR, 2LBS+1CB, TEE OFF, 95KVBIL, 3PH, 60 Hz	It should remain within limits to achieves 40% of automation of entire distribution network
2	908321101	SWGR, RMU, ODTR, 3LBS+1CB, TEE OFF, 95KVBIL, 3PH, 60 Hz	It must be as per specification (32-SDMS-11/ 32-SDMS-12)
3	908321104	SWGR, RMU, ODTR, 2LBS+2CB, TEE OFF, 95KVBIL, 3PH, 60 Hz	
4	908321124	SWGR,RMU,O/D,(3)LBS,SF6,13.8KV,3P,400 A	
5	908321126	SWGR,RMU,O/D,(4)LBS,SF6,13.8 KV,3P,400 A	
6	908321125	SWGR,RMU,O/D,(2)LBS PLUS(1)CB TEE OFF,SF6,ATS,13.8 KV,3P,400/200A	

#	Item No	RMU 33 KV	Remarks
1	908321105	SWGR, RMU, ODTR, 2LBS+1CB, TEE OFF, 200KVBIL, 3PH, 60 Hz	it should remain within limits to achieves 40% of automation of entire distribution network
2	908321103	SWGR, RMU, ODTR, 3LBS+1CB, TEE OFF, 200KVBIL, 3PH, 60 Hz	It must be as per specification (32-SDMS-11/ 32-SDMS-12)
3	908321102	SWGR, RMU, ODTR, 2LBS+2CB, TEE OFF, 200KVBIL, 3PH, 60 Hz	
4	908321127	SWGR,RMU,O/D,(3)LBS,SF6,33KV,3P,400 A	
5	908321129	SWGR,RMU,O/D,(4)LBS, SF6,33 KV,3P,400 A	
6	908321128	SWGR,RMU,O/D,(2)LBS PLUS(1) CB TEE OFF, SF6, ATS, 33KV, 3P,	

#	Item No	MRMU	Remarks
1	908321223	SWGR,MRMU,I/D,(3)LBS PLUS(1)CB TEE OFF,SF6,33 KV,3P, 630 A	It must be as per specification (32-SDMS-11/ 32-SDMS-12)
2	908321224	SWGR,MRMU,I/D,(2)LBS PLUS(1)CB TEE OFF,SF6,33 KV,3P,630 A	
3	908321225	SWGR,MRMU,I/D,(3)LBS PLUS(1)CB TEE OFF,SF6,33 KV,3P,400 A	
4	908321226	SWGR,MRMU,I/D,(2)LBS PLUS(1)CB TEE OFF,SF6,33 KV,3P,400 A	
5	908321227	SWGR,MRMU,I/D,(3)LBS PLUS(1)CB TEE OFF,SF6,13.8 KV,3P, 630 A	
6	908321228	SWGR,MRMU,I/D,(2)LBS PLUS(1)CB TEE OFF,SF6,13.8 KV,3P,630 A	
7	908321229	SWGR,MRMU,I/D,(3)LBS PLUS(1)CB TEE OFF,SF6,13.8 KV,3P,400 A	
8	908321230	SWGR,MRMU,I/D,(2)LBS PLUS(1)CB TEE OFF,SF6,13.8 KV,3P,400 A	
9	908321234	SWGR,MRMU,I/D,(2)LBS PLUS(1)CB TEE OFF,SF6,ATS,33 KV,3P,630 A	
10	908321231	SWGR,MRMU,I/D,(2)LBS PLUS(1)CB TEE OFF,SF6,ATS,33 KV,3P,400 A	
11	908321233	SWGR,MRMU,I/D,(2)LBS PLUS(1)CB TEE OFF,SF6,ATS,13.8 KV,3P,630 A	
12	908321232	SWGR,MRMU,I/D,(2)LBS PLUS(1)CB TEE OFF,SF6,ATS,13.8 KV,3P,400 A	

10 Development project & private plot plans.

10.1 Connected loads estimation

- For all such customers, an average load requirement VA/m² is considered as appropriate method for the load calculation. For customers of type C1 (normal residential dwelling) & C2 (normal commercial establishment) in plot plans, please refer to Table (2 and 4) Appendix 1 for load estimation. For other customer types, please refer to Table 19 (Load density factor method).
- Any additional loads will be considered & added as special loads.
- For street lighting, circuit breaker rating is determined by Municipality (Baladiya).
- For parks and other public areas, circuit breaker rating is provided by Municipality (Baladiya).
- The information of the building system should be provided by the municipality on the map of the approved plan which include (building percentage , number of floors , the number of units per/ floor) for each plot, and the owner of the plot plan is asked to provide this information from the municipality, exceptionally only in the case of the municipality does not provide this information, the assumption condition can be as the following table below:

Facility Type	Building Percentage	Number of Floors	Minimum number of units
Buildings (residential- commercial- mix)	60%	must be provided by municipality	Two unit/ floor + one unit for roof
Villa	60%	2.5	One unit
Schools	40%	3	One unit
Mosques	50%	2	One unit
Garden/open area	100%	1	One unit
Other facilities	60%	3	One unit

10.2 Load Estimation Methodology

For Customers' Buildings with LV Meters (from 20 A up to 800 A), calculate their Coincident Demand Load (CDL) on their Public Substation as follows :

1- For a group of (N) KWH Meters in the customer's building where all of them have same Circuit Breaker Rating (CBR) in (Amp) and same Demand Factor (DF), the equation to calculate the Coincident Demand Load (CDL) in (Amp) for this group of KWH Meters could be simplified as follows:

$$CDL\ \ on\ Substation = \left(\sum_{i=1}^{N} CBR_i \times DF_i\right) \times CF(N)$$

Where :

N = Number of all KWH Meters supplied by that Substation.

2- For a group of (N) KWH Meters in the customer's building where any one of them has different Circuit Breaker Rating (CBR) in (Amp), the equation to calculate the Coincident Demand Load (CDL) in (Amp) for this group of KWH Meters will be as follows:

a. If all Circuit Breaker rating ≤160 (Amp) the equation to calculate the Coincident Demand Load (CDL) will be as follows:

$$CDL\ on\ Substation = \sum_{i=1}^{N-1}(CBR_i \times DF_i) \times CF(N)$$

b. If Circuit Breakers rating including one or more than 160 (Amp), then the equation to calculate the Coincident Demand Load (CDL) will be as follows:

$$CDL\ on\ Substation = \left[\sum_{i=1}^{g\ number\ of\ largest\ CB} CBR_i \times DF_i\right] CF(g) + \left[\sum_{g+1}^{N} CBR_i \times DF_i \times CF(N-g)\right]$$

- For Customers' Buildings more than 800 A shall be supplied by Private Substation or by MV, calculate the Coincident Demand Load (CDL) will be as follows:

$$CDL = \left(\sum_{i=1}^{N} CL_i \times DF_i\right) \times CF(N)$$

- Calculate the Total Coincident Demand Load (CDL) for the (Development Project / Plot Plan) as follows :

$$CDL_{Total} = \sum_{i=1}^{N} CDL_i \times CF_{for\ Substations} \times CF_{for\ MV\ Feeders}$$

Where:

N = Number of all (Public Substations + Private Substations + MV RMUs) which designed to supply all Lots/Buildings within the (Development Project / Plot Plan).

$CDLi$ = Coincident Demand Load in (KVA) for the Individual element (Public Substations + Private Substations + MV RMUs) no. (i).

$CFFor\ Substations$ = Coincident Factor between (Public Substations + Private Substations + MV RMUs) = 0.9

$CFFor\ MV\ Feeders$ = Coincident Factor between (MV Feeders) = 0.9

Or (Only in case of Master Plan Stage without detailed Networks design):

N = Number of all Lots/Buildings within the (Development Project / Plot Plan).

$CDLi$ = Coincident Demand Load in (KVA) for the Individual Lot/Building no. (i).

10.3 Technical study

The developer shall submit the entire technical study prepared by the engineering or consultancy office, in accordance with the above-mentioned instructions, to the concerned electricity department, to include the following attachments:

- (3) Paper copies size (A0) and one amendable digital copy of the initial regulatory plan approved and stamped from the licensing authorities (municipality or secretariat) of scale 1/2000, to indicate the following therein:
  a. Panel number approved for the plan
  b. Owner name
  c. Total area of the plan
  d. Number of plots
  e. Type of use (residential- commercial) for each plot
  f. Building system for each plot (building percentage/ number of floors/ number of units in each floor/attachments/ etc.)
  g. Dimensions and area of each plot
  h. Dimensions of the streets with the ownership boundaries of the plan
  i. All facilities and services inside the plan (mosques/ schools/ parks/ street lighting panels/ etc.)
- Copy of the title deed for the plan in the name of the owner approved and stamped from the notary public.
- Copy of the license approved for the plan in the name of the owner and in the panel number of the plan, approved and stamped from the municipality or secretariat.
- Official letter from the Municipality indicating the location of the plan in the approved urban zone (urban development phase) (is the plan located inside or outside the approved urban zone with the determination of the urban development phase).
- (3) Paper copies size (A0) and one amendable digital copy of the plan approved and stamped from the engineering or consultancy office indicating details of the entire design of MV&LV networks, locations of distribution substations and locations of distribution cabinets necessary to supply all plots, facilities and services inside the plan.

- (3) Paper copies size (A0) and one amendable digital copy of the single line diagram indicating on it details of the entire MV network design inside the plan, approved and stamped from the engineering or consultancy office.
- (3) Paper copies and one amendable digital copy of load study forms (9, 10, 11, 12) as given in Forms, must be approved and stamped from the engineering or consultancy office. For the following :-
- Technical tables of loads calculation for all plots, facilities and services inside the plan.
- Technical tables of LV network design calculations, loading percentages on LV feeders, distribution cabinets and distribution substations, and voltage drop percentage calculations on LV feeders from the distribution substations to the meters.
- MV network design calculations, loading percentages and voltage drop percentage calculations on MV feeders and loops from the feeders to the open points.
- For the regulations and procedures of plot plan, please refereed to customer services manual.

10.4 LV Network Design

The LV network must be in line with the following:

- The network should comply with relevant design and equipment guidelines relevant for LV networks within SEC (which is outlined in Chapter 8)
- The LV network must cover all load requirements inside the plan
- It should be underground network type and radial network design with 400/230V
- Total voltage drop on LV cables from distribution cabinets should not exceed 5%
- It is allowed to use all LV feeders outgoing of mini pillar in the design of the plot plan according to the material specifications of mini pillar with latest updates, and to observe the planning standards (load percentage - voltage drop……..).
- The location of distribution cabinet should follow these guidelines:
  a. Locations of distribution cabinets are determined so that to be in the center of the plots planned to be supplied from them as possible, and to be on the outer boundaries of any piece of land on the approved streets directly.
  b. If it supplies only one plot of any type: the distribution cabinet is installed at the front fallback of this plot.
  c. If it supplies a plot of any nonresidential services and facilities with a plot or more (residential or residential/commercial): the distribution cabinet is installed at the front fallback of the nonresidential services or facility plot.

d. If is supplies more than one plot of (residential or residential/commercial) type only: the distribution cabinet is installed at the front fallback between the two plots which are located in the center of the load of the plots planned to be supplied from this cabinet.

- Loading percentage on public distribution substations, does not exceed 80% .
- Location of public distribution substation is to be between the plots supplied and not at the outer edge / boundary of the plots and should adhere to the following guidelines:

a. If only one plot is being supplied, the distribution substation can be located at the boundary of the plot
b. If the substation is intended to supply any non-residential facilities along with residential facilities, the substation should be located at the boundary of the residential plot
c. If the substation is intended to supply more than 1 residential plots, the location should be between the plots as close to the load center as possible.

- For supplying plot plan use the substation with (500 - 1000 -1500) KVA depend on CDL and design of plot plan network.
- It is allowed to use all LV outgoing (CB) in the public distribution substation (500,1000,1500) kVA in the design of the plot plan according to the material specifications of the substations (with latest updates), and to observe the planning standards (load percentage - voltage drop……).
- Dedicated (private) distribution substations are to be used when supplying to a single plot of more than 800A connected load and when the coincident demand load of the plot does not exceed 4MVA and the suitable rating of the dedicated distribution substation should be such that it fully serves the entire CDL of the plot without exceeding 100% of its rating.
- Dedicated distribution substation may be located within the plot that is being supplied by it, at the outer boundaries.

10.5 MV Network Design

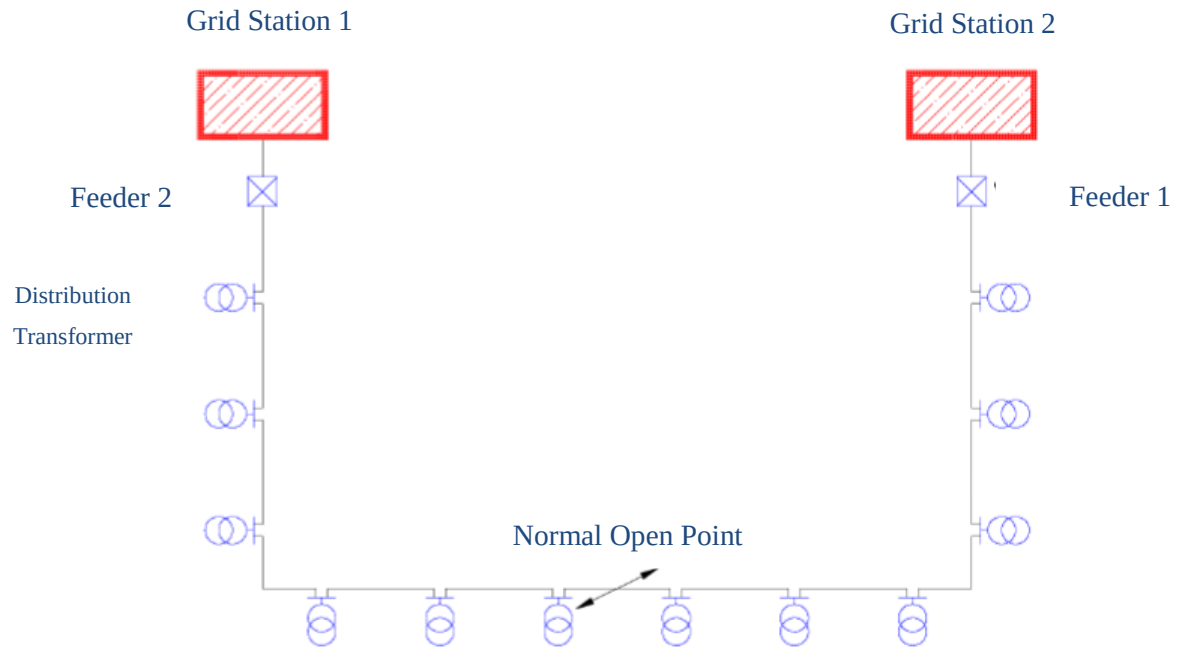
Development project

The following criteria need to be adhered to while designing MV network inside the new Development project:-

- All load requirements must be covered.
- Single loop design should be adopted taking into account n-1 design criteria (as outlined in figure 35 below). Tee lop can be used for large projects that the developer establish grid station to utilize the capacity available at the station.
- Voltage drop within the system should not exceed 5%.

- The rating and loading of equipment used will be based on (Medium Voltage Connection Planing)
- The maximum sustained load of 13.8 kv single loop is 7.6 MVA for cable 3×500mm² Al.
- The maximum sustained load of 33 kv single loop is 16.6 MVA for cable 3×400mm² Al.

Figure 35: Standard Design of MV Network for Plot Plan



Relevant formulas for calculation of load on MV feeders:

$$CDL_{on\ MV\ Single\ Loop} = \sum_{i=1}^{N} CDL_i \times CF_{for\ Substations}$$

Where

$\boldsymbol{CDL_{on\ MV\ Single\ Loop}}$ = CDL on MV single loop

$\boldsymbol{CDL_I}$ = CDL calculated for station (i) supplied from the loop

$\boldsymbol{CDL_{for\ substations}}$ = Coincident Factor between peak CDL of substations (public distribution, private distribution, MV switchgears) supplied from the loop = 0.9

N = Number of substations (public distribution, private distribution, MV switchgears) supplied from the loop

$$Loading\ \%_{on\ MV\ Single\ Loop} = \frac{CDL\ _{on\ MV\ Single\ Loop}}{Rating_{of\ MV\ Cable}} \times 100$$

Where

$\boldsymbol{Rating_{of\ MV\ Cable}}$= De-rated Capacity of MV cables

Connecting of Plot Plan MV Network to SEC Supply Source

Connection of plot plan MV network to SEC supply sources is treated on temporary operating basis as opposed to permanent planned. This is due to the fact that during the time of connection, there is no actual load from the plot plan and to account for the eventuality of delay in growth of actual load in the plot plan. Therefore the design of plot plan MV network as following:-

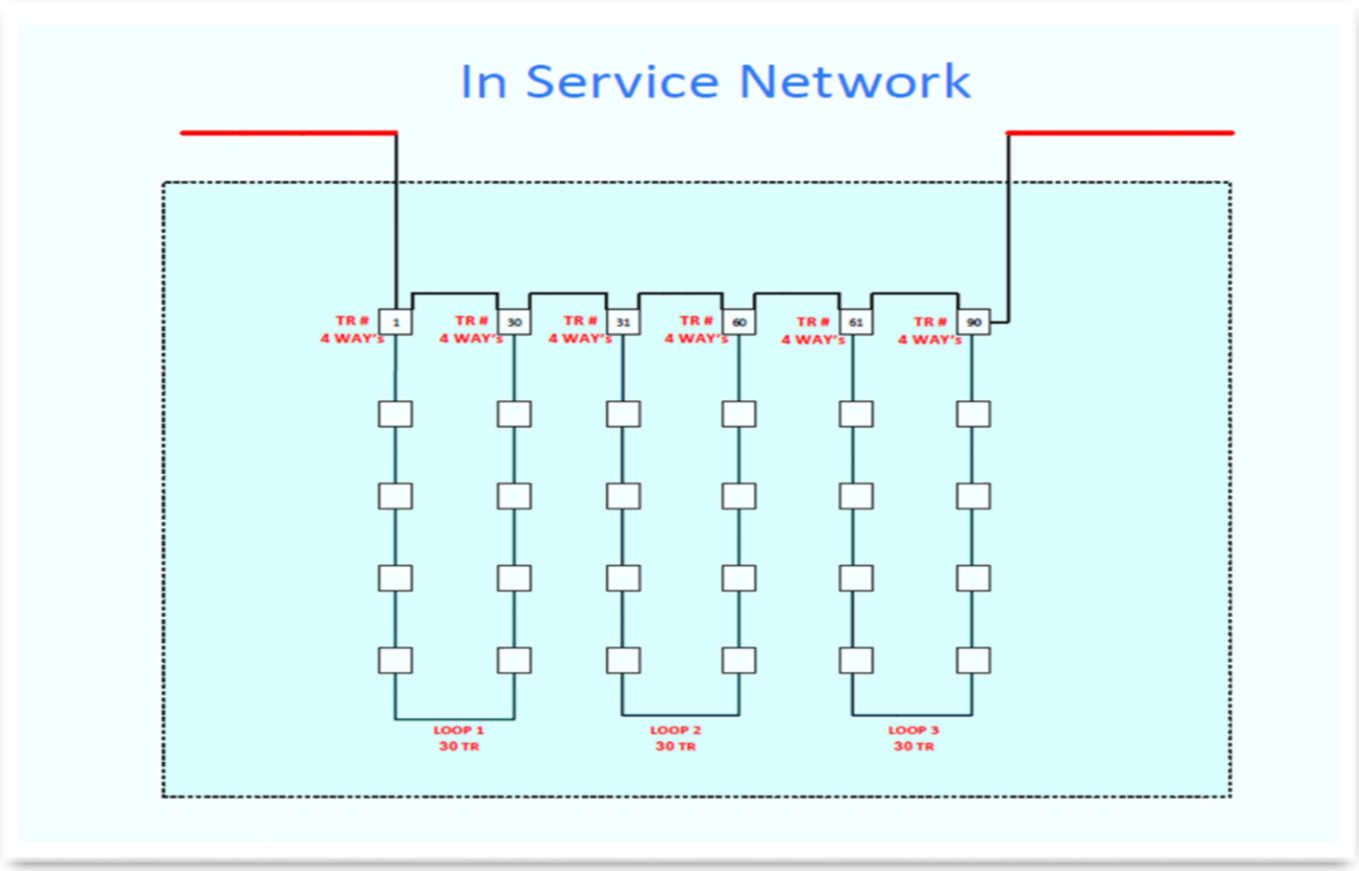
1. The electrical networks are designing Simple Loop according to the following requirements:

   a. The number of distribution substations does not exceed 30.
   b. Installation of 4 ways RMUs within the plot plan every 10-distribution substation.
   c. If the number of distribution substations more than 30, the connection of loops with each other by distributing the stations to circuit with appropriate number schematically according to the study submitted for the planned load
   d. Each single loop containing 30 distribution stations. is connected with an single loop according to the figure 36 below:

2. The number of Single Loops needed to feed the loads of all plots , utilities and services within the plot plan is determined based on all previous standards.

In this way, supply from grid station is considered temporary only until the completion of the emergence of the actual loads on MV loops in the plan in the future, so that the loads on it can be controlled and its supply can be boosted in the future, according to reinforcement plan

Figure 36: MV Network Design for plot plan.



Route of LV& MV Cables

- MV cables are laid in the approved paved streets (under paving) and not under the pavements.
- LV & MV cables are not laid in pedestrians, exceptionally only, if the pedestrians width not less than 6 meters, they can be laid therein ,Taking into consider the requirements of the Municipality.
- Reduce, as possible, the crossing of streets when laying MV cables.
- Route of MV feeders necessary for the plan, are planned so that the beginnings of all feeders (terminals of MV single loops inside the plan) will be at one point at the edge of the plan, the nearest to the source of supply, which is determined by the company

10.6 Method for Determining Need for Dedicated Grid Station for Private plot Plan

If the total area of the private plan is greater than 600,000 m² or if the total CDL of the plan is greater than 25 MVA, there is a need to evaluate the location of a separate transmission grid

station for supply to the plan. Either of the following conditions is required to be met for the above:

- Case 1: Grid station is already approved in the area where the plot plan is located and the location for the grid station is identified
- Case 2: Grid station is planned as per capital budget for the next 5 years according to load forecast plan but location hasn't been identified
- Case 3: If need for grid station is established using 10% of CDL of private plot plan using the following process:
- Calculate CDL of all MV single loops inside the plan

$$CDL_{For\ Plot\ Plan} = \sum_{i=1}^{N} CDL_i \times CF_{on\ MV\ Single\ Loop}$$

Where

$\boldsymbol{CDL_{For\ Plot\ Plan}}$= CDL for the entire plan

$\boldsymbol{CDL_I}$ = CDL calculated for the single loop No. (i) of the plan

N = Number of single loops for the plan

$\boldsymbol{CF_{on\ MV\ Single\ Loop}}$= Coincident Factor (CF) between CDLs of MV single loops inside the plan = 0.9

- Calculate future load forecast for the zone for the each of the next 8 years

  Total forecasted load for zone / area (year i) = Total forecasted peak load on grid stations in the zone / area (year i) + Total forecasted spot loads in the zone / area (year i) + Total forecasted load transfers in the zone / area (year i) + [10% x CDL for plot plan]

- Calculate forecasted loading percentage for the zone / area for each year

  Loading Percentage % = Forecasted Load for (year i) / Forecasted firm capacity for (year i) x 100

- If loading percentage in any year exceeds 100%, this means that transmission grid station will be required in light of private plot plan. If the year when loading percentage exceeds 100% is (year i), the grid station will need to be planned for approval in year (i-3), assuming a 3 year time window for completion of all works to build and energize the transmission grid station.

10.7 Revision of Technical Study

The Technical Study will be evaluated by SEC Distribution ED network planning team before approval of the submitted technical study in the checklist form 13 in **Forms**

11 Network Planning Strategy

The recommended network planning strategy should have two key pillars:

1. Integrated and 5 years planning.
2. Coordination and alignment with various functions

This is highlighted in the figure below:

11.1 Dimensions of Network Planning Strategy

Integrated and 5 years planning

To ensure alignment with evolving demand conditions, it is proposed that SEC Distribution should adopt a multi-stage planning approach as per which planning is undertaken separately. This would imply that for any given Sectors or plans would need to be developed for 5-year planning horizon

The key inputs to developing the 5 years plan are target KPIs for network performance , load forecast and Future Assets, the 5 years network planning process will also result in development of the 5 year plan; in other words ‘how will the Distribution network look like 5 years from today’.

The 5 years planning will ensure definition and prioritization of all projects for the 5-year Plan. This will include integration ,reinforcement,…etc projects with CAPEX estimates of projects required. This would facilitate CAPEX approval from SEC. Once projects (and hence 5 year CAPEX Plan) has been approved will be updated to reflect the changes in projects and their prioritization in yearly Paln.

In addition, the technical details of yearly network planning are covered in stage gate processes for each project.

In terms of timing, the both 5 year and yearly network planning will be update every year

Furthermore, the network plan would be integrated with other similar planning processes within SEC Distribution, such as Load Forecasting, Integrated CAPEX Planning and Material Planning

Coordination and alignment with various functions

The second pillar of the proposed network planning strategy is coordination. For SEC Distribution, the network plan for the ED will act as a key input and coordination point between network planning, operations and National Grid (as detailed below).

The network planning team, will need to assume a proactive role in ensuring that all required inputs are obtained from Electricity Department (ED), operations and National Grid teams to develop network plan. Additionally, the defined network plan needs to be verified and cross-

checked with the above teams to ensure that all inputs are captured. The details of the checks and its timings are mentioned in the detailed process steps in the subsequent sections.

To enable this coordination, it is proposed to establish a working committee in each (ED).

11.2 Yearly Capital Planning

The Yearly network planning process will intail Prepare the Projects of Distribution networks for the Sectors.This will be consolidated into project roadmap which will be executed over the Next year. This process will include Review and Evaluate the Scope of work, Projects Prioroties and Cost Estimation within Distribution Capital Plan Committee.

i. Process Inputs & Output

The following inputs are needed for the process:

a. 5 Year Plan roadmap for the Sectors

b. New plot plans to be added

c. Target KPIs set for the Sectors

d. Bulk customers demand The expected output of the process are:

- Project roadmap for the year (including connection of plot plans):
  a. Timeline of execution of MDN substations (where relevant)
  b. Timeline of execution and Address of feeders
  c. Roadmap for reinforcement, Replacement and integration projects (transformers, feeders and other equipment)
  d. Roadmap for new connection projects (transformers, meters and other equipment)

1- Project Material Planning

For each ED following activities should be followed:

1. After finalization of project roadmap for next year and main assets number of projects for next year for each project type (new connections, replacement, reinforcement and integration)
2. The material planning process for the next year, taking into consideration material consumption and Capex Plan for related projects.
3. Align with SEC HQ Corporate Planning function to finalize CAPEX (as well as project roadmap) and share with the planning team in Sectors for project monitoring

Output Format

The year network plan will be prepared and consist of the following sections:

The following general guidelines need to be followed:

1. The 1-year network plan needs to be submitted as excel file
2. There will be 1 submission which will be submitted by the DED
3. The network plan will be submitted to DED who will approve and forward to HQ for final approval
4. DED will be responsible for preparing all documents necessary for obtaining approval on CAPEX spending from HQ Corporate Planning team
5. The template is provided as following sheets:
    a. GS & MDN SS (1-year): for information on grid stations & MDN substations which will be integrate for the next year
    b. Feeder (1-year): for information on feeders planned for the next year
    c. CAPEX Plan (1-year): for information related to implementation and CAPEX requirements of the proposed projects

For each section of the 1-year network plan report, the following specific guidelines should be followed (the output templates are provided in a separate file):

Grid station & MDN substations plan:

- In this sub-section, detailed plan on status and expected progress will be provided for all new grid stations and MDN substations (including capacity increases) that are planned or under implementation (this will include new and on going projects in next years.
    a. Current project status
    b. Earlier expected date of completion
    c. New expected date of completion
    d. Justification for change in date of completion (if applicable)
    e. Dependencies & risks

CAPEX Requirements:

- This section will have CAPEX requirements for feeders, other projects and reinforcement projects.
- The CAPEX will be based on value of similar projects historically and will include the following cost items:
    a. Equipment or material cost (if unit rate)
    b. Contractor installation cost (if unit rate)
- The CAPEX will be checked Yearly by DED

11.3 Load Forecasting Guidelines

This section as a guideline for developing a medium range load forecast for distribution sector, covering Grid stations which contain outoing for distribution network (High voltage) and main distribution substations (Medium voltage) all the main components of distribution network:

Issues related to Load Forecasting

This section highlights some important issues related to load forecast process such as forecasting accuracy, forecasting range and various methodologies used for development of load forecast.

1- Forecasting Accuracy

Load forecasting is an important function in planning and operation of an efficient electric power system. The better the forecast, the better would be the justification to invest as capital cost.

It could result in delaying power supply to new customers as well as exhibit poor quality of power supply to the existing customers.

In view of the above, an error margin of ± 10% could be considered acceptable in all forecasting levels.

2- VARIANCE ANALYSIS

Variance analysis is the study of deviations of actual versus forecasted. This analysis is used to improve the quality of the forecast.

After agreed loads on the substations, Variance Analysis shall be applied and conducted individually at the Electricity Department level and Sector level against the actual forecasted load this process conducted once at the end of each year.

The formula of the Variance used, would be:

$$\text{Variance} = \frac{(Actual\ load - Forcasted\ Load)}{Actual\ load} \times 100$$

Values of the actual load obtained from the annual agreed load meeting which held between Distribution business line and national grid.

For those Electricity Departments whom exceed the Acceptable Margin (10%) root cause will be identified and action plan will be, The percentage based on Forecasting Accuracy at this Guidelines.

- Steps for conducting Root cause analysis on variance
- Root cause analysis will be identified based on the major factor causing variance

1. Justification and action plan will be given in accordance to the identified Root because leading to the variance.
2. All information mention shall be reflected by filling the dedicated variance analysis template for each operating area.

- Root Cause contributing the variance shall be conducted by the Network Planning Division Managers in each ED, reviewed by Electricity Department managers and Network Planning Division Managers in each DED, then approved by Distribution Engineering Department Managers of the regions.

Template for collecting data For each region(Variance analysis form 11 in Forms).

3- Forecasting Range

The load forecast can be developed in next 5 years range.

For the SEC Distribution System, where accurate information of city development in the kingdom is not available, it is advisable for the time being to go for 5 years range forecasting. This will be in line with the recommended Network Planning Strategy processes.

One purpose of the 5 years planning is to assure that lead-time requirements are met for the different types of projects. The output of the 5 years planning process is a set of decisions and specification for future change to the system.

4- Demand Forecast Methodologies

Several methods are being used for the electric load forecasting and no single forecasting technique is best for all applications, and trending methods are recommended in certain circumstance.

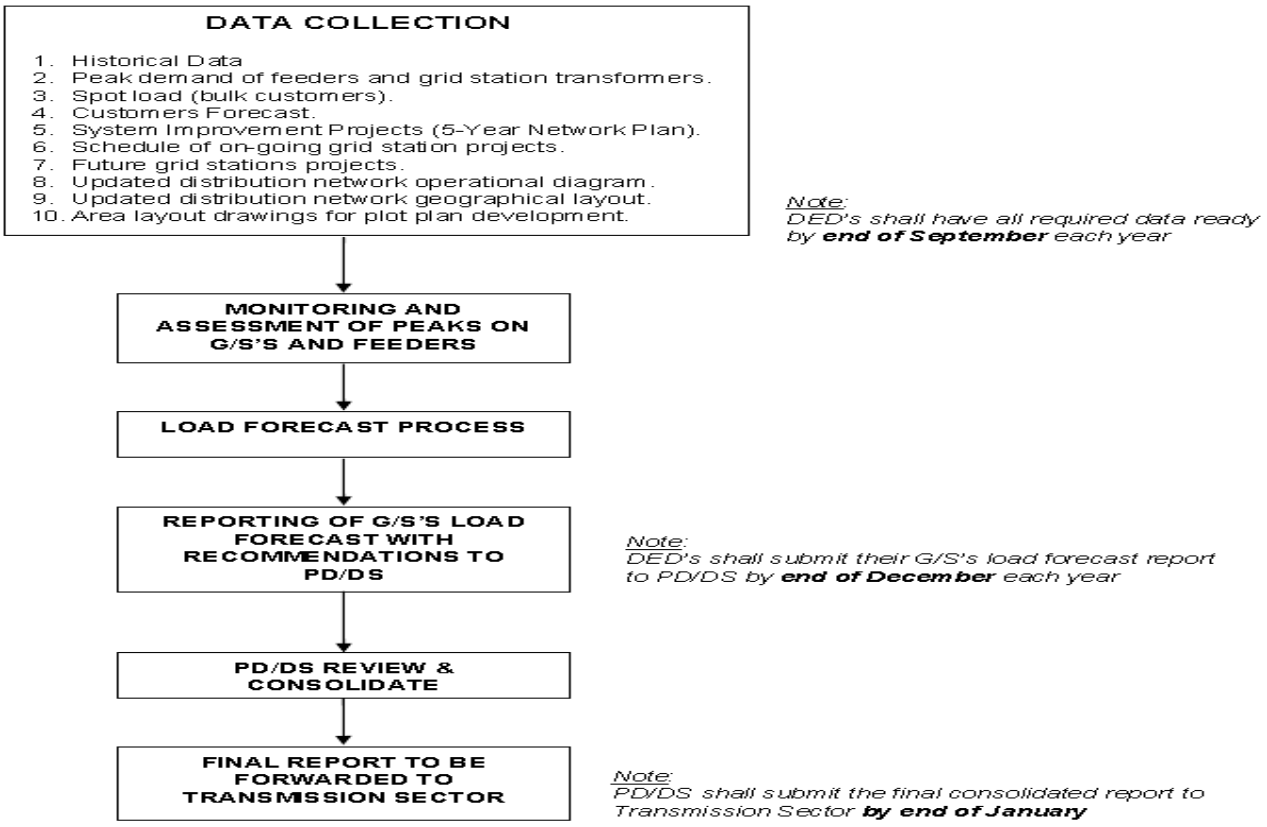
Trending Methods:

Trending methods work with historical load data, extrapolating past load growth patterns into the future. The most common trending method and the method most often thought of, as representative of trending in general, is polynomial curve fitting, using multiple regression to fit a polynomial function to historical peak load data and then extrapolating that function into the future to provide the forecast. The modified multiple regression method to consider some dynamic factors such as load transfer and unusual spot loads, is recommended to be used in SEC. This method will be explained in detailed in subsequent sections.

Flowchart and Timing of Load Forecasting Process

The following process will be followed for load forecasting:

Figure 37: SEC Load Forecasting Process



Data Collection:

Data collection is a vital activity in the load forecast process as correction of information is

MAJOR ACTIVITIES OF LOAD FORECAST PROCESS AND THEIR CORRESPONDING DUE DATA

ACTIVITIES	ACTION BY	SUBMITTED		January	February	March	April	May	June	July	August	September	October	November	December	January
		DATE	TO													
Data collection	Ele. Depts/ DEDs	End of September	DEDs													
Analysis of peak on substations and feeders	Ele. Depts/ DEDs															
Load forecasting process	DEDs															
Reporting of substation's load forecast with recommendations.	DEDs	End of December	PD/DS													
Review and consolidate the final report	PD/DS															
Submittal of final load forecast report.	PD/DS	End of January	Transmission Sector													

Note:
Above time table is suitable for Normal Scenario. For Special Scenario, collection of data for peak load of feeders and substations shall start from November of previous year. Other activities and corresponding due date indicated in above table will remain the same.

required to have a realistic output. Different types of data are required for the load forecast process.

Assessment of Substations Peak:

In this activity, recorded peak readings of grid stations, main distribution substation, are analyzed and the maximum peaks are then assessed. Detailed methods/ techniques of peak assessment are explained in subsequent sections.

Load Forecast Process and Analysis:

Different parameters are considered in load forecast process. These parameters are incorporated through an excel-based program.

Reporting of Substations Load Forecast:

In this activity, load forecast reports of grid station and main distribution substation are prepared based on the output of load forecast process and analysis. then sent these reports to Planning Corporate Department/Distribution Services (PD/DS) for review and consolidation. Details of formats and sequence of report is explained in subsequent sections.

Report Review and Consolidation:

Load forecast reports shall be reviewed and consolidated by PD/DS in order to have a comprehensive final report that shall be submitted to Nationl Grid.

Submitting of Final Report to Nationl Grid:

After review by PD/DS, final report of load forecast of transmission substations is to be submitted to Nationl Grid by the mid of January of the following year.

Required Input Data

Following information is essentially required for the development of 5 Year Load Forecast Reports:

a. Historical data.
b. Peak demand of the transformers of main distribution substation, and grid substation.
c. Spot load (bulk customers).
d. Schedule of on-going grid station and main distribution substation projects.
e. Future grid stations and main distribution substation projects.
f. Area layout drawings for plot plan development.

Historical data:

Historical information about peak load of feeders, main distribution substations and grid stations is very important in the load forecast process. This data gives a picture of how was the trend of peak load and substations in the past years, through which future demand load can be forecasted

considering other factors, such as bulk loads, load transfer between main distribution substations, and grid substations due to network modifications.

A considerable body of research has shown that when working with typical distribution network, the most recent six years of data give slightly better results than any other historical sample, , or more years of data.

Peak demand of the transformers of main distribution substation, and grid substation:

This data is required throughout the year on a routine basis. The load data, which comprises of substation Power Transformers

Spot load (bulk customers):

Customer having coincident demand load (CDL) grater than 4 MVA is considered as a bulk load. All such customers shall be taken into account during load forecast because of their considerable effects on the future demand of the distribution network.

Identified bulk customers are to be filled by Electricity Departments. Following data shall be included:

a. Customer's name.
b. Type of load (industrial, commercial, … etc.)
c. Approved coordination certificate and power supply date.
d. Existing load in MVA (contracted and demand) (if any).
e. Supply voltage level.
f. Year of energization for existing customers.
g. Source of supply (substation and feeders).
h. Ultimate load of the customer.
i. Future load requirement in MVA (contracted and demand) along the span of the forecast period.

Distribution system improvement projects (5-Year Network Plan):

System improvement projects of distribution network during the forecast period shall be identified and considered in load forecast. These projects lead to load transfer between primary feeders and substations.

System improvement projects can be identified based on the last year load forecast, which helps in identifying overloaded feeders, main distribution substations, and grid stations. Overall study to relieve main distribution substations shall be carried out in coordination between DED and Electricity Departments considering future substation projects.

Schedule of on-going grid station and main distribution substation projects:

This information indicates schedule completion of on-going (under construction) grid station and main distribution substation projects which assists to plan the completion of the associated Distribution System projects at the proper time for immediate utilization.

Updated distribution network geographical layout:

Distribution network geographical layout shall be prepared and updated by Electricity Departments. These layouts shall indicate substations, primary feeders, transformers and scale of drawing.

For planning purposes, it is preferred to have these layouts in scale of 1:5000 or above. Updated layouts shall be kept ready up on the request of DED or shared electronically.

Distribution network geographical layouts help in visualizing the network and to make the most economical alternative system improvement projects.

Area layout drawings for plot plan development:

Electricity Departments shall provide layout drawings showing all area development/new plot plans during the forecast period. Area development plans provide the basic information for the development of network plans.

Details of the new plot plans, such as total number of lots, anticipated total demand, source of supply and approved budget year.

Load Forecasting Process & Analysis

Main Parameters of Load Forecast:

Main parameters considered during preparation of load forecast are followings:

- Normal growth
- Spot loads (Bulk Customer)
- Load transfer due to network plan and system improvement projects.

Explanation on how to incorporate these factors in load forecast process would be demonstrated in section below.

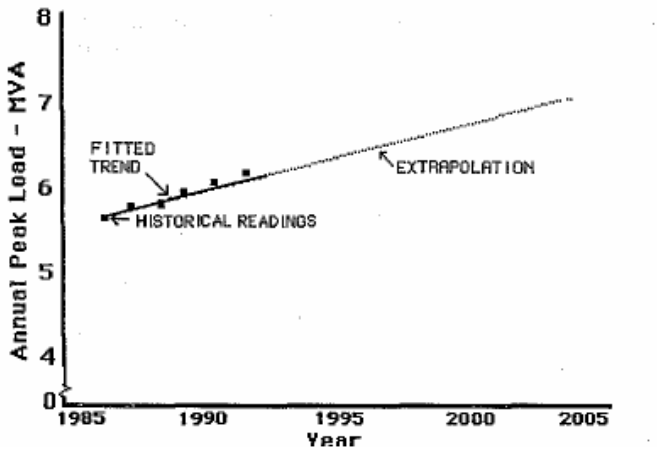
Load Forecast Methods and Techniques:

There are different methodologies and techniques used for load forecast. Since electrical distribution network is a dynamic system (changeable due to load transfer on the level of feeders and substations), there is no pure technique that can be directly applied to predict loads in future years. However, modified and customized techniques can be useful for this purpose.

In order to forecast loads on distribution networks of Saudi Electricity Company (SEC), a modified multiple regression method is recommended to be utilized.

Multiple regression method, which is a type of trending methods, is used to find out growth factor by analyzing the trend of historical load data and then extrapolating past load growth patterns into the future. Trending encompasses a number of different forecasting methods that apply the same basic concept -- predict future peak load based on extrapolation of past historical loads. Many mathematical procedures have been applied to perform this projection, but all share a fundamental concept; they base their forecast on historical load data alone (Figure below).

Figure 38: Trending Method of Forecasting



In general, the curve fit is applied to the annual peak load data. There are two reasons for this. First, annual peak load is the value most important to planning, since peak load most strongly impacts capacity requirements. Second, annual peak load data for facilities such as substations is usually fairly easy to obtain.

It is called "modified multiple regression method" because it incorporates some other factors, such as load transfer and spot load, in order to forecast future loads. Following formula is developed for this purpose:

Forecasted load for next year (N+1) = [(current year (N) peak * (1+normal growth factor)) + Spot load ± load transfer] …. (1) And for year N+2

Forecasted load for year (N+2) = [(forecasted load for year (N+1) * (1+normal growth factor)] + Spot load ± load transfer)] .. (2)

And so on … for following years: The above formula can be used to predict load on feeders or substations.

Steps of Load Forecasting:

1. Growth Factor Calculation:

$$G.F = \left[\left[\left(\frac{A}{B}\right)^{\frac{1}{n}}\right] - 1\right] x\ 100$$

Where,

G.F. = Historical trends in growth (% Growth)

A = Current year peak.

B = Peak of the year (n), which precedes forecasted year by n years.

N = No. of preceding years – No. of historical data (preferably 6 years for 5-year load forecast)

Since loads of substations are sometimes experienced frequent change from year to another year due to load transfer, a growth factor is recommended to be calculated on zone or district levels. Then, it can be reflected on the feeders or substations.

Derived growth factor on zone or district levels usually incorporates spot loads of past years. Therefore, it is more effective to subtract spot loads from historical data before finding growth factor. This requires maintaining a database of energized spot loads in each year.

2. Incorporating Load Transfer:

Proposed load transfer from overloaded substations to the nearby lightly loaded substations or to the new substations shall then be entered.

Formulas 2 & 3 shows that load transfer can be positive or negative. Positive sign means that some loads would be transferred to the feeder or substation while negative sign means shifting some loads from the substation.

3. Incorporating Spot Loads:

Spot loads (greater than 4 MVA) shall also be incorporated. A load of 4 MVA can be considered as a spot load in some areas while it can be considered as a normal growth in some other area. This can be judged be area responsible engineer.

Spot load can be allocated in one single year or it might be distributed for more than one year depending on its nature and type (i.e. industrial, commercial, residential …). For example, industrial load most probably comes in one year while commercial load might require many years to materialize. Considerable loads of normal customers, which are due to new plot plans development can also be considered as a spot load. Normal increase of new or existing customers would be covered by the normal growth.

Load Forecasting Reports

In this section contents and formatting of two types of reports (substations load forecast reports) are described.

Substations Load Forecast Report

DED's shall prepare substations load forecast report and submitted it to Planning Department\Distribution Services for their consolidation. This report shall include following parts:

- Summary of forecasted demand in MVA for each substation belongs to each Electricity Department
- Grid stations and main distribution substations, which exceeded 100% of their firm capacity during current year peak period
- Grid stations and main distribution substations, which are expected to exceed 100% of their firm capacity during next 5 years plan period.
- List of bulk customers coincident demand load (CDL) > 4.0 MVA, which are expected to be energized during forecast period.
- 5-Year load forecast of main distribution substations, incorporating any important remarks or relieving projects for overloaded substations.
- 5-Year load forecast of grid stations, incorporating any important remarks or relieving projects for overloaded substations.
- List of required grid stations
- List of required main distribution substations

PD/DS shall review reports submitted by DED's and consolidate them in one report as 5-Year Load Forecast of all Operating Areas. This report shall be then submitted to the Higher Management and concerned departments with a summary of most important highlights and required future projects.

Load Forecasting Review

Review of load forecasting will be conducted by the Distribution HQ Load Forecasting team then send it to the National Grid

Zone Definition Guidelines

Zone is the basic unit of Network Planning. A zone is a geographical region in which the load of transmission substations can be transferred by distribution switching actions or by other arrangements (permanently or temporarily). In general, a zone shall be bounded by physical obstacle, for example a main road, mountains, etc.

1- Benefits of Zones

The zones are defined for the following benefits:

Ease of Network Planning Process:

The distribution system / network planning can be conducted with greater ease due to manageable scope of a zone vs. a full city, in particular for large cities like Riyadh, Jeddah, etc. Furthermore, division of zones will enable the ED to maintain a close view on network KPIs

Efficiency of Operations & Maintenance:

Within a zone, the operation of the network shall be more smooth since, in case of fault occurrence or the operation staff shall require to shift the load or NOP then it shall be within limited area of zone. This will make the job easy and the contingencies can be handled effectively.

In other words, while handling contingency events, as the limits of network operations are confined within the specific area of the zone, it will make the process of response handling speedy.

Preparation of Load Forecast Reports:

The zone-wise load forecast shall indicate the particular area's loading conditions, then it shall be easy to keep the zone in normal loading condition by transferring the load to other substations which shall be present in the same zone or prepare plan for a new substation in the zone.

Improvement of System Reliability:

Zoning will make the system more simple. As discussed earlier, the boundaries will be clearly identified, so that the distribution network / system will be limited and no hard situations like crossing of roads, mountains and military areas will be faced. This will significantly enhance accessibility within a zone for operations and maintenance purposes.

Ease of Design of New Projects:

A planning engineer shall identify / check a particular zone's loading positions (either overloaded or under loaded). By zoning, a planning engineer shall identify requirements for new substations in an easy manner.

Load Balancing of Distribution Network:

Zones will provide constraints in terms of allowable substations for load transfer and reduce their numbers. This will make it easy for planning engineer to do the job of transferring loads to balance the system. In case of non-availability of firm capacity within the zone, the load may be transferred from other nearby zone temporarily, and shall be transferred back again within the zone when capacity is available.

2- Criteria for Zones

The following criteria have to be followed while defining zones:

- The existing network configuration will be considered and all substations within the zone shall be interlinked
- At least on transmission substation or one MDN substation should be present in a zone. The remote areas shall be exception to this rule. If there is a plan to cancel / remove the primary distribution substation in a particular zone, the firm capacity should be cancelled and load kept
- Municipality / baladiya regulations and natural / geographical conditions shall be considered for determining zone boundaries
- Every zone shall be supplied from more than one source, from different transmission substations if possible
- The load should be preferably be transferred within the zone
- Zone may be determined in accordance with the nature of load (Holy places, industrial cities, military areas, dedicated substations and housing schemes).
- zones may be revised if required with an exception that an adequate justification is made for the change.

GOVERNANCE PROCESS FOR UPDATE OF DPS

DPS will be studied and updated in a periodic manner when needed. The responsibility for conducting the update will be on the HQ Distribution Corporate Planning team.

The reasons for updating the DPS may be many-fold and may include reasons such as:

a. Update in equipment standards and addition of new equipment.

b. Updates due to new conditions / standards from WERA

c. Detailing / Improvement of processes and / or guidelines by Corporate Planning team.

On an on-going basis, the HQ Distribution Corporate Planning team will maintain repository of all updates to be made to DPS. Given the diverse nature of potential updates, the change requests will be recorded by the team and quarterly reports will be made to the Head of HQ Distribution Corporate Planning team outlining all the update requirements for DPS.

The following processes will need to be followed for assessing updates for each of the above categories:

1. **Update in equipment standards and addition of new equipment:**

- Any proactive changes to equipment standards or addition of new equipment will be initiated by HQ Distribution O&M Support (Standardization) team.
- These changes may be received by the HQ Distribution Corporate Planning team at any point of time during the year.
- In May of each year, all changes received in the previous time period will be collated by the HQ Distribution Corporate Planning team.
- The HQ Distribution Corporate Planning team will initiate meetings with HQ Distribution Technical Support (Standardization) team to confirm the changes in standards and also assess whether any additional updates are required to Distribution equipment.
- After all changes are aligned, the HQ Distribution Corporate Planning team will update the relevant sections of DPS.
- The updated DPS is approved by Head of HQ Distribution Corporate Planning team before circulation to network planning engineers of all EDs.

2. **Updates due to new conditions / standards from WERA.**

- These requests will be initiated by WERA and will be treated as change requests only after there is alignment between SEC and WERA to implement them and there is approval from SEC senior management. All such change requests will need to be approved by Head of HQ Distribution Corporate Planning team before implementation

- Once the changes are confirmed, the HQ Distribution Corporate Planning team will update the relevant sections of DPS.
- The updated DPS is approved by Head of HQ Distribution Corporate Planning team before circulation to network planning engineers of all EDs

3. Detailing / Improvement of processes and / or guidelines by Corporate Planning team:

- These updates are initiated by the HQ Distribution Corporate Planning team
- Planning Engineers from the regions can also identify areas for improvement or further detailing. However, such requests will need to be communicated with the HQ Distribution Network Planning team who will decide on whether they require changes in DPS
- The changes will be drafted by the HQ Distribution Corporate Planning team who may seek inputs from EDs and DEDs
- These change activities can be undertaken at any time during the year and will need to be approved by Head of HQ Distribution Corporate Planning team
- The HQ Distribution Network Planning team will update the relevant sections of DPS
- The updated DPS is approved by Head of HQ Distribution Corporate Planning team before circulation to network planning engineers of all EDs

In all the 3 cases, the following guidelines will be followed regarding update of DPS:

- DPS will be updated According Changes to planning standards and / or guidelines may be issued at any point of time (for example, urgent updates to equipment specifications, WERA guidelines) but they will be reflected in DPS.
- Change requests will be collected by HQ Distribution Corporate Planning team.
- All changes need to be approved by Head of HQ Distribution Corporate Planning team before they can be reflected in the DPS.
- The update will be undertaken by HQ Distribution Corporate Planning team.
- Updated DPS will be approved by Head of SEC Distribution Services and Head of SEC Distribution.
- Updated DPS will be circulated to all planning engineers in EDs and subsequently updated in SEC intranet.

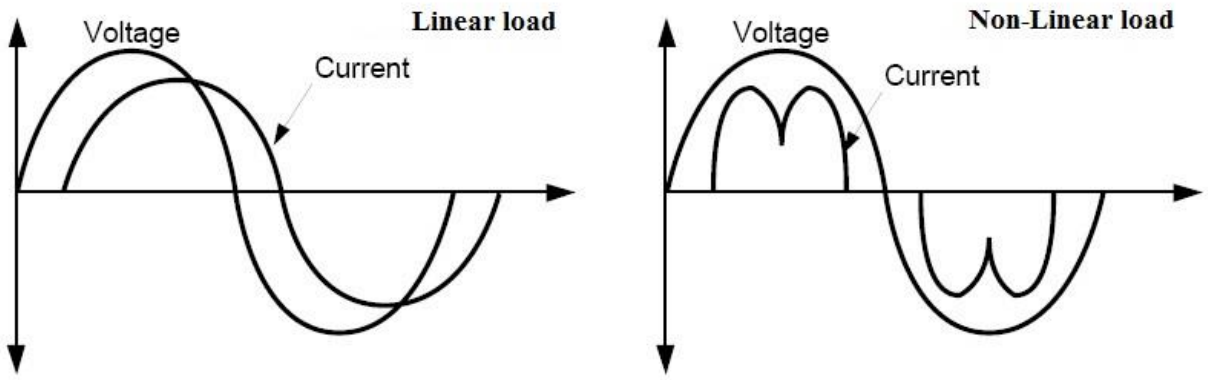
APPENDIX

POWER QUALITY

Harmonics

Harmonics are sinusoidal voltages and currents with frequencies that are integer multiples of the frequency at which the supply system operates.

Harmonic disturbances are generally caused by equipment with non-linear voltage – current characteristics or by periodic and line-synchronized switching of loads. Such equipment may be regarded as sources of harmonic currents.



The harmonic current from the different sources produces harmonic voltage drops across the impedance of the network.

As a result of cable capacitance, line inductance and the power factor correction capacitors, parallel or series resonance may occur in the network and cause a harmonic voltage amplification even at a remote point from the distorting load. The waveforms proposed are the result of the summation of different harmonic orders of one or several harmonic sources

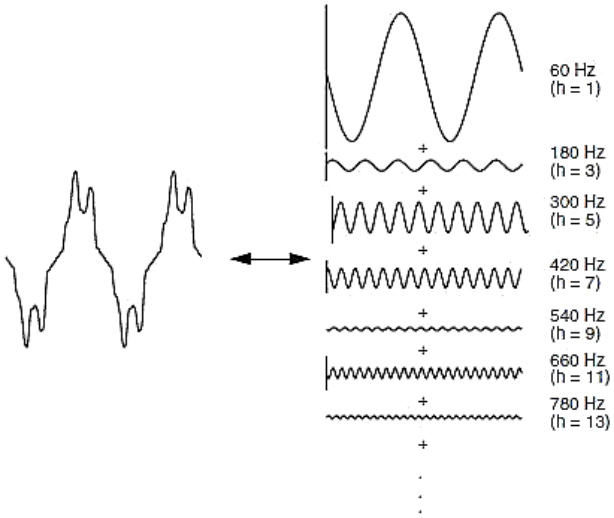


It should be noted that:

- Distortion increases closer to the load.
- Most distortion is periodic.

- Harmonic distortion is caused by non-linear devices in the power system.
- Non-linear loads appear to be sources of harmonic currents in shunt with and injecting harmonic currents into the power system.
- Any periodic, distorted waveform can be expressed as a sum of sinusoids. When a waveform is identical from one cycle to the next, it can be represented as a sum of pure sine waves in which the frequency of each sinusoid is an integer multiple of the fundamental frequency of the distorted wave. This multiple is called a harmonic of the fundamental, hence the name of this subject matter. The sum of sinusoids is referred to as a Fourier series.
- When both the positive and negative half cycles of a waveform have identical shapes, the Fourier series contains only odd harmonics.
- In the presence of harmonic distortion, the power system no longer operates in a sinusoidal condition.
- A distorted waveform in power systems contains only odd harmonics.

Figure 39: Fourier Series Representation of Distorted Waveform



Sources:

Harmonic currents are generated to a small extent by generation, transmission and distribution equipment and to a greater extent by industrial and residential loads.

Power electronics based equipment is a major contributor of harmonics in the power system. These devices and loads can usually be modeled as current sources that inject harmonic currents into the power system. Voltage distortion results as these currents cause nonlinear voltage drops across the system impedance. Harmonic distortion is a growing concern for many customers and for the overall power system, due to increasing application of power electronics equipment.

Significant harmonic currents in a network can be generated by non-linear loads e.g.

- Controlled and uncontrolled rectifiers, especially with capacitive smoothing (for example used in television, indirect and direct static frequency converters, and self-ballasted lamps).
- phase controlled equipment, some types of computers and UPS equipment

Effects:

- The long time exposure to relatively high harmonic distortion conditions may cause some serious effects on the equipment.
- Even very high short term harmonics distortion, e.g. resonance condition, may cause dielectric breakdown due to over voltages.
- Harmonics can lead to overloading. Hence overheating increases dielectric stress and the power loss.
- Capacitors for power factor correction often act as sinks for a particular order of harmonic currents. In this case, it can lead to capacitor over current.
- Non-sinusoidal power supplies result in reduction of torque of induction motors.
- Increase in interference with telephone, communication circuits.
- Can cause errors in reading of induction type energy meters, which are calibrated for pure sinusoidal AC power.
- High order harmonics cause voltage stresses.
- Can cause additional losses.
- Can cause overheating of rotating equipment, transformers and conductors.
- High levels of reactive harmonic current injection may cause abnormal rms voltage or very distorted wave shape.
- Premature failure or operation of protective devices (e.g. relays).
- Power electronics' devices can mis-operate and cause a malfunction of the customer's process.

Limits:

Two facts must be considered. One is that the number of harmonic sources is increasing. The other is that the proportion of purely resistive loads, which function as damping elements, is decreasing in relation to the overall load. Therefore increasing harmonic levels are to be expected in power supply systems until the sources of harmonic emissions are brought under effective limits.

Indicative values of planning levels are shown in section 1.1.3 tables (3-4-5).

Control:

Mitigating harmonics for the network user begins at the disturbance source. The following may be the choice according to the particular circumstances:

- Embedded solution: e.g. PWM (Pulse Width Modulation) technology used in modern power electronic communities.
- A properly sized delta-connected transformer will provide a circulating path for these harmonics, reducing their effect upstream from the transformer (toward the power source

and other loads common to it). Single-phase rectified input switching power supplies are rich in third harmonic current but contain significant higher-order harmonics.

- By special transformer circuitry. One example is the use of a zigzag transformer or a Scott- or T-connected transformer.
- To install harmonic filters close to the harmonic producing loads.

Interharmonics:

Between the harmonics of the power frequency voltage and current, further frequencies can be observed which are not an integer multiple of the fundamental. They can appear as discrete frequencies or as a wide-band spectrum.

Sources:

The main sources are:

- frequency converters
- switch mode power supplies
- adjustable speed drives
- arc welding machines
- arc furnaces
- power supplies to traction systems

Effects:

- Can give rise to flicker.
- Generate additional energy losses.
- Disturb the operation of fluorescent lamps and electronic equipment such as television receivers.
- Risk of unpredictable resonant effects, which can amplify the voltage distortion and lead to overloading or disturbance of equipment.
- Production of acoustic noise.

Harmonic Filter:

Filter: An equipment generally constituted of reactors, capacitors and resistors if required, tuned to present a known impedance over a given frequency range

Tuned Filter: A filter with a tuning frequency, which differs by no more than 10% from the frequency which is to be filtered.

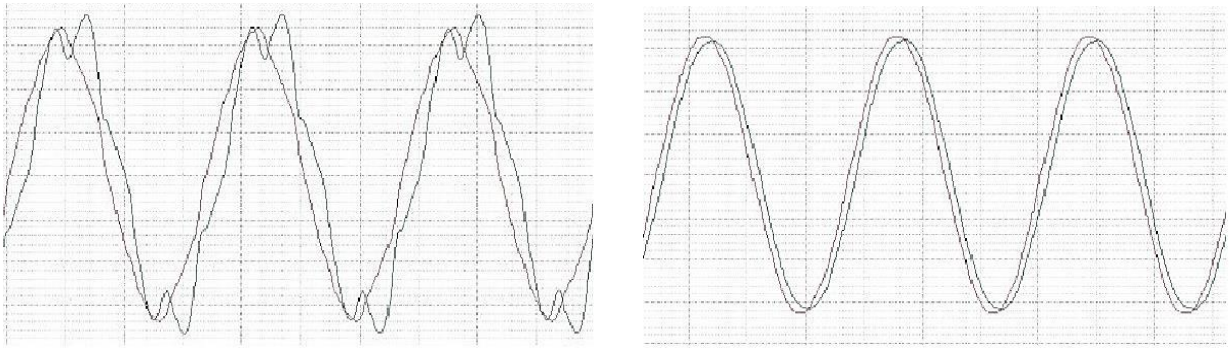
Detuned filter: A filter with a tuning frequency more than 10% below the lowest harmonic frequency with considerable current/voltage amplitude

Filters are composed of resistances R, inductances L and capacitances C selected such that the circuit they form absorbs current at selected harmonic frequencies. This current is thereby prevented from propagating into the network.

The harmonic producing device can generally be viewed as a source of harmonic current. The objective of the harmonic filter is to shunt some of the harmonic current from the load into the filter, thereby reducing the amount of harmonic current that flows into the power system.

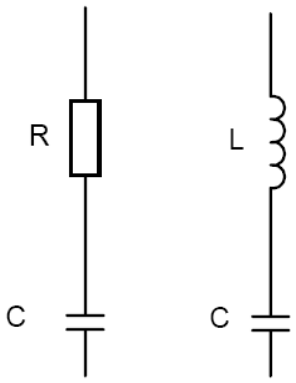
Harmonic filters are designed to control the level of voltage and current distortion generated by all the elements of the equipment to which they are connected, including susceptible equipment, which often generates distortion by itself.

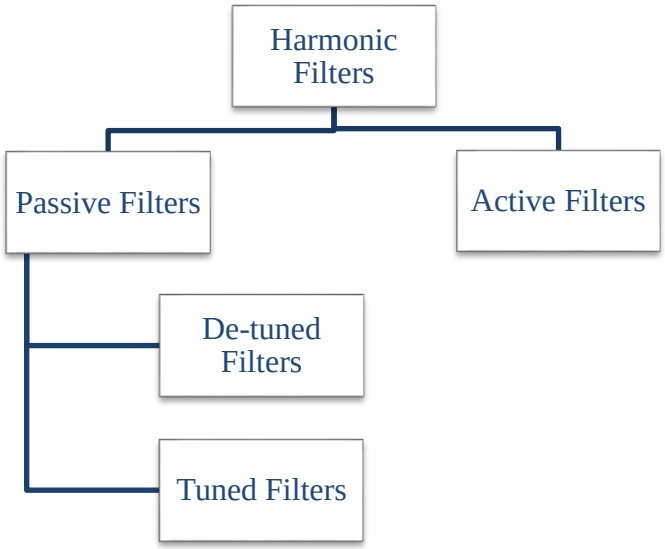
Figure 40: Voltage and Current Waveforms Without Use of Filters



Filters consist of active or passive circuit elements. The simplest type of shunt harmonic filter is a series inductance/capacitance (LC) circuit. More complex harmonic filters may involve multiple LC circuits, some of which may also include a resistor.

Figure 41: 1st and 2nd Order Filters





- Passive Filters: Passive filters are reactor based systems basically used for the suppression of harmonics and maintenance of healthy power factor. This is the preferred industry choice as the equipment acts as the sink for certain harmonic current orders.
- Active Filters: Active filters are IGBT based power electronic devices. Mostly used for harmonic current fluctuating situations, thus the response time is the key factor for characterizing its performance.

The classification of de-tuned filters and tuned filters basically depends on the tuning frequency of the filter reactor & capacitor circuit and the selection of harmonic filter type depends on the level & order of harmonics present in the distribution network. Key design considerations for harmonic filter include the following:

- Reactive power (kVar) requirements
- Harmonic limitations
- Normal system conditions, including ambient harmonics
- Normal harmonic filter conditions
- Contingency system conditions, including ambient harmonics
- Contingency harmonic filter conditions

Harmonic filters may be located at individual devices or at a common bus that feeds many loads. They may be located at low voltage or at medium voltage. The alternatives in a given application should be evaluated based on meeting the acceptable harmonic voltages and currents and the effect of the resulting harmonic load flows on the affected equipment and conductors (e.g., losses, heating).

Harmonic filter is tuned to the desired frequency according to:

$$f_{tuned} = \frac{1}{2\pi\sqrt{LC}}$$

The design of a harmonic filter requires information about the power system and the environment in which the harmonic filter will be installed. In addition to harmonic filtering, the filter equipment will provide the system with capacitive reactive power

that will improve the power factor.

The capacitive reactance for the filter tuned to the *h* harmonic at power frequency is calculated by:

$$X_C = \left(\frac{h^2}{h^2 - 1}\right) X_{eff}$$

Where:

$$X_{eff} = \frac{kV^2}{Q_{eff}}$$

The inductive reactance for the filter at power frequency is calculated by:

$$X_L = \frac{X_C}{h^2}$$

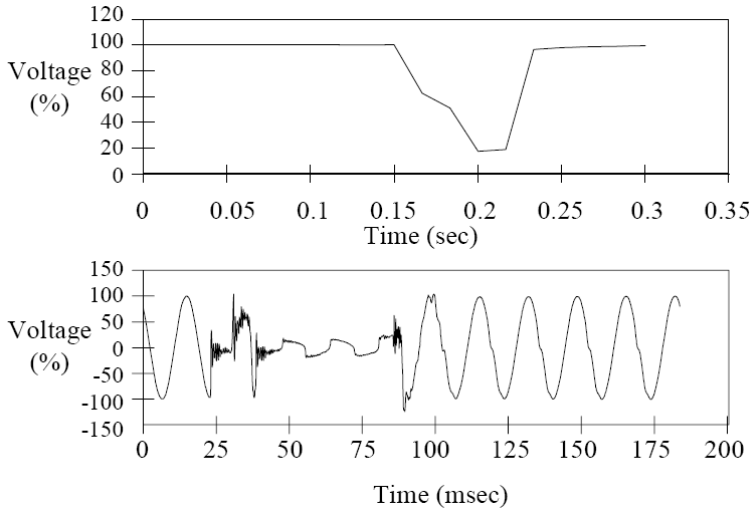
where

X_{eff} effective reactance of the harmonic filter
Q_{eff} effective reactive power (MVar) of the harmonic filter
V nominal system line-to-line voltage

Voltage Dips:

Sudden reduction of the voltage at a particular point on an electricity supply system below a specified dip threshold followed by its recovery after a brief interval. A dip is associated with the occurrence and termination of a short circuit or other extreme current increase on the system or installations connected to it. A voltage dip is a two-dimensional electromagnetic disturbance, the level of which is determined by both voltage and time duration. Duration of voltage dip is the time between the instant at which the voltage at a particular point on an electricity supply system falls below the start threshold and the instant at which it rises to the end threshold. The thresholds adopted are 90% of reference voltage for the start and end of the voltage dip, with durations extending to 01 min.

Voltage dips are unpredictable, largely random events arising mainly from electrical faults on the power supply system or large installations.



Causes:

The primary source of voltage dips observed on the public network is the electrical short circuit occurring at any point on the electricity supply system.

The short circuit causes a very large increase in current, and this, in turn, gives rise to large voltage drops in the impedances of the supply system. Short circuit faults are an unavoidable occurrence on electricity systems. The short circuit can occur between phases, phase and neutral, or phase and earth. Any number of phases can be involved. At the point of the short circuit, the voltage effectively collapses to zero. Simultaneously, at almost every other point on the system the voltage is reduced to the same or, more generally, a less extent. Supply systems are equipped with protective devices to disconnect the short circuit from the source of energy. As soon as that disconnection takes place, there is an immediate recovery of the voltage, approximately to its previous value, at every point except those disconnected. Some faults are self-clearing: the short circuit disappears and the voltage recovers before disconnection can take place. The sudden reduction of voltage, followed by voltage recovery, as just described, is the phenomenon known as voltage dip.

The switching of large loads, energizing of transformers, starting of large motors and the fluctuations of great magnitude that are characteristic of some loads can all produce large changes in current similar in effect to a short circuit current. Although the effect is generally less severe at the point of occurrence, the resulting changes in voltage observed at certain locations can be indistinguishable from those arising from short circuits. In that case they also are categorized as voltage dips.

Unless a self-clearing fault is involved, the duration of voltage dips is governed by the speed of operation of the protective devices.

The magnitude of the voltage dip is governed by the position of the observation point in relation to the site of the short circuit and the source(s) of supply.

Effects:

- Motor drives, including variable speed drives, are particularly susceptible because the load still requires energy that is no longer available except from the inertia of the drive. In processes where several drives are involved, individual motor control units may sense the loss of voltage and shut down the drive at a different voltage level from its peers and at a different rate of deceleration, resulting in complete loss of process control.
- Data processing and control equipment is also very sensitive to voltage dips and can suffer from data loss and extended downtime.
- Voltage dips can cause equipment to perform in a manner other than that which is intended.
- Voltage dips cause a temporary stoppage of the energy flow to the equipment. This leads to a degradation of performance in a manner that varies with the type of equipment involved, possibly extending to a complete cessation of operation.
- Modern manufacturing methods often involve complex continuous processes utilizing many devices acting together. A failure or removal from service of any one device, in response to a voltage dip, can necessitate stopping the entire process, with the consequence of loss of product and damage or serious fouling of equipment.
- AC relays and contactors can drop out when the voltage is reduced below about 80% of nominal for a duration of more than one cycle. The consequences vary with the application, but can be very severe in safety or financial terms.
- Often, the dip is sensed by electronic process controllers equipped with fault-detection circuitry that initiates shutdown of other, less sensitive loads. Additionally, many control systems use relay logic and contactors that can be highly sensitive to dips.
- A slight speed change of induction machinery and a slight reduction in output from a capacitor bank can occur during a dip.
- The visible light output of some lighting devices may be reduced briefly during a dip

Solutions:

As most of these events are caused by circuit faults, improving system operation management skills and constructing robust supply systems are always the fundamental procedure to decrease the effects of these unpredictable events.

There are embedded solutions to improve the sensitive load immunity for riding through these events. Following are the solutions at the end user level, by the application of devices e.g.

- Uninterruptible power supply (UPS)
- Superconducting magnetic energy storage (SMES)
- Dynamic voltage restorer (DVR)
- Static var compensator (SVC)

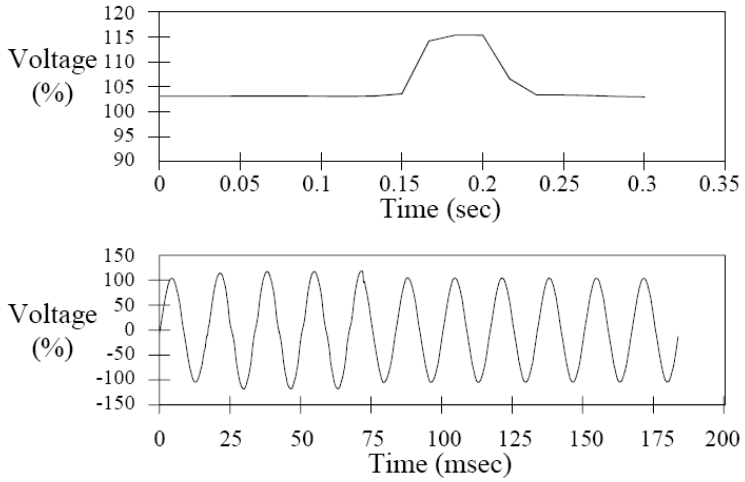
- Ferroresonant transformers (constant voltage transformer) (CVT)
- Magnetic synthesizer
- Active series compensator
- Motor-generator set with flywheel
- Superconductor magnetic energy storage device
- Static transfer switch

Voltage Swells:

Sudden increase of the voltage at a point in an electrical system followed by voltage recovery after a short period, usually from a few cycles to a few seconds. The swell threshold is greater than 110% of reference voltage

Voltage swell phenomenon may occur to be unpredictable and random. Depending upon the magnitude and duration, voltage swell may affect different types of load differently for the same voltage swell event.

Voltage swells are much less common than voltage dips.



Causes:

- Energizing capacitor banks.
- If a resonance condition is created.
- Ferroresonance
- Sudden loss of load on the MV network.

Effects:

An increase in voltage applied to equipment above its nominal rating may cause failure of the components depending upon the magnitude and frequency of occurrence.

- Degradation of IT equipment.
- Reduction in life of filament lamps.

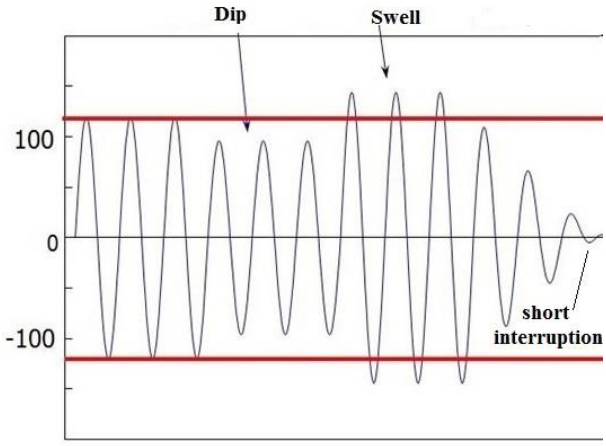
- Electronic devices, including ASDs, computers, and electronic controllers, may show immediate failure modes during these conditions.
- Transformers, cables, busses, switchgear, CTs, PTs and rotating machinery may suffer reduced equipment life over time.
- A temporary increase in voltage on some protective relays may result in unwanted operations while others will not be affected.
- Frequent voltage swells on a capacitor bank can cause the individual cans to bulge while output is increased from the bank.
- Clamping type surge protective devices (e.g. varistors, silicon avalanche diodes) may be destroyed by swells exceeding their maximum continuous operating voltage rating.

Solutions:

As most of these events are caused by circuit faults, improving system operation management skills and constructing robust supply systems are always the fundamental procedure to decrease the effects of these unpredictable events.

There are embedded solutions to improve the sensitive load immunity for riding through these events. Following are the solutions at the end user level, by the application of devices e.g.

- Uninterruptible power supply (UPS)
- Superconducting magnetic energy storage (SMES)
- Dynamic voltage restorer (DVR)
- Static var compensator (SVC)
- Ferroresonant transformers (constant voltage transformer) (CVT)



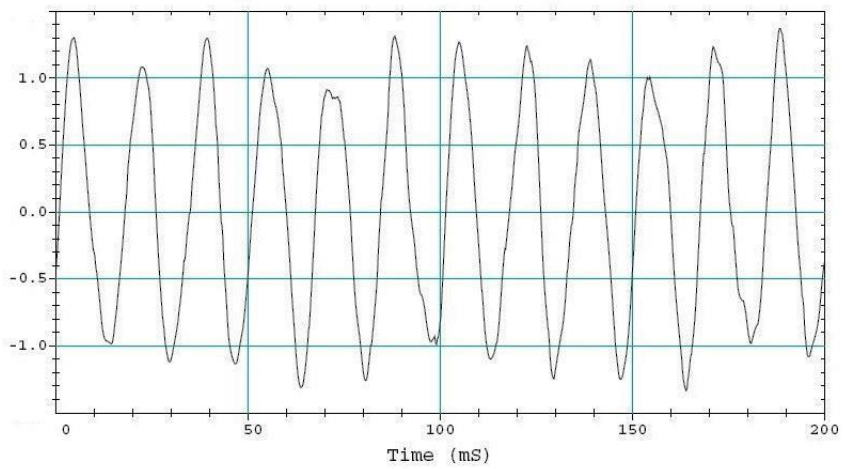
Voltage Fluctuations:

Series of voltage changes or a cyclic variation of the voltage envelope. Voltage fluctuations are produced by fluctuating loads, operation of transformer tap changers and other operational adjustments of the supply system or equipment connected to it. Voltage fluctuations can cause flicker. Voltage fluctuations are normally within 10% magnitude.

For non-repetitive voltage variation, or voltage dips, such as those associated with motor-starting, welding equipment or power system switching, the voltage variation shall not exceed 7% of the fundamental nominal voltage under normal circumstances. Such variations shall not occur more frequently than 3 times per day.

No Customer shall connect equipment to the power system, which causes voltage fluctuation at the Customer interface in excess of these requirements. The SEC shall ensure that the power supply, at each Customer's interface, conforms to these requirements.

Figure 42: Voltage Fluctuations



Sources:

Fluctuations caused by domestic appliances are not generally significant and are mainly produced by:

- continuously but randomly varying large loads such as:
  a. resistance welding machines
  b. rolling mills
  c. large motors with varying loads
  d. arc furnaces
  e. arc welding plant
- single on/off switching of loads (e.g. motors)
- step voltage changes (due to tap voltage changers of transformers)

These industrially-produced fluctuations can affect a large number of customers. Such equipment operates continuously or infrequently.

Effects:

- Degradation of performances in equipment using storage devices (e.g. capacitors)
- loss of function in control systems
- instability of internal voltages and currents in equipment
- increased ripple

The main disadvantage is flicker. Additionally, voltage fluctuations can cause a number of harmful technical effects such as data errors, memory loss, equipment shutdown, motors stalling and reduced motor life, resulting in disruption to production processes and substantial costs. However considering the fact that voltage fluctuations are normally within 10% magnitude, most of these above mentioned effects are more typical of voltage dips or swells.

Rapid Voltage Changes:

It is expressed by the relative steady-state voltage change and/or by a maximum relative r.m.s. voltage change aggregated over several cycles. Rapid voltage changes even within the normal operational voltage tolerances are considered as a disturbing phenomenon. Individual customers' installations should not produce significant voltage variations even if they are tolerable from the flicker point of view. Rapid voltage changes are often caused by start-ups, inrush currents or switching operation of equipment.

Limit for LV Customers:

Under normal circumstances, the value of rapid voltage changes is limited to 3% of nominal supply voltage. However, rapid voltage changes exceeding 3% can occur infrequently on the public supply network.

Limit for MV Customers:

No. of changes n	Rapid voltage changes
$n \leq 4$ per day	5%
$n \leq 2$ per hour and > 4 per day	4%
$2 < n \leq 10$ per hour	3%

Flicker:

Periodic fluctuations in voltage, at fluctuation frequencies below the fundamental frequency. These are generally expressed as percentage variations, relative to the fundamental voltage.

Voltage fluctuation cause changes of the luminance of lamps which can create the visual phenomenon called flicker. Above a certain threshold, flicker becomes annoying. The annoyance grows very rapidly with the amplitude of the fluctuation. At certain repetition rates, even very small amplitudes can be annoying.

Intensity of flicker annoyance, flicker severity is calculated with respect to both short and long term effects.

The short term severity level, denoted by Pst, is determined for a 10-minute period. The long-term severity level, denoted by P_{lt} , is calculated for a two-hour period.

The severity of flicker can be measured with a flicker meter.

Flicker is considered to be an annoying problem for the customers. Most of the time, it does not have a high financial impact. However, at high levels it can cause inconvenience to people when frequent flickering of lights and computer screens occurs at their work places or homes

Limits for LV:

Short-term: $P_{st} = 1.0$
Long-term: $P_{lt} = 0.8$

Limits for MV:

short-term: $P_{st} = 0.9$
long-term: $P_{lt} = 0.7$

Solutions:

Mitigating flicker for the network user begins at the disturbance source. It is always accomplished by controlling fluctuating power drawn by the disturbance load, e.g., electric arc furnace and elevator.

- Use of higher voltage level supply as agreed between system operators and end users
- Static Var compensators (SVC)
- Static synchronous compensators (STATCOM) or Static Var generation (SVG)

In cases where SVCs, STATCOMs, or SVGs are used, response time is the key factor for characterizing its performance

Power Quality Measurement and Monitoring:

The following measuring instruments may be used and selected as per the specific objectives of the analysis:

- Power analyzer
- Flicker meter
- Event indicator
- Oscilloscope
- Power quality monitor
- Spectrum analyzer

Devices such as digital fault recorders, energy meters, protection relays may include power quality functions.

Harmonics' measurements shall be made at least up to the 50th order.

Power quality monitoring is necessary to characterize electromagnetic phenomena at a particular location. The objective may be as simple as verifying steady state voltage regulation at a service entrance, or may be as complex as analyzing the harmonic current flows within a distribution network.

The primary reason to monitor power quality is economic, particularly if critical process loads are being adversely affected by electromagnetic phenomena. Effects on equipment and process operations can include mis-operation, damage, process disruption, and other such anomalies. Such disruptions are costly since a profit-based operation is interrupted unexpectedly and must be restored to continue production. In addition, equipment damage and subsequent repair cost both money and time. Product damage can also result from electromagnetic phenomena requiring that the damaged product be either recycled or discarded, both of which are economic issues. Equipment compatibility problems can create safety hazards resulting from equipment mis-operation or failure.

Problems related to equipment mis-operation can be assessed if customer disturbance reports are kept. These logs describe the event inside the facility: what equipment was affected, how it was affected, what were the weather conditions, and what losses were incurred. A sample form is outlined below:

Figure 43: Sample Power Quality Disturbance Recording Form

Date:		Time:	
Customer:			
Address:			
Description of Disturbance:			

Sequence of events:

Time of disturbance	Description of disturbance

Restoration events:

Restoration time	Equipment back to Service

Equipment affected:

Equipment type	Manufacturer	Equipment rating

At time of submission of application for new bulk MV connection, the single line diagram shall also illustrate the arrangement for power quality, for SEC's comments and approval.

National Grid has is now installing Advanced Metering Infrastructure (AMI) at its HV/MV grid stations; (specification no.40-TMSS-03 "AMI"). That system also includes a power quality meter. So one will be able to measure the parameters at grid end also.

The following customers should be treated as priority:-

DRAFT

- Steel arc furnaces
- Steel re-rolling mills
- Arc welding / spot welding

The following roadmap can be adopted:-

- In 1st phase, power quality parameters to be measured and recorded at interface with bulk MV customers.
- In 2nd phase, power quality parameters to be measured and recorded at interface with bulk LV customers.
- In next phase, power quality parameters to be measured and recorded at interface with random LV customers.

Simultaneously, there is need to create awareness among the customers. e.g. reduction in harmonics is also financially beneficial for the customer itself.

APPENDIX 1

Table 1: Facility Category C1: Loads Of Residential Buildings -220 V Phase To Phase

Circuit Breaker Rating(AMP)	Total Connected (KVA)Load	Constructed Area Building of (m^2)
250	91	801
	93	825
	96	850
	98	875
	101	900
	104	925
	106	950
	109	975
	112	1000
300	113	1001
	114	1025
	117	1050
	120	1075
	122	1100
	125	1125
	128	1150
	130	1175
	133	1200
400	134	1201
	144	1300
	154	1400
	166	1500
500	167	1501
	176	1600
	186	1700
	197	1800
	208	1900
	218	2000
600	219	2001
	229	2100
	240	2200
	250	2300
	261	2400
	272	2500
800	273	2501
	283	2600
	293	2700
	304	2800
	315	2900
	325	3000
	346	3200
	367	3400

Circuit Breaker Rating(AMP)	Total Connected (KVA)Load	Constructed Area Building of (m^2)
30	3	25
	6	50
	10	75
	13	100
40	14	101
	15	110
	16	125
50	17	126
	19	150
	22	175
	26	200
70	27	201
	29	225
	32	250
	34	275
	37	300
100	38	301
	40	325
	42	350
	45	375
125	46	376
	48	400
	50	425
	53	450
	54	460
	56	475
150	57	476
	58	500
	61	525
	64	550
	66	575
	69	600
200	70	601
	72	625
	74	650
	77	675
	80	700
	82	725
	85	750
	88	775
	90	800

Table 2: Facility Category C1: Loads Of Residential Buildings - 400/230 V

Circuit Breaker Rating(AMP)	Total Connected (KVA)Load	Constructed Area Building of (m^2)
150	120	1075
	122	1100
	125	1125
200	126	1126
	128	1150
	130	1175
	133	1200
	144	1300
	154	1400
	166	1500
250	167	1501
	176	1600
	186	1700
	197	1800
	208	1900
300	209	1901
	218	2000
	229	2100
	240	2200
	250	2300
400	251	2301
	261	2400
	272	2500
	283	2600
	293	2700
	304	2800
	315	2900
	325	3000
500	326	3001
	346	3200
	367	3400
	376	3500
	389	3600
600	390	3601
	410	3800
	432	4000
	454	4200
	475	4400
800	476	4401
	497	4600
	518	4800
	539	5000
	560	5200
	581	5400
	602	5600
	623	5800
	648	6000

Circuit Breaker Rating(AMP)	Total Connected (KVA)Load	Constructed Area Building of (m^2)
20	3	25
	6	50
	10	75
	13	100
	16	125
30	17	126
	19	150
	22	175
40	23	176
	26	200
	29	225
	32	250
50	33	251
	34	275
	37	300
	40	325
70	41	326
	42	350
	45	375
	48	400
	50	425
	53	450
	56	475
100	57	476
	58	500
	61	525
	64	550
	66	575
	69	600
	72	625
	74	650
	77	675
	80	700
	82	725
125	83	726
	85	750
	88	775
	90	800
	93	825
	96	850
	98	875
	101	900
150	102	901
	104	925
	106	950
	109	975
	112	1000
	114	1025
	117	1050

Table 3: Facility Category C2: Loads Of Commercial Buildings - 220 V Phase To Phase

Circuit Breaker Rating(AMP)	Total Connected (KVA)Load	Constructed Area Building of (m^2)
300	108	626
	111	650
	115	675
	120	700
	124	725
	128	750
400	129	751
	133	775
	137	800
	146	850
	154	900
	162	950
	166	1000
500	167	1001
	179	1050
	188	1100
	197	1150
	205	1200
	214	1250
600	215	1251
	222	1300
	239	1400
	256	1500
800	257	1501
	274	1600
	290	1700
	307	1800
	342	2000
	358	2100
	363	2200

Circuit Breaker Rating(AMP)	Total Connected (KVA)Load	Constructed Area Building of (m^2)
30	5	25
	8	50
	10	55
	13	75
50	14	76
	18	100
	22	125
70	23	126
	26	150
	30	175
	34	200
100	35	201
	38	225
	43	250
150	44	251
	47	275
	51	300
	56	325
	60	350
	64	375
200	65	376
	69	400
	73	425
	77	450
	82	475
	86	500
250	87	501
	90	525
	94	550
	98	575
	102	600
	107	625

Table 4: Facility Category C2: Loads Of Commercial Buildings - 400/230 V

Circuit Breaker Rating(AMP)	Total Connected (KVA)Load	Constructed Area Building of (m^2)
250	167	1001
	175	1025
	179	1050
	184	1075
	192	1125
300	193	1126
	205	1200
	222	1300
	239	1400
400	240	1401
	256	1500
	274	1600
	290	1700
	307	1800
	316	1850
500	317	1851
	342	2000
	358	2100
	375	2200
	393	2300
600	394	2301
	410	2400
	426	2500
	444	2600
	461	2700
800	462	2701
	478	2800
	495	2900
	512	3000
	529	3100
	546	3200
	563	3300
	580	3400
	597	3500
	614	3600
	631	3700
	640	3800

Circuit Breaker Rating(AMP)	Total Connected (KVA)Load	Constructed Area Building of (m^2)
20	5	25
	8	50
	13	75
30	14	76
	18	100
	22	125
40	23	126
	26	150
	30	175
50	31	176
	34	200
	38	225
70	39	226
	43	250
	47	275
	51	300
	56	325
100	57	326
	60	350
	64	375
	69	400
	73	425
	77	450
	82	475
125	83	476
	86	500
	90	525
	94	550
	98	575
	102	600
150	103	601
	107	625
	111	650
	115	675
	120	700
	124	725
200	125	726
	128	750
	133	775
	137	800
	141	825
	146	850
	150	875
	154	900
	158	925
	159	926
	162	950
	166	1000

Table 5: Individual equipment demand factors

S/N	Type of Load	Demand Factors Used by SEC

		Residential	Commercial	Industrial	Agr. Farms
1	Central A/Cs	0.9	0.9	0.9	0.9
2	Window Type A/Cs	0.6	0.6	0.7	0.7
3	Lighting (Interior / Exterior)	1.0	1.0	1.0	1.0
4	Refrigeration / Cooling	0.6	0.6	0.6	0.6
5	Fans / Blowers	0.2	0.2	0.2	0.2
6	Equipment Used in Kitchens	0.2	0.2	0.2	-
7	Water Heaters	0.2	0.2	0.2	-
8	Laundry Equipment	0.2	0.2	0.2	-
9	Appliances Used for Recreation	0.2	0.2	-	-
10	Appliances Used for Services	0.2	0.2	0.2	-
11	Equipment Used in Office / Labs	-	0.2	0.2	-
12	Welding Equipment	-	0.15	0.20	-
13	Electric Motors Used for Crafts, Workshops & Service Centers	-	0.25	0.25	-
14	Electric Motors Used for Batch Work, Fluctuating of Multiple Production	-	-	0.4	0.4
15	Electric Motors Used for Continuous Process and Mass Production	-	-	0.6	-
16	Process Heating Using Ovens	-	-	0.35	-
17	Process Heating Using Furnaces	-	-	0.7	-
18	Miscellaneous (not covered above)	0.1	0.1	0.1	0.1

Demand factors are based on IEEE STD 241-1974 and Electric Utility Engineering Reference Book by Westinghouse.

APPENDIX 2

1. Example on applying correction factor

Example (1):

The Continuous Current Ratings for cable 4x300mm² Al LV is 360 A at Laying Conditions , buried under 0.8 m with soil thermal resistivity of 1.5 °C.m/w and ground temperature of 30 degrees Celsius. What is the rating of the cable according to SEC standard condition?

The formula for calculating the corrected cable rating is:

$$\begin{aligned} &Corrected\ Cable\ Rating \\ &= Cable\ Rating \times Burial\ Depth\ Correction\ Factor \\ &\times Soil\ Thermal\ Resistivity\ Correction\ Factor \\ &\times Ground\ Temperature\ Correction\ Factor \end{aligned}$$

As per the relevant SEC standard condition table 8 and correction factors in tables 9, 10 and 11, the following values are to be used:

Burial depth correction factor = 1.02

Soil thermal resistivity correction factor = 0.88

Ground temperature correction factor= 0.96

The corrected rating is: 360 x 1.02 x 0.88 x 0.96 = 310 A.

2. Example of load estimation

Example (1):

Calculate building area of a residential plot of raw area 600 m² and building percentage 60%, consists of (3)floors and roof of area (40% of the floor area).

Building area for the individual floor (m²) = area of the individual plot × floors building percentage = 600 × 60% = 360 m²

Roof area (m²) = surface area × attachment roof building percentage = 360 × 40% = 144 m²

Plot building area (m²) = Building area for the individual floor × number of floors + roof area = 360 × 3 + 144 = 1224 m²

Example (2):

Calculate building CL for a mosque of building area 2000 m²

CL for the individual unit (KVA) = building area of the individual unit (m²) × load density factor (VA/m2) ÷ 1000

From the table 21 load density factor for the mosque (C9) = 148 ((VA/m²)

= (2000 × 148) ÷ 1000 = 296 KVA

Example (3):

Calculate connected load of normal residential building type C1 with covered area of 200 m² and height of 5 meters connected at 230/400V

For a normal residential building (type C1), the connected load is 26 KVA

However, the height of the building is more than standard height (which is 3.5m). Hence, there will be additional cost of AC cooling, which is mentioned in Section 2.4.5

Additional volume (m³) = [Total Height (m) - Standard Height (3.5 m)] X Covered Area (m²)

= (5m-3.5m) x 200m² = 300 m²

Additional AC Load (VA) =(24 VA/m³) × Additional volume (m³)

= 24 x 300 = 720 VA = 7.2 KVA

So, total connected load of the building is 34.2 KVA (26 + 7.2)

Example (4):

Calculate connected load of commercial building type C1 with covered area of 5000m²
This is a normal commercial building (type C2) whose area exceeds the limits given in Table Hence, the formula will be applied.

$$Total\ Connected\ Load\ (KVA) = Built\ up\ Area\ (m^2) \times Load\ Density\ (VA/m^2) / 1000$$

By using the load density for Commercial Customers = 172 VA/m²
Total Connected Load = 5000 x 172 = 860 KVA

Example (5)

Calculate connected load of normal residential building type C1 on 400/230 V with covered area of 200m² with central AC. The declared AC load by customer is 24 KVA

From the tables it can be ascertained that the connected load for a residential building is 26 KVA

AC load is estimated using the formula as 200m² x 81 VA/m² = 16.2 KVA

This gives the non-AC load to be 9.8 KVA

In this case, the declared AC load by customer is higher than the calculated AC load and therefore, will be used for calculations. However, if the declared load is missing or less than 16.2 KVA, the figure of 16.2 KVA will be used
Demand load is calculated using the following formula:

Connected Load = (Non-AC Connected Load + Central AC Load
= 9.8 + 24 = 33.5 KVA
Connected load of 33.5 as per Table, this gives the circuit breaker rating as 50A

3. EXAMPLES for CDL CALCULATION

Example (1):

Calculate CDL for residential plot of raw area 600 m² and building percentage 60% consists of three floors, each floor contains two units and plot has one roof unit (40% of the floor area).
Building area for the individual floor (m²) = area of the individual plot × floors building percentage = 600 × 60% = 360 m²
roof area (m²) = surface area × roof building percentage = 360 × 40% = 144 m²
Plot building area (m²) = Individual floor building area × number of floors + roof building area = 360 × 3 + 144 = 1224 m²
Individual unit building area in the floors (m2) = Floors building area ÷ number of units = 360 × 3/6 = 180 m²
Attachment unit building area (m2) = Attachment floor building area ÷ number of units = 144 ÷ 1 = 144 m²
From relevant tables related to load estimation circuit breaker rating is determined for the area of each unit
Circuit breaker rating for the residential unit of area (180) m² = 40 A
Circuit breaker rating for the roof of area (144) m² = 30 A
Total circuit breaker ratings for all units = 6 × 40 + 30 = 270 A

$$CDL = \left(\sum_{i=1}^{N}(CBR_i \times DF_i)\right) \times CF(N)$$

CDL = [(270 × 0.5) + × 0.636 = 85.86 A

Example (2):

Calculate CDL for a plot allocated for a school of raw area 5000 m² and building percentage 60%, consists of two floors.

Individual floor building area (m²) = individual plot area × floors building percentage

= 5000 × 60% = 3000 m²

Plot building area (m²) = Individual floor building area × (number of floors)

= 3000 × 2 = 6000 m²

CL for the individual unit (KVA)

= Individual unit building area (m2) × load density factor (VA/m²) ÷ 1000

From the relevant table, load density factor for the school C8 = 144 VA/m2

= (6000 × 144) ÷ 1000 = 864 KVA

CL 864 KVA (1247) A is greater than (800 A)

CDL is calculated from

$$CDL = \left(\sum_{i=1}^{N}(Cl_i \times DF_i)\right) \times CF(N)$$

CDL = 864 × 0.7 × 1 = 604.8 KVA

Example (3):

Calculate CDL for a plot consists of 16 residential units with CB rating 40 A for each unit on 230/400 V.

CDL is calculated from the equation:

$$CDL_{on\ Network\ Element} = \left(\sum_{i=1}^{N}(CBR_i \times DF_i)\right) \times CF(N)$$

= 40 × 16 × 0.5 × 0.602 × 192.64 A = 133.46 KVA

Example (4):

Calculate CDL for a plot consists of 16 units (10 unit residential with CB rating 40 A for each unit and 6 unit Commercial with CB rating 50 A for each unit) on 230/400 V.

CDL is calculated from the equation:

$$CDL_{on\ Network\ Element} = \left(\sum_{i=1}^{N}(CBR_i \times DF_i)\right) \times CF(N)$$

= ((40 × 10 × 0.5) + (50 × 6 × 0.6)) 0.602 = 228.76 A = 158.5 KVA

Example (5):

Calculate CDL for a plot consists of 5 residential units (1 unit with CB rating 400 A and 4 unit with CB rating 70 for each one) on 230/400 V.

CDL is calculated from the equation:

$$CDL = \left[CBR_{Largest\ \text{Circuit Breaker}} \times DF_{Largest\ \text{Circuit Breaker}}\right] + \left[\sum_{i=1}^{N-1} CBR_i \times DF_i \times CF(N-1)\right]$$

= 400 × 0.5 + (4×70×0.5×0.668) = 293.5 A = 203.3 KVA

Example (6):

Calculate CDL for a plot consists of 5 residential units (2 unit with CB rating 400 A and 3 unit with CB rating 70 for each one) on 230/400 V.

CDL is calculated from the equation:

$$CDL = \left[\sum_{i=1}^{g\ number\ of\ largest\ \text{CB}} CBR_i \times DF_i\right] CF(g) + \left[\sum_{g+1}^{N} CBR_i \times DF_i \times CF(N-g)\right]$$

= (2×400 × 0.5 × 0.723)+ (3×70×0.5×0.688) = 361.4 A = 250.4 KVA

Example (7):

Calculate CDL for a plot consists of 5 residential units (unit CB rating 400 A + unit CB rating 300 A + 3 unit with CB rating 70 for each one) on 230/400 V.

CDL is calculated from the equation:

$$CDL = \left[\sum_{i=1}^{g\ number\ of\ largest\ \text{CB}} CBR_i \times DF_i\right] CF(g) + \left[\sum_{g+1}^{N} CBR_i \times DF_i \times CF(N-g)\right]$$

= (400 × 0.5) + (3×70×0.5+ 300× 0.5) ×0.668 = 370.34 A = 265.6KVA

4. Examples of Voltage Drop Calculation

Example (1)

The customer CDL is 150KVA on 230/ 400 V. The connection is directly from Distribution Substation through 300mm² cable of length 75m. What is the voltage drop?

The K-value for 300mm² cable is 10132

The simplified formula for voltage drop calculation is:

$$VD_{\%} = \frac{KVA \times L}{K}$$

VD for cable = 150 x 75 / 10132 = 1.11% (within allowed limit)

Example (2)

The customer CDL is 170 KVA on 230/ 400 V. The connection is directly from Distribution Substation through 300mm² cable of length 320m. What is the voltage drop?

The K-value for 300mm² cable is 10132

The simplified formula for voltage drop calculation is:

$$VD_{\%} = \frac{KVA \times L}{K}$$

VD for cable = 170x 320 / 10132 = 5.36% (which is above the allowed limit)

Example (3)

The customer CDL is 50 KVA on 230/ 400 V. The connection is from Distribution Substation to DP (through 300mm² cable) of length 80m and from DP to customer service cable of 70 mm² of length 40m. What is the voltage drop?

The K-value for 300mm² cable is 10132

The K-value for 70mm² cable is 3003

The simplified formula for voltage drop calculation is:

$$VD_{\%} = \frac{KVA \times L}{K}$$

Firm capacity of cable 300 mm² is 248 A = 171.8 KVA

VD for LV main feeder = 171.8x80/10132 =1.36%

VD for service cable = 50x40/3003 = 0.67%

Total VD = 2.026% (within allowed limit)

Example (4)

The customer CDL is 100 KVA on 230/ 400 V. The connection is from Distribution Substation to DP (through 300mm² cable) of length 250 m and from DP to customer service cable of 185 mm² of length 80 m. What is the voltage drop?

The K-value for 300mm² cable is 10132

The K-value for 185 mm² cable is 7040

The simplified formula for voltage drop calculation is:

$$VD_{\%} = \frac{KVA \times L}{K}$$

Firm capacity of cable 300 mm² is 248 A = 171.8 KVA

VD for LV main feeder = 171.8x250/10132 = 4.24%

VD for service cable = 100x80/7040 = 1.14 %

Total VD = 5.38% In this example, the VD to customer is higher than limit of 5%. Hence, this connection design has to be changed

Example (5)

CDL Customer of load 50 KVA, connected at 230/ 400V and is connected using 120mm² main feeder (of 80m) and 50 mm² service drop (of 45m). What is the voltage drop for each element?

The K –value for 120mm² conductor is 5064

The K –value for 50 mm² conductor is 2217

The formula for Voltage Drop Calculation is

$$VD_{\%} = \frac{KVA \times L}{K}$$

VD for main LV feeder = 110x80/4165 = 2.11%

VD for service drop = 110x45/4165= 1.19%

The total VD is 3.30%, which is also within the limit of 5%

5. Examples of Underground LV Connection Design

Example (1):

Calculate CDL and suitable supply method for a plot consists of 16 residential units with CB rating 50 A for each unit through underground network on 230/ 400 V

CDL is calculated from the equation:

$$CDL_{on\ Network\ Element} = \left(\sum_{i=1}^{N}(CBR_i \times DF_i)\right) \times CF(N)$$

= 30 × 16 × 0.5 × 0.602 × = 144.5 A = 100 KVA

Suitable supply from DP, main LV feeder will be 300mm² with cable of 185mm² LV cable to customer

Example (2):

Calculate the loading percentage on an aluminum feeder of size 4 × 300 mm^2 to supply CDL 160 KVA

Loading percentage is calculated from the equation:

$$Loading\ \%_{on\ Network\ Element} = \frac{CDL_{\ on\ Network\ Element}}{Rating_{of\ Network\ Element}} \times 100$$

Rating of an aluminum cable of size 4 × 300m m^2 = 215 KVA

Loading % = (160 ÷ 215) ×100 = 74% (within allowed limit)

Example (3):

What is the rating of the distribution substation required to supply one commercial unit with covered building area 5000 m^2. on 400/230 V of building area 8000 m^2.

CL for the individual unit (KVA) = Individual unit building area (m^2) × load density factor (VA/m^2) ÷ 1000

Commercial unit area 5000 m^2 is out of the area tables, therefore, calculation will depend on the load density factor from the Chapter 5 for commercial facilities (C2) = 172 (VA/m^2)

CL for the individual unit (KVA)

= (5000×172) ÷ 1000 = 860 KVA (1241) A is greater than (800 A)
Supply will be from a private distribution substation of rating 1000 KVA and loading 86%

Example (4):

What is the rating of the MV Switchgear required to supply a commercial mall with CL 12000 KVA on 13.8 KV:

$$CDL = \left(\sum_{i=1}^{N}(CBR_i \times DF_i)\right) \times CF(N)$$

= 12000 × 0.6 × 1= 7200 KVA

CDL in Ampere = 301 A

Required MV Switchgear rating is 400 A and loading percentage 75%

Example (5):

What is the suitable rating of the distribution substation to supply three residential plots, each plot will be supplied by 10 CBs, rating of each CB is 100 A. on 230/ 400 V

Total CB ratings for the individual plot = 10 ×100 = 1000 A

Total CB ratings for the three plots = 1000 × 3 = 3000 A

CDL is calculated from the equation:

$$CDL_{on\ Network\ Element} = \left(\sum_{i=1}^{N}(CBR_i \times DF_i)\right) \times CF(N)$$

= 3000 × 0.5 × 0.598 = 897 A, which corresponds to CDL of 621 KVA

Suitable rating of the public distribution substation to supply them is 1000 KVA and the loading percentage 62 %.

Example (6)

The connected load for customer is 800 KVA on 230/ 400 V. The customer is a commercial customer with a single meter. Determine the connection design.

Coincident Demand Load = 800 x 0.6 = 480 KVA.

Supply using dedicated distribution substation 500 KVA .

6. Examples of LV Overhead Connection Design

Example (1):

Calculate CDL and suitable supply method for a plot consists of 10 residential units with CB rating 40 A for each unit through overhead network at 230/400V.

CDL is calculated from the equation:

$$CDL_{on\ Network\ Element} = \left(\sum_{i=1}^{N}(CBR_i \times DF_i)\right) \times CF(N)$$

= 50 × 10 × 0.5 × 0.619 × = 155 A = 107 KVA

This can be supplied from PMT of 200KVA with conductors of 120mm² or using direct underground feeder of 185mm².

Example (2):

Calculate the loading percentage on an aluminum conductor of size 120 mm^2 to supply CDL 110 KVA at 230/ 400V.

Loading percentage is calculated from the equation:

$$Loading\ \%_{on\ Network\ Element} = \frac{CDL_{\ on\ Network\ Element}}{Rating_{of\ Network\ Element}} \times 100$$

Rating of an aluminum cable of size 120 mm^2 = 139 KVA

Loading % = (110 ÷ 139) ×100 = 79%

Example (3)

Determine the connection design for fuel stations load is circuit breaker size of 400A at 230/400V (overhead).

CDL=CBR×D.F×CF = 400×0.6×1=240 A

This can be supplied from PMT of 200KVA with direct underground feeder of 300mm. AL.

Example (5)

A customer applied for electricity conduction for residential building consisting of 8 units. The Built-up area of each unit is 300 m². The nearest electricity transformer (200 KVA & maximum demand 100 A) at a distance of 200 m with voltage (230V/400V).
Firstly, the load is calculated for the customers using the following steps:

1. Load calculation of the customer is made according to the built-up area of the building:

Built-up area m^2	Circuit breaker (A)
300	*50*

2. $CDL = (\sum_{i=1}^{N}(CBR_i \times DF_i)) \times CF(N)$

= 50×8×0.5×0.629 = 126 A= 87 (KVA)

Secondly, study of electricity supply needs to be conducted:

1. Calculation of supply possibility from the transformer using the

CDL for transformer and new customer = 126+100= 226 A= 156 KVA

$$Loading\ \%_{on\ PMT} = \frac{CDL\ (KVA)_{on\ PMT}}{Rating_{PMT}} \times 100$$

= (156/200)×100= 78%

2. Voltage drop is estimated using the following formula:

V.D.% = $\frac{KVA*L}{K}$

Supply source: Direct feeder from PMT through LV.Aluminum conductor of size 120 mm^2
VD % = 87 x 200 / 5064 = 3.43%
The voltage drop percentage should not exceeds 5%, alternate supply options should be considered.

7. EXAMPLES FOR Calculation Voltage Drop (M V NETWORK)

Example (1)

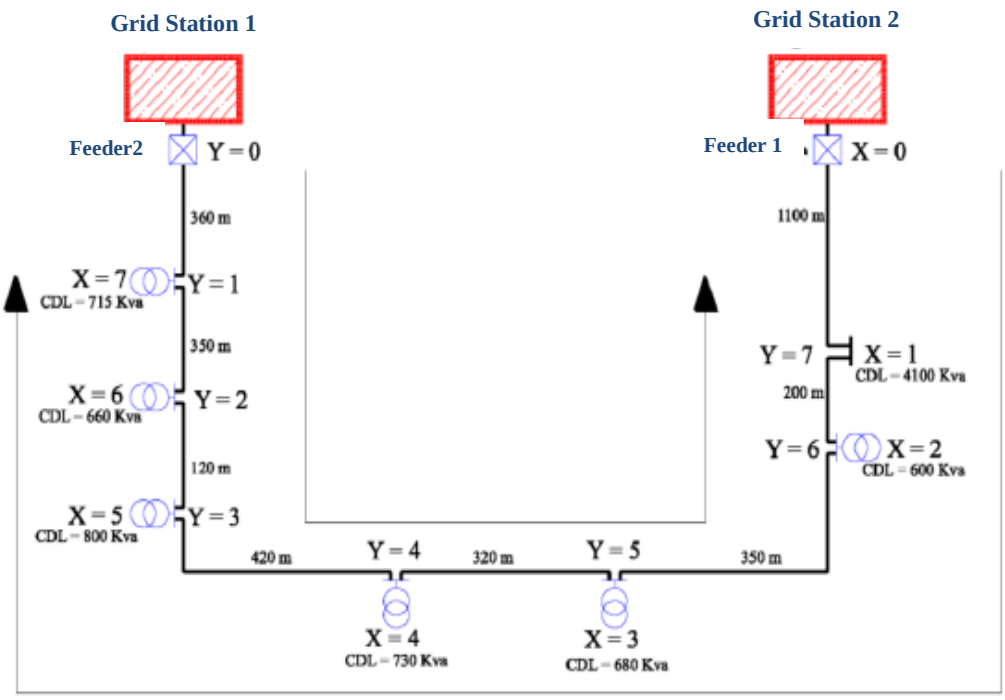
What is the voltage drop for underground feeder of length 6 km with a load of 5MVA and voltage of 13.8 kV. The conductor type is (3X500mm²). AL
Voltage drop is calculated using the following formula:

$$\textbf{V.D.\%} = \frac{KVA*L}{K}$$

VD % = 5000 x 6 / 15252 = 1.9 %

Example (2):

Calculate CDL on each segment of MV cable between two substations in the single loop according to the following figure:



- CDL is calculated on each segment of MV cable between two substations according to the following equation:

$$CDL_{(x,x+1)} = \sum_{i=x+1}^{N} CDL_i \times CF_{For\ Substations}$$

First: CDL is calculated for each segment in Feeder 1 from the beginning of the feeder (X=0) to the last substation in the loop (X=7).

CDL on the segment (X=0, X=1)

= (4100 + 600 + 680 + 730 + 800 + + 660 + 715) × 0.9 = 7456 KVA

CDL for the rest of the segment is according to the table below:

Segment	(0,1)	(1,2)	(2,3)	(3,4)	(4,5)	(5,6)	(6,7)
CDL	7456	3767	3227	2615	1958	1238	644

Second: CDL is calculated for each segment in Feeder 2 from the beginning of the feeder (Y=0) to the last substation in the loop (Y=7)

CDL on the segment (Y=0, Y=1)

= (715 + 660 + 800 + 730 + 680 + + 600 + 4100) × 0.9 = 7456 KVA

CDL for the rest of the segment is according to the table below:

Segment	(0,1)	(1,2)	(2,3)	(3,4)	(4,5)	(5,6)	(6,7)
CDL	7456	6813	6219	5499	4842	4230	3690

Example (3):

Calculate Voltage Drop percentage on each segment of MV cable between two substations in case of using cable size 3×500mm^2 Al, 13.8 KV for the same loop in Example (2).

- VD % is calculated for each segment according to the following equation:

$$VD\ \%_{(x,x+1)} = \frac{CDL\ (KVA)_{(x,x+1)} \times L_{(x,x+1)}}{K_{MV\ Cable}}$$

First: VD % is calculated for each segment in Feeder 1 from the beginning of the feeder (X=0) to the lase substation in the loop (X=7)

K factor for the aluminum cable size 3×500mm^2, 13.8 KV = 15252

VD % is calculated on segment (0,1) = (7456 KVA × 1.1 km) ÷ 15252 = 0.54%

VD for the rest of the segment is according to the table below:

Segment	(0,1)	(1,2)	(2,3)	(3,4)	(4,5)	(5,6)	(6,7)
CDL	7456	3767	3227	2615	1958	1238	644
Distance (m)	1100	200	350	320	420	120	350
VD%	0.54	0.05	0.07	0.05	0.05	0.01	0.01

Second: CDL is calculated for each segment in Feeder 2 from the beginning of the feeder (Y=0) to the last substation in the loop (Y=7).

VD% (0,1) = (7456 KVA × 0.36 km) ÷ 15252 = 0.18%

Segment	(0,1)	(1,2)	(2,3)	(3,4)	(4,5)	(5,6)	(6,7)
CDL	7456	6813	6219	5499	4842	4230	3690
Distance (m)	360	350	120	420	320	350	200
VD%	0.18	0.16	0.05	0.15	0.10	0.10	0.05

Example (4):

Calculate total VD % from MV single loop in Example(3)

Total VD % is calculated from the beginning of the loop inside the plan to the last distribution substation on the loop according to the following equation:

$$VD\%_{Total} = \sum_{x=0}^{N} VD\%_{(x,x+1)}$$

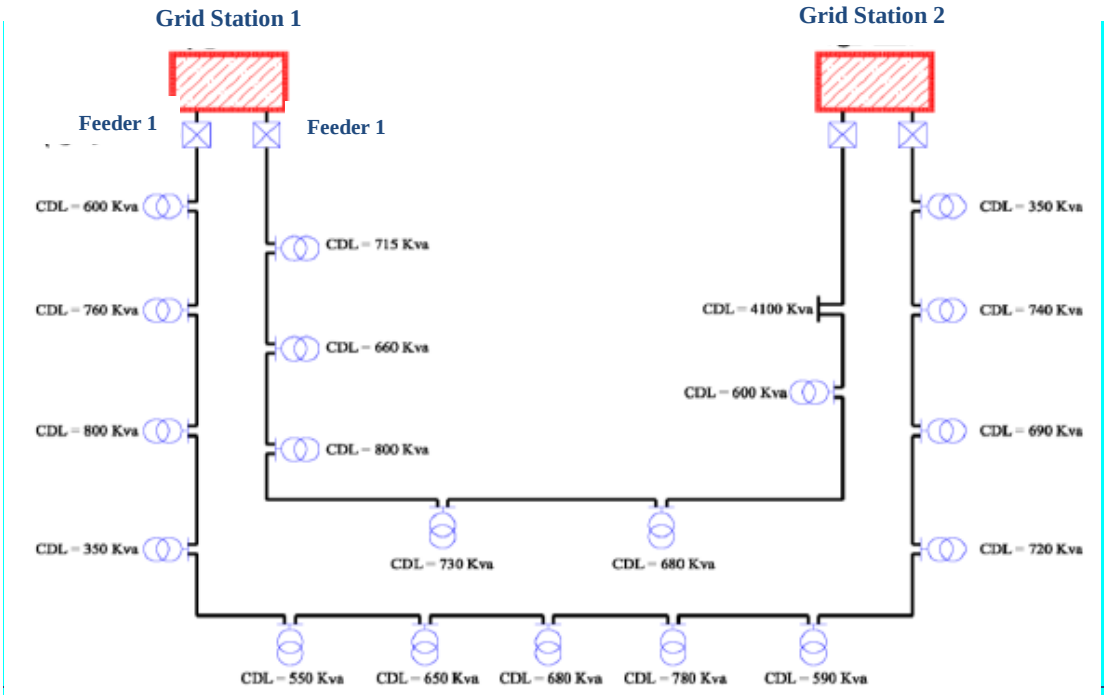
First: VD % is calculated for each segment in Feeder 1 from the beginning of the feeder (X=0) to the lase substation in the loop (X=7)

= (0.53+ 0.05 + 0.07 + 0.05 + 0.05 + 0.01 + 0.01) = 0.79%

Second: CDL is calculated for each segment in Feeder 2 from the beginning of the feeder (Y=0) to the last substation in the loop (Y=7)
= (0.18 + 0.16 + 0.05 + 0.15 + 0.1 + 0.1 + 0.05) = 0.78%

Example (5):

Calculate CDL for plan according to the figure below:



CDL is calculated for the MV single loop from the equation

$$CDL_{on\ MV\ Single\ Loop} = \sum_{i=x+1}^{N} CDL_i \times CF_{For\ Substations}$$

CDL for the first MV single loop = (350 + 740 + 690 + 720 + 590 + 780 + 680 + 680 + 650 + 550 + 350 + 800 + 760 + 600)×0.9 = 7434 KVA

CDL for the second MV single loop = (4100 + 600 + 680 + 730 + 800 + 660 + 715) × 0.9 = 7456 KVA

CDL is calculated for the entire plan according to the following equation:

$$CDL_{for\ Plot\ Plan} = \sum_{i=x+1}^{N} CDL_i \times CF_{on\ MV\ Single\ Loop}$$

CDL for the entire plan = CDL (for the first MV single loop + for the second MV single loop)×0.9 = (7456 + 7434) × 0.9 = 13401 KVA

8. Example illustrates the manual calculation method for voltage regulator

Assume the following overhead circuit: 13.8 kV



Step 1

A voltage drop analysis is performed using the voltage drop calculation guidelines:

$$\textbf{V.D.\%} = \frac{KVA * L}{K}$$

Voltage drop from Grid Station to Node A:

$$\frac{5000 \times 5}{4952} = 5.05\%$$

Voltage drop from Node A to Node B:

$$\frac{3500 \times 2}{4952} = 1.41\%$$

Voltage drop from Node B to C:

$$\frac{1500 \times 3}{4952} = 0.91\%$$

The voltage drops from the G/S to the nodes is the sum of the segment voltage drops:

Voltage drop from G/S to Node A:	5.05%	
Voltage drop from G/S to Node B:	5 .05+ 1.41	= 6.46%
Voltage drop from G/S to Node C:	5.05 + 1.41 + 0.91	= 7.37%

The light load peak demand is estimated to be 50% of the peak demand and is used to calculate the light load voltage drops:

Voltage drop from G/S to Node A:	0.5 × 5.05	= 2.52%
Voltage drop from G/S to Node B:	0.5 × 6.46	= 3.23%
Voltage drop from G/S to Node C:	0.5 × 7.37	= 3.68%

If the grid station bus is set to maintain a constant voltage out of 14200 volts (102.9%), then peak load voltages at the primary of the customer's substation will be:

	Peak load	Light load
Voltages at G/S:	102.9%	102.9%
Voltages at Node A:	97.9%	100.4%
Voltages at Node B:	96.4%	99.7%
Voltages at Node C:	95.5%	99.2%

Step 2

The minimum customer voltage should be at 95%. For this to happen, the primary side voltage needs to be 102.5% (95% + 2.5% to account for distribution transformer voltage drop + 5% to account for secondary service voltage drop.

In this example, to cover the voltage drop of the entire line (at peak load), the primary voltage at the grid station needs to be 109.87% (102.5% + 7.37%) which is higher than the permissible service voltage limits (which is set at +/- 5%). Conversely, if we set primary voltage at grid station at 105% (which is the maximum permissible limit), the voltage at node A will be 99.95% (105% - 5.05%) which will result in voltage at customer to be 92.45% (99.95% - 2.5% - 5%) which is lower than permissible limits.

To correct this situation, a voltage regulator may be placed between grid station and node A which is set at 105%. On the other hand, placing the voltage regulator between node A and B or between node B and C will not take into account the voltage drop at node A.

Step 3

The voltage drop at peak load and light load should be calculated to ensure there is no over-voltage. In this example, the voltage profile will look as follows:

	Peak load	Light load
G/S Bus	102.9%	102.9%
Node A	105%	105%
Node B	103.6%	104.3%
Node C	102.7%	103.8%

9. Example Illustrating Manual Calculation Method for capacitor

Assume the following overhead circuit: where the voltage profile as example shown at previous section.



Peak	102.9%	97.9%	96.4%	95.5%
Light	102.9%	100.4%	99.7%	99.2%

The calculation done based on pf = 0.85

The MVAR flow in segment as below



If we install capacitor bank 1800 kVAR to compensate kVAR in segment AB then the MVA will change as:



Note: After installing 1800 kVAR at node B, MVA in segment G/S-A reduced from 5.0 MVA to 4.3 MVA and MVA in segment A-B reduced from 3.5 MVA to 2.97 MVA with the same active load.

VD at peak load

VD from G/S to Node A	(4300 × 5) / 4952	= 4.3%
VD from Node A to Node B	(2970 × 2) / 4952	= 1.2%
VD from Node B to Node C	(1500 x 3) / 4952	= 0.9%
VD at Node C		= 6.4%

During light load, if we keep 1800 kVAR capacitor the power factor become leading so that we keep only 900 kVAR and remove 900 kVAR to improve power factor but still lagging.



VD from G/S to point A = (2160 × 5) / 4952 = 2.2%
VD from point A to point B = (1487 × 2) / 4952 = 0.6%
VD from point B to point C = (750 × 3) / 4953 = 0.45

VD from G/S to Node A	(2160 × 5) / 4952	= 2.2%
VD from Node A to Node B	(1487 × 2) / 4952	= 0.6%
VD from Node B to Node C	(750x 3) / 4952	= 0.45%
VD at Node C		= 3.25%

	G/S	A	B	C
Voltage during peak load	102.9%	98.6%	97.4%	96.5%
Light load (900 KVAR)	102.9%	100.7%	100.1%	99.7%

Note:

1. The design of capacitor must meet light load to avoid over voltage.
2. During voltage drop occur we use first capacitor method then voltage regulator.

Forms

From number	Form name
DPS Form-01	Load declarations by customers – SEC enquiry
DPS Form-02	Coincident dmand load calculation (CDL)
DPS Form-03	Site visit check list
DPS Form-04	Substation or meter room check list
DPS Form-05	Voltagedrop calculation
DPS Form-06	Reinforcment form
DPS Form-07	Inetegration form
DPS Form-08	Replacement form
DPS Form -09	Load calculation for plot plans
DPS Form-10	Check list for plot plans
DPS Form-11	Variance analysis

Document Revision Log Template	Doc Code	Date	Rev#	الشركة السعودية للكهرباء Saudi Electricity Company
	DPS	NOV.,2023	REV 2	

Ver#	Document Name	Description of Change	Rev Dt	Change done by	
				Name	Badge
DPS REV 2	DISTRIBUTION PLANNING STANDARD	Update item 7.5.1 UNDERGROUND connection schemes	NOV.,2023	ALSADI ABDULLAH	96559
DPS REV 2	DISTRIBUTION PLANNING STANDARD	UPDATE ITEM 7.5.2 RURAL SUPPLY	NOV.,2023	ALSADI ABDULLAH	96559
DPS REV 2	DISTRIBUTION PLANNING STANDARD	UPDATE ITEM 8.3 NETWORK REPLACEMENT	NOV.,2023	SAUD A. ALMULHEM	84003
DPS REV 2	DISTRIBUTION PLANNING STANDARD	ADD ITEM 9.4 DISTRIBUTION AUTOMATION CRITERIA AND SMART SWITCHES/EQUIPMENT	NOV.,2023	ABDULLAH B. ALBARGI	69955
DPS REV 2	DISTRIBUTION PLANNING STANDARD	UPDATE ITEM 11.1 DIMENSIONS OF NETWORK PLANNING STRATEGY	NOV.,2023	SAUD A. ALMULHEM	84003
DPS REV 2	DISTRIBUTION PLANNING STANDARD	UPDATE ITEM 11.3 LOAD FORECASTING GUIDELINES AND ADD VARIANCE ANALYSIS	NOV.,2023	BASSAM O. BASUNAID	69780
DPS REV 2	DISTRIBUTION PLANNING STANDARD	UPDATE GOVERNANCE PROCESS FOR UPDATE OF DPS	NOV.,2023	ALSADI ABDULLAH	96559

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